

AICC
25.300.1

CC-CMS Interface Reference (RESTful)

Issue Date **2025-12-09**



Copyright © Huawei Technologies Co., Ltd. 2025. All rights reserved.

No part of this document may be reproduced or transmitted in any form or by any means without prior written consent of Huawei Technologies Co., Ltd.

Trademarks and Permissions



HUAWEI and other Huawei trademarks are trademarks of Huawei Technologies Co., Ltd.

All other trademarks and trade names mentioned in this document are the property of their respective holders.

Notice

The purchased products, services and features are stipulated by the contract made between Huawei and the customer. All or part of the products, services and features described in this document may not be within the purchase scope or the usage scope. Unless otherwise specified in the contract, all statements, information, and recommendations in this document are provided "AS IS" without warranties, guarantees or representations of any kind, either express or implied.

The information in this document is subject to change without notice. Every effort has been made in the preparation of this document to ensure accuracy of the contents, but all statements, information, and recommendations in this document do not constitute a warranty of any kind, express or implied.

Huawei Technologies Co., Ltd.

Address: Huawei Industrial Base
Bantian, Longgang
Shenzhen 518129
People's Republic of China

Website: <https://www.huawei.com>

Email: support@huawei.com

Contents

Contents.....	1
1 About This Document	5
2 Change History	6
3 Overview.....	29
3.1 Function Description.....	29
3.2 Interface Parameter Usage Principle	30
3.3 Interface Encoding Description	30
4 Interface Description	30
4.1 Real-Time Data Query	31
4.1.1 Real-Time Interfaces for VDNs.....	31
4.1.1.1 Querying Information About System Access Codes in a Specified VDN.....	31
4.1.1.2 Querying Agent Rest Information in a Specified VDN	34
4.1.1.3 Querying Agent Phone Numbers in a Specified VDN.....	39
4.1.1.4 Querying Information About All Agents in a Specified VDN	42
4.1.1.5 Querying Information About the Current Status of All Agents in a Specified VDN	46
4.1.1.6 Querying Information About Static Configurations of All Agents in a Specified VDN	49
4.1.1.7 Querying the IVR Status Details of a Specified VDN	54
4.1.1.8 Querying Information About IVR Flow Access Codes in a Specified VCC.....	57
4.1.1.9 Querying Information about Each Skill Queue in a Specified VCC.....	60
4.1.1.10 Querying All Rest Reason Codes in a Specific VDN	63
4.1.1.11 Querying Information About Calls of a Specified Access Code	66
4.1.1.12 Querying Call Information in a Specified VDN.....	69
4.1.1.13 Querying Call Information in a Specified VDN (V2).....	73
4.1.1.14 Querying the Call track Data in a Specific VDN.....	77
4.1.1.15 Querying Information About Static Configurations of All Agents in a Specified VDN (Not Supported by CTI 3.6)	80
4.1.1.16 Querying Information About All Agents in a Specified VDN (Not Supported by CTI 3.6)	85
4.1.1.17 Obtaining Real-Time Monitoring Indicators by VDN	87
4.1.2 Real-Time Interfaces for Skill Queues	92
4.1.2.1 Querying Signed-in Agents in Skill Queues in Batches.....	92
4.1.2.2 Querying Extended Information About Skill Queues in Batches	95
4.1.2.3 Querying the Call Statistics of Skill Queues in the Current 5 Minutes in Batches	99
4.1.2.4 Querying Information About the Status of a Specified Skill Queue	103
4.1.2.5 Querying Information About Skill Queue Configurations in Batches.....	106
4.1.2.6 Querying the Statistics Details of Skill Queues in Batches.....	109
4.1.2.7 Querying All Calls Queuing in a Skill Queue in Batches	115

4.1.2.8 Querying the Number of Calls that Each Queue Answers from Other Queues in Batches When Calls are Queuing in Multiple Skill Queues	119
4.1.2.9 Querying the Number of Agents in Each State in a Skill Group	123
4.1.2.10 Querying Details About Signed-in Agents in a Skill Queue.....	126
4.1.2.11 Querying Service Quality for Skill Queues in Batches	130
4.1.2.12 Obtaining Real-Time Monitoring Indicators by Skill Queue	133
4.1.3 Real-Time Interfaces for Agents.....	141
4.1.3.1 Querying Skill Queues to Which Agents Sign In in Batches	141
4.1.3.2 Querying Agent Information in Batches	145
4.1.3.3 Querying the Current Call ID of a Specified Agent.....	149
4.1.3.4 Querying Call Information About All Agents in Batches.....	151
4.1.3.5 Querying Private Calls of a Specified Agent.....	156
4.1.3.6 Querying the Number of Agents in the Agent State in batches	158
4.1.3.7 Querying Details of a Specified Agent.....	162
4.1.3.8 Querying Agent Rest Information	166
4.1.3.9 Checking Whether an Agent Is Processing a Call.....	170
4.1.3.10 Querying System Status by Agent	172
4.1.3.11 Obtaining Real-Time Monitoring Indicators by Agent	175
4.1.3.12 Obtaining Agent Operation Details	180
4.1.3.13 Querying Agent Information by Agent Status in Pagination Mode.....	185
4.1.4 Real-Time Interfaces for Call Details.....	188
4.1.4.1 Querying Basic Information About a Call with a Specified Call ID.....	188
4.1.4.2 Querying Private Call Details of a Specified Agent.....	193
4.1.4.3 Querying CDRs of a Specified Call ID	197
4.2 Historical Data Query	202
4.2.1 Obtaining Historical VCC Monitoring Indicators	202
4.2.2 Querying Information About Missed Calls in a Specified VCC.....	207
4.2.3 Obtaining Historical IVR Monitoring Indicators.....	210
4.2.4 Obtaining Historical Monitoring Indicators by Skill Queue.....	214
4.2.5 Obtaining Historical Monitoring Indicators by Agent.....	220
4.2.6 Obtaining Historical Monitoring Indicators in a Specified Time Range by Skill.....	226
4.2.7 Obtaining Historical Monitoring Indicators in a Specified Month Range by Skill	230
4.2.8 Obtaining Historical Monitoring Indicators in a Specified Time Range by Agent	234
4.2.9 Obtaining Historical Monitoring Indicators in a Specified Month by Agent	239
4.2.10 Collecting Statistics on Result Sets in Daily Settlement Table for Inbound Calls.....	243
4.2.11 Collecting Statistics on Result Sets in Daily Settlement Table for Outbound Calls	253
4.2.12 Obtaining Agent Rest Monitoring Indicators of the Current Day or the Previous Day	266
4.3 Configuration Data Query.....	269
4.3.1 Querying Information about All VDNs in a Specified Call Center.....	269

4.3.2 Querying Information About All Agents by VDN ID	272
4.3.3 Obtaining All Called Routes in a VDN.....	275
4.3.4 Querying Information About All Agents by Tenant Space ID	278
4.3.5 Querying Skill Groups in a Specified VDN	281
4.3.6 Querying Information About All Tenants Accessible to an Access User	283
4.4 Inspection Relationship Maintenance Interfaces.....	286
4.4.1 Querying a User Group Member.....	286
4.4.2 Deleting Members from a Group in Batches.....	290
4.4.3 Adding Members to a Group in Batches	292
4.4.4 Querying All User Groups.....	294
4.4.5 Adding a User Group.....	297
4.4.6 Deleting a User Group	300
4.4.7 Updating User Group Information	302
4.4.8 Setting Inspection Relationships.....	304
4.4.9 Deleting Configured Inspection Relationships.....	306
4.4.10 Adding an Agent to a User Group.....	308
4.4.11 Updating Agent Names	311
4.5 Querying Outbound Call Task Lists	314
4.5.1 Querying the Outbound Call Task List of a Tenant.....	314
4.5.2 Querying the Daily Settlement Statistics of a Specified Task	317
4.5.3 Querying the Real-Time Statistics of a Specified Task	320
4.6 Interfaces for Querying System Indicators	322
4.6.1 Querying UAP Performance Information	322
4.6.2 Querying Key Information About Control Blocks.....	328
4.6.3 Querying the Registration Status of a Terminal.....	333
4.6.4 Interface for Querying UAP Performance Indicator Statistics.....	337
4.6.5 Interface for Querying UAP Terminal Statistics	340
4.6.6 Interface for Querying CTI System Indicators.....	342
4.6.7 Querying the Time of the CTI Platform Server	345
4.6.8 Querying the Current License Usage in the System.....	347
4.7 Checking Cluster Health Status.....	351
4.8 Interface for Uploading Logs	353
4.9 Downloading Inspection Recording Files	354
5 Appendix	356
5.1 Indicator Code Reference	356
5.1.1 Skill Queue Monitoring Indicators.....	356
5.1.2 Agent Monitoring Indicators.....	363
5.1.3 VDN Monitoring Indicators	373
5.1.4 IVR Monitoring Indicators.....	381

5.2 Error Code Reference	381
5.3 CTI Platform Agent Status Code Reference	382
5.4 Description of Call Media Types	382
5.5 Description of Call Submedia Types	384
5.6 Description of Call Types	384
5.7 Agent Type Description	389
5.8 Call Features	390
5.9 Reasons for Leaving the Device of the Call.....	390
5.10 Cause Code.....	393
5.11 Device Types.....	393
5.12 Agent Operation Types.....	393
5.13 Interfaces of Earlier Versions (Versions Earlier Than 8.13.0)	398
5.13.1 Querying Details of a Specified Agent.....	398
5.13.2 Querying System Status by Agent.....	401
5.13.3 Querying the Agent Information in Rest State.....	403
5.13.4 Querying Statistics Details of Skill Queues in Batches.....	406
5.13.5 Querying Details About Agents in Configured Skill Queues.....	410

1 About This Document

Overview

This document describes the CC-CMS interfaces of the WebService type provided by the CTI platform and its usage instructions for third-party service customization personnel.

Intended Audience

This document is intended for third-party business customization personnel.

Symbol Conventions

The symbols that may be found in this document are defined as follows.

Symbol	Description
 DANGER	Indicates an imminently hazardous situation which, if not avoided, will result in death or serious injury.
 WARNING	Indicates a potentially hazardous situation which, if not avoided, could result in death or serious injury.
 CAUTION	Indicates a potentially hazardous situation which, if not avoided, may result in minor or moderate injury.
 NOTICE	Indicates a potentially hazardous situation which, if not avoided, could result in equipment damage, data loss, performance deterioration, or unanticipated results. NOTICE is used to address practices not related to personal injury.
 NOTE	Calls attention to important information, best practices and tips. NOTE is used to address information not related to personal injury, equipment damage, and environment deterioration.

Privacy statement

This product provides secondary development capabilities and some of the interfaces are involved in processing customers' personal data (such as mobile numbers and consumption records). You are allowed to provide or develop corresponding functions based on the business requirements and within the purpose and scope permitted by laws and regulations. During the customization development, you are obligated to take considerable measures to ensure that the personal data of customers and content for communication are fully protected. The measures include but are not limited to the following:

- Collect and process individual data only for required businesses.
- Do not provide hidden channels for accessing the system without being authenticated based on normal mechanisms.

- Do not record sensitive data that may cause economic loss to customers, such as user password (plaintext or ciphertext) and bank account, in logs.

2 Change History

Issue 16 (2025-09-30)

This issue is released with AICC 25.300.0. The following interfaces are modified.

No.	Interface	Type	Remarks
1	All interfaces in the real-time interfaces for skill queues	Modified	The value range of the skill queue ID is extended.
2	https://ip:port/rest/cmsapp/v1/openapi/vdn/queryAgentByStatus	New	Description of large agent ID adaptation is added.

7

Issue 16 (2025-06-30)

This issue is released with AICC 25.200.0. The following interfaces are modified.

No.	Interface	Type	Remarks
1	https://ip:port/rest/cmsapp/v1/openapi/calldata/querybasiccallinfo https://ip:port/rest/cmsapp/v1/openapi/calldata/querydetailcallinfo https://ip:port/rest/cmsapp/v1/openapi/configures/skills https://ip:port/rest/cmsapp/v1/openapi/agentgroup/queryagentcalls https://ip:port/rest/cmsapp/v1/openapi/agent/agentoprinfo	New	The SubMediaType field is added to responses.

Issue 15 (2025-03-31)

This issue is released with AICC 25.100.0. The following interfaces are modified.

No.	Interface	Type	Remarks
1	https://ip:port/rest/\${service.name}/v1/ClientLog/uploadClientLog	New	The interface for uploading logs is added.

Issue 14 (2025-01-15)

This issue is released with AICC 24.400.0 and has no changes.

Issue 13 (2024-09-30)

This issue is released with AICC 24.300.0. The following interfaces are modified.

No.	Interface	Type	Remarks
1	https://ip:port/rest/cmsapp/v1/openapi/hisindex/vdn https://ip:port/rest/cmsapp/v1/openapi/hisindex/ivr https://ip:port/rest/cmsapp/v1/openapi/hisindex/skill https://ip:port/rest/cmsapp/v1/openapi/hisindex/agent	Modified	The description of mandatory parameters for the timeGrain field is added.

Issue 12 (2024-06-30)

This issue is released with AICC 24.200.0. The following interfaces are modified.

No.	Interface	Type	Remarks
1	https://ip:port/rest/cmsapp/v1/openapi/hisindex/agent https://ip:port/rest/cmsapp/v1/openapi/hisindex/vdn https://ip:port/rest/cmsapp/v1/openapi/hisindex/ivr https://ip:port/rest/cmsapp/v1/openapi/hisindex/skill	Modified	Due to limited performance, the real-time cache data is not queried by default. Only the data generated 5 minutes before the current time can be viewed. For example, if the current time is 14:32, the interface can obtain only the data generated before 14:30.

Issue 11 (2023-11-15)

This issue is released with AICC 23.300.0. The following interfaces are modified.

No.	Interface	Type	Remarks
1	https://ip:port/rest/cmsapp/v1/openapi/vdn/querycalls https://ip:port/rest/cmsapp/v2/openapi/vdn/querycalls https://ip:port/rest/cmsapp/v1/openapi/agent/agentoprinfo https://ip:port/rest/cmsapp/v1/openapi/calldata/querycalldetailbyid	Modified	The retention period of the queried data is added in Scenario .

Issue 10 (2023-07-30)

This issue is released with AICC 23.200.0. The following interfaces are modified.

No.	Interface	Type	Remarks
1	https://ip:port/rest/cmsapp/v1/openapi/calldata/querybasiccallinfo	Modified	The description of the request body parameter callId is added. The mediaBillType (multimedia report type) parameter is added to the response body.
2	https://ip:port/rest/cmsapp/v1/openapi/hisindex/vdn	Modified	The description of the response body parameter val is added.

No.	Interface	Type	Remarks
3	https://ip:port/rest/cmsapp/v1/openapi/hisindex/ivr	Modified	The description of the response body parameter val is added.
4	https://ip:port/rest/cmsapp/v1/openapi/hisindex/skill	Modified	The description of the response body parameter val is added.

Issue 09 (2023-04-30)

This issue is released with AICC 23.100.0. The following interfaces are modified.

No.	Interface	Type	Remarks
1	https://ip:port/rest/cmsapp/v1/openapi/hisindex/vdn	Modified	The following agent monitoring indicator parameters are added to the interface for obtaining historical monitoring indicators by VCC: IDX_COUNT_SYSTEM_INBOUND: number of inbound calls on the platform IDX_COUNT_SYSTEM_IN_SUCC: number of connected calls on the platform IDX_COUNT_AGENT_ASSIGNED: number of manual connected calls on the platform
2	https://ip:port/rest/cmsapp/v1/openapi/hisindex/agent	Modified	The following VCC monitoring indicator parameters are added to the interface for obtaining historical monitoring indicators by agent: IDX_COUNT_AGENT_ARRANGE: number of wrap-up times IDX_DURATION_AGENT_ARRANGE: wrap-up duration IDX_COUNT_AGENT_PLAY: number of voice playback times IDX_DURATION_AGENT_PLAY: voice playback duration
3	https://ip:port/rest/cmsapp/v1/openapi/vdn/querycalls	Modified	The mediaType (media type) and subMediaType (submedia type) parameters are added to the response body.

No .	Interface	Type	Remarks
4	https://ip:port/rest/cmsapp/v2/openapi/vdn/querycalls	Modified	The mediaType (media type) and subMediaType (submedia type) parameters are added to the response body.
5	https://ip:port/rest/cmsapp/v1/openapi/vdn/querycalltracedetail	Modified	The subMediaType (submedia type) parameter is added to the response body.
6	https://ip:port/rest/cmsapp/v1/callin/queryMissedCalls	Modified	The mediaType (media type) and subMediaType (submedia type) parameters are added to the response body.

Issue 08 (2023-01-30)

This issue is released with AICC 22.200.0. The following interfaces are modified.

No .	Interface	Type	Remarks
1	The Internal Encapsulation Interface subsection is added for some interfaces.	New	The Internal Encapsulation Interface subsection is added for interfaces that invokes the BMS module to query data, and the name of the internally invoked interface is added.
2	https://ip:port/rest/cmsapp/v1/openapi/vdn/queryagentreststate	Modified	The description that the value must be greater than or equal to 0 is added to offset . The data type of resultDesc in the response body is changed from array to object. The description of waitRestAgentNum in resultDesc is changed from "number of agents who wait for a rest" is changed to "number of agents who apply for a rest." The

No	Interface	Type	Remarks
.			description of beginTime in the agent rest details is changed from "this parameter indicates the rest start time for resting agents, and the request rest time for agents who request for a rest or are allowed to have a rest" to "this parameter indicates the rest start time for resting agents, and the start time of applying for or approving the rest for agents not resting." The unit of applyRestLength in the agent rest details is changed from minute to second.
3	https://ip:port/rest/cmsapp/v1/openapi/vdn/allagentphones	Modified	The description that the value must be greater than or equal to 0 is added to offset . The description of phone in resultDesc is changed from "phone number used by an agent to sign in" to "agent phone number."
4	https://ip:port/rest/cmsapp/v1/openapi/vdn/totalagentabstractex3	Modified	The description that the value ranges from 1 to 100 in CTI 3.6 is deleted for limit . The description that the value must be greater than or equal to 0 is added to offset .

No	Interface	Type	Remarks
5	https://ip:port/rest/cmsapp/v1/openapi/vdn/totalagentdynamicinfo	Modified	The description that the value must be greater than or equal to 0 is added to offset .
6	https://ip:port/rest/cmsapp/v1/openapi/vdn/queryagentstaticconfinfo	Modified	The description that the value must be greater than or equal to 0 is added to offset .
7	https://ip:port/rest/cmsapp/v1/openapi/vdn/queryivrstatusinfo	Modified	The value range of the request parameter flowId is changed from "1-100000" to "1-1500."
8	https://ip:port/rest/cmsapp/v1/openapi/vdn/queryivrprocessinnoinfo	Modified	The response body example is modified by deleting the data and count attributes. The description of the description attribute in resultDesc is changed from "flow description" to "access code function description."
9	https://ip:port/rest/cmsapp/v1/openapi/vdn/queryaccesscodecallinfo	Modified	The value range (1-100000) of innolds in the request body is deleted.
10	https://ip:port/rest/cmsapp/v1/openapi/vdn/querytotalagentinfo	Modified	The description of the response parameter resultDesc is modified. The description that the value of this parameter is -1 is deleted. The value range of the agent weight is changed from "33, 34, 35,

No	Interface	Type	Remarks
.			
			and 36 to 128" to "33-256."
11	https://ip:port/rest/cmsapp/v1/openapi/realindex/agent/agentsinskill	Modified	The value range (1-2000) of the skill queue ID is added to the skillIds parameter description in the request message body.
12	https://ip:port/rest/cmsapp/v1/openapi/skill/totalskillstatusex3	Modified	The value range (1-2000) of the skill queue ID is added to the skillIds parameter description in the request message body.
13	https://ip:port/rest/cmsapp/v1/openapi/skill/skillcfginfo	Modified	The descriptions of maxWaitTime and maxTalkTime in resultDatas are changed from "time" to "duration."
14	https://ip:port/rest/cmsapp/v1/openapi/skill/stateinfo	Modified	The value range (1-2000) of the skill queue ID is added to the skillIds parameter description in the request message body. The description "including wrap-up state" is added to agentWorkNums in resultDesc . The description of totalCallNums is changed from "number of calls processed per minute" to "total number of processed calls, including the calls that are being processed."

No	Interface	Type	Remarks
15	https://ip:port/rest/cmsapp/v1/openapi/skill/batchskilloverload	Modified	The data type of queueId in overloadDetails is changed from string to integer.
16	https://ip:port/rest/cmsapp/v1/openapi/skill/agentstatusbyskill	Modified	The value range (1–2000) of the skill queue ID is added to the description of the skillIds array in the request. The description "number of business representatives" is changed to "number of agents" for each parameter in resultDesc .
17	https://ip:port/rest/cmsapp/v1/openapi/skill/servicequality	Modified	The description of in20sCallNum in the resultDesc object is changed to "Total number of calls that are connected in 20 seconds in the current 5 minutes in a skill queue."
18	https://ip:port/rest/cmsapp/v1/openapi/agentgroup/siteagentcallandskillinfobyagentids	Modified	The maximum length of the input parameter agentIds is changed to 6000, both in CTI 3.6 and CTI 3.8.
19	https://ip:port/rest/cmsapp/v1/openapi/agentgroup/agentstatusinfobyagentids	Modified	The maximum length of the input parameter agentIds is changed to 6000, both in CTI 3.6 and CTI 3.8. The description of agentRelease in resultDesc is changed from "number of times calls are proactively

No	Interface	Type	Remarks
.			
			released" to "number of times an agent proactively releases calls." The description of currentPrivateCallNum is changed from "number of private calls of an agent" to "current number of private calls of an agent."
20	https://ip:port/rest/cmsapp/v1/openapi/agentgroup/queryagentcalls	Modified	The description that the maximum length is 1000 in the Oracle database is deleted for the input parameter agentIds .
21	https://ip:port/rest/cmsapp/v1/openapi/agent/agentprivatecall	Modified	The descriptions of private calls and how to make private calls are added to the scenario description.
22	https://ip:port/rest/cmsapp/v1/openapi/agentgroup/teamagentstatusinfo	Modified	The maximum length of the input parameter agentIds is changed to 6000, both in CTI 3.6 and CTI 3.8. The description "number of business representatives" is changed to "number of agents" for each parameter in resultDesc .
23	https://ip:port/rest/cmsapp/v1/openapi/agentgroup/agentrestinfos	Modified	The description that this parameter and offset must exist at the same time or neither of them exists is added to the limit

No	Interface	Type	Remarks
.			parameter in the message body. The description of allowRestNum in resultDesc is changed from "number of agents that are allowed to rest" to "number of agents who are allowed to have a rest but still at work." The description of onRestNum is changed from "number of agents who take a rest" to "number of agents in resting state." The description of the waitRestNum parameter is changed from "number of agents that are waiting for a rest" to "number of agents who apply for a rest." The mapping table is added to cantRestReason in agentsrestdetails. The description of beginTime is changed to "this parameter indicates the rest start time for resting agents, and the start time of applying for or approving the rest for agents not resting." The time format is timestamp.
24	https://ip:port/rest/cmsapp/v1/openapi/agentgroup/systemstate	Modified	The description that the value of offset is greater than or equal to 0 is added. The following

No	Interface	Type	Remarks
.			descriptions are changed for the parameters in resultDesc: "busy" is changed to "busy state" for busyAgent; "idle" is changed to "idle state" for idleAgent; and "total number of login agents" is changed to "number of signed-in agents" for totalLoginAgent.
25	https://ip:port/rest/cmsapp/v1/openapi/calldata/querydetailcallinfo	Modified	The interface name and scenario description are changed from "querying details of a specified call ID" to "querying details of a private call of a specified agent." The description of callId in the message body is changed from "Call ID. The value is the call ID in a call and contains a maximum of 20 characters." to "Call ID. The value is the private call ID of a specified agent and contains a maximum of 20 characters. For details about how to obtain the value, see the interface for querying private calls of a specified agent." The description of waitTime in resultDesc is changed from "duration that a call waits in a queue" to "waiting

No	Interface	Type	Remarks
.			duration in a queue."
26	https://ip:port/rest/cmsapp/v1/openapi/rindex/vdn	Modified	The maximum length of the vdnIds array in the request body is changed from 2 to 100, and the value range (1–5000) is added to vdn .
27	https://ip:port/rest/cmsapp/v1/openapi/agentgroup/rindex	Modified	The description of indexIds in the request body is modified by adding the description that the maximum number of indicators is 4.
28	https://ip:port/rest/cmsapp/v1/openapi/skill/rindex	Modified	The description of indexIds in the request body is modified by adding the description that the maximum number of indicators is 12.
29	https://ip:port/rest/cmsapp/v1/openapi/vdn/queryAgentByStatus	Modified	The value range of the cclId request parameter is changed to 1–65535, and that of vdn is changed to 1–5000. The description that the value must be greater than or equal to 0 is added to offset . The parameter value mapping link of status in the request body is changed to CTI Platform Agent Status Code Reference , and that of currentState in data in the response is also

No	Interface	Type	Remarks
.			changed to CTI Platform Agent Status Code Reference .
30	https://ip:port/rest/cmsapp/v1/openapi/querymonitor/systemstate	Modified	The following descriptions are changed for the parameters in resultDatas : "busy" is changed to "busy state" for busyagent ; "idle" is changed to "idle state" for idleagent ; and "total number of login agents" is changed to "number of signed-in agents" for totalloginagent .
31	https://ip:port/rest/cmsapp/v1/openapi/agentgroup/agentrestinfo	Modified	The description of allowrestnum in resultDatas is changed from "number of agents that are allowed to rest" to "number of agents who are allowed to have a rest but still at work." The description of onrestnum is changed from "number of agents who take a rest" to "number of agents in resting state." The description of the waitrestnum parameter is changed from "number of agents that are waiting for a rest" to "number of agents who apply for a rest." The mapping table is added to cantrestreason of agentsrestdetails .

No	Interface	Type	Remarks
.			
			The description of begintime is changed to "this parameter indicates the rest start time for resting agents, and the start time of applying for or approving the rest for agents not resting." The time format is timestamp.
32	https://ip:port/rest/cmsapp/v1/openapi/realindex/skill/stateinfo	Modified	The data type of resultDatas in the response body is changed from long to integer. The description of totalCallNums is changed from "number of calls processed per minute" to "total number of processed calls, including the calls that are being processed."
33	https://ip:port/rest/cmsapp/v1/openapi/agentgroup/agentbyconfiguredskill	Modified	The result mapping description is added to the status parameter in resultDatas .
34	Interfaces that contain the input parameter offset and the description "Value of offset = Value of limit x (Value of page - 1)"	Modified	The description of "value range" is changed to "value formula". The description that the value must be greater than or equal to 0 is added.
35	https://ip:port/rest/cmsapp/v1/openapi/agentgroup/agentrestinfos	Modified	The following description is added for limit and offset : This field takes effect only in CTI 3.6.

Issue 07 (2022-07-30)

This issue is released with AICC 22.100.0. The following interfaces are modified.

No.	Interface	Type	Remarks
1	https://ip:port/rest/cmsapp/v1/openapi/hisindex/vdn	Modified	The following indicators are added: IDX_UNIQUE_COUNT_IVR_INBOUND IDX_UNIQUE_COUNT_IVR_INBOUND_CONNECTED IDX_UNIQUE_COUNT_IVR_OUTBOUND IDX_UNIQUE_COUNT_IVR_OUTBOUND_CONNECTED
2	https://ip:port/rest/cmsapp/v1/openapi/qcrelationconf/batchdeletegroupmember https://ip:port/rest/cmsapp/v1/openapi/qcrelationconf/batchaddgroupmember https://ip:port/rest/cmsapp/v1/openapi/qcrelationconf/createusergroup https://ip:port/rest/cmsapp/v1/openapi/qcrelationconf/querygroupmemberbycondition https://ip:port/rest/cmsapp/v1/openapi/qcrelationconf/joinusergroup https://ip:port/rest/cmsapp/v1/openapi/qcrelationconf/updateagentgroup https://ip:port/rest/cmsapp/v1/openapi/agent/agentoprinfo https://ip:port/rest/cmsapp/v1/openapi/hisindex/agent https://ip:port/rest/cmsapp/v1/openapi/hisindex/agentforday https://ip:port/rest/cmsapp/v1/openapi/hisindex/agentformonth	Modified	Values of the request parameters workNo , workNos , agentId , agentIds and currentAgentId that indicate the agent ID are restricted to 0 to 999999.
3	https://ip:port/rest/cmsapp/v1/openapi/hisindex/agent https://ip:port/rest/cmsapp/v1/openapi/hisindex/agentforday https://ip:port/rest/cmsapp/v1/openapi/hisindex/agentformonth	Modified	Two monitoring indicators are added: IDX_COUNT_TOTAL_AGENT_CALL_OUT (total number of outbound calls) IDX_RATE_AGENT_SUCC_CALL_OUT (success rate of outbound calls)
4	https://ip:port/rest/cmsapp/v1/callinday/querymanualindexbycondition	Modified	The following response parameters are added: inCallAllCostMinute (total charging duration of each call, in minutes) The called numbers in the calleeNos parameter (an array of called numbers) and the call types in the

No.	Interface	Type	Remarks
			callTypes parameter (an array of integers that indicate call types) can contain only digits.
5	https://ip:port/rest/cmsapp/v1/calloutday/querymanualindexbycondition	Modified	The following response parameters are added: outCallAllCostMinute (total charging duration of each call, in minutes) The calling numbers in the callerNos parameter (an array of calling numbers) and the call types in the callTypes parameter (an array of integers that indicate call types) can contain only digits.
6	https://ip:port/rest/cmsapp/v2/openapi/vdn/querycalls	New	(V2) Interface for querying call information of a specified VCC.
7	https://ip:port/rest/cmsapp/v1/openapi/vdn/queryAgentByStatus	New	Interface for querying agent information in pagination mode based on the agent status.
8	https://ip:port/rest/cmsapp/v1/openapi/hisindex/skill https://ip:port/rest/cmsapp/v1/openapi/hisindex/skillforday https://ip:port/rest/cmsapp/v1/openapi/hisindex/skillformonth	Modified	Interface for obtaining historical monitoring indicators by skill. Interface for obtaining historical monitoring indicators by skill on a specified date. Interface for obtaining historical monitoring indicators by skill in a specified month. The following monitoring indicator is added for the preceding three interfaces: IDX_COUNT_SKILL_INCOMING_CALL (Number of manual inbound call requests)
9	https://ip:port/rest/cmsapp/v1/openapi/agent/agentoprinfo	Modified	Interface for obtaining agent operation details. The operateType parameter in the request is changed from mandatory to optional.
10	https://ip:port/rest/cmsapp/v1/openapi/hisindex/vdn https://ip:port/rest/cmsapp/v1/openapi/hisindex/ivr https://ip:port/rest/cmsapp/v1/openapi/hisindex/skill https://ip:port/rest/cmsapp/v1/openapi/hisindex/agent https://ip:port/rest/cmsapp/v1/openapi/hisindex/skillforday https://ip:port/rest/cmsapp/v1/openapi/hisindex/skillformonth	Modified	The value 55 (H5 video) is added to the mediaType parameter in the request.

N o.	Interface	Type	Remarks
	https://ip:port/rest/cmsapp/v1/openapi/hisindex/agentform https://ip:port/rest/cmsapp/v1/openapi/hisindex/agentformonth		

Issue 06 (2022-01-30)

This issue is released with AICC 8.17.0. The following interfaces are modified.

N o.	Interface	Type	Remarks
1	https://ip:port/rest/cmsapp/v1/openapi/skillgroups/{ccld}/{vdnld}	New	The interface for querying skill groups in a specified VCC is added.
2	https://ip:port/rest/cmsapp/v1/openapi/uap/queryUapPerfInfo https://ip:port/rest/cmsapp/v1/openapi/uap/queryterminalstatus https://ip:port/rest/cmsapp/v1/openapi/uap/dsppidkey https://ip:port/rest/cmsapp/v1/openapi/agent/agentopinfo https://ip:port/rest/cmsapp/v1/callinday/querymanualindexesbycondition https://ip:port/rest/cmsapp/v1/calloutday/querymanualindexesbycondition https://ip:port/rest/cmsapp/v1/openapi/tenant/querytenantdetail	Modified	The optional request parameter ccld is added.
3	https://ip:port/rest/cmsapp/v1/openapi/skill/agentstatusbyskill	Modified	The optional request parameter isSkillAllMatch (whether agent skills are fully matched) is added.
4	https://ip:port/rest/cmsapp/v1/openapi/hisindex/vdn https://ip:port/rest/cmsapp/v1/openapi/hisindex/ivr https://ip:port/rest/cmsapp/v1/openapi/hisindex/agent https://ip:port/rest/cmsapp/v1/openapi/hisindex/skill	Modified	The optional request parameter enableOffset is added.
5	https://ip:port/rest/cmsapp/v1/openapi/vdn/querycalls https://ip:port/rest/cmsapp/v1/openapi/vdn/queryagentreststate https://ip:port/rest/cmsapp/v1/openapi/qcrelationconf/querygroupmemberbycondition https://ip:port/rest/cmsapp/v1/openapi/uap/perfindex	Modified	The pagination parameter verification is added: The value of limit cannot be less than or equal to 0. The value of offset cannot be less than 0.

N o.	Interface	Type	Remarks
	https://ip:port/rest/cmsapp/v1/openapi/uap/terminalindex		
6	https://ip:port/rest/cmsapp/v1/openapi/agent/agentprofile	Modified	The time range specified by the mandatory input parameters beginTime and endTime is changed from an integer multiple of 30 minutes to an integer multiple of 15 minutes.
7	https://ip:port/rest/cmsapp/v1/openapi/hisindex/agent	Modified	The agentIds parameter verification is added: agentIds must belong to the VCC specified by the input parameter.
8	https://ip:port/rest/cmsapp/v1/openapi/hisindex/skill	Modified	Querying IDX_COUNT_SKILL_CONNECTED_IN(N) can be queried by minute.
9	https://ip:port/rest/cmsapp/v1/openapi/hisindex/skill https://ip:port/rest/cmsapp/v1/openapi/hisindex/skillfor day https://ip:port/rest/cmsapp/v1/openapi/hisindex/skillfor month	Modified	The following indicators are added: IDX_COUNT_SKILL_USER_ABANDON_QUEUE IDX_COUNT_SKILL_USER_ABANDON_AGENT IDX_DURATION_SKILL_ABANDON_QUEUE IDX_DURATION_SKILL_ABANDON_ACK IDX_DURATION_SKILL_WAIT_SUCCESS_CONNECTED_QUEUE IDX_DURATION_SKILL_SUCCESS_ACK IDX_DURATION_SKILL_ALL_ACK IDX_COUNT_SKILL_WAIT_FAIL_CONNECTED_QUEUE IDX_DURATION_SKILL_ALL_QUEUE IDX_DURATION_SKILL_CONNECTED
10	https://ip:port/rest/cmsapp/v1/openapi/hisindex/agentfor day	Modified	The logic for querying the IDX_DURATION_AGENT_AVG_CALL_OUT indicator is modified. The logic for querying the IDX_DURATION_AGENT_AVG_CALL_IN indicator is modified.

Issue 05 (2021-10-15)

This issue is released with AICC 8.16.0. The following interfaces are modified.

No.	Interface	Type	Remarks
1	https://ip:port/rest/cmsapp/v1/openapi/hisindex/vdn https://ip:port/rest/cmsapp/v1/openapi/hisindex/ivr https://ip:port/rest/cmsapp/v1/openapi/hisindex/skill https://ip:port/rest/cmsapp/v1/openapi/hisindex/agent https://ip:port/rest/cmsapp/v1/openapi/hisindex/skillforday https://ip:port/rest/cmsapp/v1/openapi/hisindex/skillformonth https://ip:port/rest/cmsapp/v1/openapi/hisindex/agentforday https://ip:port/rest/cmsapp/v1/openapi/hisindex/agentformonth	Modified	The optional request parameters mediaType and subMediaType are added.
2	https://ip:port/rest/cmsapp/v1/openapi/uap/queryUapPerfInfo	New	The interface for querying key information about control blocks is added.
3	https://ip:port/rest/cmsapp/v1/openapi/cti/sysindex	New	The interface for querying CTI system indicators is added.
4	https://ip:port/rest/cmsapp/v1/openapi/uap/perfindex	New	The interface for querying UAP performance indicator statistics is added.
5	https://ip:port/rest/cmsapp/v1/openapi/uap/terminalindex	New	The interface for querying UAP terminal statistics is added.
6	https://ip:port/rest/cmsapp/v1/openapi/skill/agentstatusbyskill	Modified	The response parameter callOut is added.

Issue 04 (2021-08-23)

This issue is released with AICC 8.15.1. The following interface is modified.

No.	Interface	Type	Remarks
1	https://ip:port/rest/cmsapp/v1/openapi/hisindex/vdn	Modified	The request parameter calleeNo is added.

Issue 03 (2021-07-20)

This issue is released with AICC 8.15.0. The following interfaces are modified.

No.	Interface	Type	Remarks
1	https://ip:port/rest/cmsapp/v1/openapi/vdn/querytotalagentinfo	New	The interface for querying all static configuratio

No	Interface	Type	Remarks
			n information about agents in a VCC is added.
2	https://ip:port/rest/cmsapp/v1/openapi/vdn/totalagentstate	New	The interface for querying the current states of all agents in a VCC is added.
3	https://ip:port/rest/cmsapp/v1/openapi/vdn/queryagentreststate	Modified	The input parameters limit and offset are changed to mandatory, and the maximum value of limit is changed to 1000.
4	https://ip:port/rest/cmsapp/v1/openapi/vdn/allagentphones	Modified	The input parameters limit and offset are changed to mandatory, and the maximum value of limit is changed to 1000.
5	https://ip:port/rest/cmsapp/v1/openapi/vdn/totalagentabstractex3	Modified	The input parameters limit and offset are changed to mandatory, and the maximum value of limit is changed to 1000.

No .	Interface	Type	Remarks
6	https://ip:port/rest/cmsapp/v1/openapi/vdn/totalagentdynamicinfo	Modified	The input parameters limit and offset are changed to mandatory, and the maximum value of limit is changed to 100.
7	https://ip:port/rest/cmsapp/v1/openapi/vdn/queryagentstaticconfinfo	Modified	The input parameters limit and offset are changed to mandatory, and the maximum value of limit is changed to 100.
8	https://ip:port/rest/cmsapp/v1/openapi/agentgroup/agentstatusinfobyagentids	Modified	The maximum length of the input parameter agentIds is 6000 in CTI 3.8 and 1000 in the Oracle database.
9	https://ip:port/rest/cmsapp/v1/openapi/agentgroup/queryagentcalls	Modified	The maximum length of the input parameter agentIds is 3000 in CTI 3.8 and 1000 in the Oracle database.
10	https://ip:port/rest/cmsapp/v1/openapi/agentgroup/siteagentcallandskillinfobyagentids	Modified	The maximum length of the input

No	Interface	Type	Remarks
			parameter agentIds is 6000 in CTI 3.8 and 1000 in the Oracle database.
11	https://ip:port/rest/cmsapp/v1/openapi/agentgroup/teamagentstatusinfo	Modified	The maximum length of the input parameter agentIds is 6000 in CTI 3.8 and 1000 in the Oracle database.

Issue 02 (2021-05-15)

This issue is released with AICC 8.14.1. The following interfaces are modified.

No.	Interface	Type	Remarks
1	https://ip:port/rest/cmsapp/v1/openapi/qcrelationconf/joinusergroup	New	The interface for adding an agent to a user group is added.
2	https://ip:port/rest/cmsapp/v1/openapi/qcrelationconf/updateagentgroup	New	The interface for updating an agent name is added.

Issue 01 (2021-03-31)

This issue is released with AICC 8.14.0. The following interfaces are modified.

No.	Interface	Type	Remarks
1	https://ip:port/rest/cmsapp/v1/openapi/tenant/querytenants	New	The interface for querying information about all tenants accessible to an access user is added.
2	https://ip:port/rest/cmsapp/v1/openapi/tenant/querytenantdetail	New	The interface for querying information about all agents under a tenant is added.
3	https://ip:port/rest/cmsapp/v1/callinday/querymanualindexbycondition	Modified	The reserved field for outbound call statistics is added to the response.

N o.	Interface	Type	Remarks
4	https://ip:port/rest/cmsapp/v1/calloutday/querymanualin dexesbycondition	Modifi ed	The reserved field for inbound call statistics is added to the response.
5	https://ip:port/rest/cmsapp/v1/openapi/hisindex/skillforda y	Modifi ed	The customized indicator IDX_SKILL_RESERVED_COL(N) is added.
6	https://ip:port/rest/cmsapp/v1/openapi/hisindex/skillform onth	Modifi ed	The customized indicator IDX_SKILL_RESERVED_COL(N) is added.
7	https://ip:port/rest/cmsapp/v1/openapi/hisindex/agentfor day	Modifi ed	The customized indicator IDX_AGENT_RESERVED_COL(N) is added.
8	https://ip:port/rest/cmsapp/v1/openapi/hisindex/agentfor month	Modifi ed	The customized indicator IDX_AGENT_RESERVED_COL(N) is added.
9	https://ip:port/rest/cmsapp/v1/openapi/uap/queryUapPerf Info	New	The interface for querying UAP performance information is added.
10	https://ip:port/rest/cmsapp/v1/openapi/uap/querytermina lstatus	New	The interface for querying the registration status of a terminal is added.
11	https://ip:port/rest/cmsapp/v1/openapi/hisindex/vdn	Modifi ed	The anonymous call flag is added to the input parameters.
12	https://ip:port/rest/cmsapp/v1/openapi/hisindex/vdn	Modifi ed	The customized indicator IDX_COUNT_TOTAL_CUSTOMER is added.
13	https://ip:port/rest/cmsapp/v1/openapi/hisindex/skill	Modifi ed	The customized indicator IDX_COUNT_SKILL_TOTAL_CUSTOME R is added.
14	https://ip:port/rest/cmsapp/v1/openapi/hisindex/agent	Modifi ed	The customized indicator IDX_COUNT_AGENT_TOTAL_CUSTO MER is added.
15	https://ip:port/rest/cmsapp/v1/openapi/vdn/querycalls	New	The interface for querying call information in a specific VCC is added.
16	https://ip:port/rest/cmsapp/v1/openapi/vdn/querycalltrac edetail	New	The interface for querying the call track data in a specific VCC is added.
17	https://ip:port/rest/cmsapp/v1/openapi/fromisales/queryt asks	New	The interface for querying the outbound call task list of a tenant is added.

N o.	Interface	Type	Remarks
18	https://ip:port/rest/cmsapp/v1/openapi/fromisales/dailystatistic	New	The interface for querying the daily settlement statistics of a specified task is added.
19	https://ip:port/rest/cmsapp/v1/openapi/fromisales/realstatistic	New	The interface for querying the real-time statistics of a specified task is added.
20	https://ip:port/rest/cmsapp/v1/openapi/hisindex/vdn	Modified	The customized indicator IDX_COUNT_IVR_ONLY_INBOUND is added.
21	https://ip:port/rest/cmsapp/v1/openapi/hisindex/vdn	Modified	The customized indicator IDX_COUNT_IVR_ONLY_INBOUND_CONNECTED is added.
22	https://ip:port/rest/cmsapp/v1/openapi/hisindex/vdn	Modified	The customized indicator IDX_DURATION_IVR_ONLY_INBOUND is added.
23	https://ip:port/rest/cmsapp/v1/openapi/hisindex/vdn	Modified	The customized indicator IDX_COUNT_CALL_OUT_CUSTOMER is added.
24	https://ip:port/rest/cmsapp/v1/openapi/hisindex/agent	Modified	The customized indicator IDX_COUNT_AGENT_CALL_OUT_CUSTOMER is added.
25	https://ip:port/rest/cmsapp/v1/openapi/hisindex/agent	Modified	The customized indicator IDX_COUNT_AGENT_CONNECTED_IN_(N) is added.

3 Overview

3.1 Function Description

The original CMSGateway of an enterprise functions as an open platform that provides monitoring capability interfaces for third-party partners. The CMSGateway provides RESTful interfaces for third-party partners to develop their own monitoring systems.

The CMSGateway interfaces can implement the following functions:

- Data query: queries information such as skill queues, VDNs, and called route by call center ID, VDN ID, and skill ID.



- Currently, interfaces can be accessed only in HTTPS mode.

3.2 Interface Parameter Usage Principle

When using the WebService interfaces provided by the CMS, third-party services must follow the usage principles described in [Table 1](#).

Table 1 Interface parameter usage principle

Type	Description
Mandatory	Mandatory parameters must be contained in interface invoking request messages sent by services to the CTI platform in any case.
Optional	Whether to use optional parameters can be determined based on business requirements.

3.3 Interface Encoding Description

The web service interfaces use the HTTPS+JSON encoding mode. The **Content-Type: application/json; charset=UTF-8** message header information must be configured in HTTPS requests (applicable only to the POST and PUT methods).

4 Interface Description

- In AICC 8.13.0 and later versions, the following structures returned by all third-party interfaces are not recommended. The following structures are used to be compatible with versions earlier than 8.13.0. In 8.13.0 and later versions, **resultCode** instead of **result** is used as the result code, and **resultDesc** instead of **resultDatas** is used as the data returned by the interface.

Parameter	Type	Description
result	String	Query result. The options are as follows: 0 : success Other values: failure
resultDatas	List <JSONObject>	Object array in the returned result after a successful query.

- The NSLB server IP address and port number involved in the request URLs of all interfaces in the following sections vary according to the deployment mode.
 - SaaS public cloud scenario: The NSLB server IP address is the IP address of the outer NSLB server (providing external access to virtualized services), and the port number is the port number provided for external services (corresponding to **aicc-webproxy** in NSLB routing rules). The default port number is 28050.
 - OP integrated deployment and OP independent deployment scenarios: The NSLB server IP address is the IP address of the NSLB server corresponding to the CC-CMS, and the port number is the port number provided for external services (corresponding to **aicc-openproxy** in NSLB routing rules). The default port number is 28080.

4.1 Real-Time Data Query

4.1.1 Real-Time Interfaces for VDNs

4.1.1.1 Querying Information About System Access Codes in a Specified VDN

Description

This interface is invoked to query system access codes of a specified VDN. (VDN: virtual directory number. A tenant corresponds to a VDN after being created and configured.)

For details about system access codes, see *(For Engineer) ICD VxxxRxxxCxx Product Documentation 01 (ICD)*.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Request URL

https://ip:port/rest/cmsapp/v1/openapi/vdn/queryaccesscodeinfo

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed to application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the request message body

No.	Parameter	Value Type	Mandatory	Description
1	cclid	Integer	Yes	ID of a call center. The value ranges from 1 to 65535.

Table 2 Parameters in the request message body

No.	Parameter	Value Type	Mandatory	Description
2	vdn	Integer	Yes	ID of a VDN. The value ranges from 1 to 5000.

Response Description

Table 3 Parameters in the response message body

No.	Parameter	Value Type	Description
1	result	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
2	resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
3	resultDatas	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
4	resultDesc	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 .

Table 4 Parameters of resultDesc/resultDatas

No.	Parameter	Value Type	Description
2.1	acId	Integer	Access code ID.
2.2	mediatypeId	Integer	Media type. For details about this object, see Table 1 .
2.3	accessCode	String	Access code.
2.4	codeDescription	String	Description of an access code.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "ccld":1,
  "vdn":170
}
```

- Response parameters

```
{
  "result": "0",
  "resultCode": "0100000",
  "resultDatas": [
    {
      "acld": 289,
      "mediatypeld": 5,
      "accessCode": "650114",
      "codeDescription": "Voice access"
    },
    {
      "acld": 288,
      "mediatypeld": 5,
      "accessCode": "650115",
      "codeDescription": "Voice access"
    }
  ],
  "resultDesc": [
    {
      "acld": 289,
      "mediatypeld": 5,
      "accessCode": "650114",
      "codeDescription": "Voice access"
    },
    {
      "acld": 288,
      "mediatypeld": 5,
      "accessCode": "650115",
      "codeDescription": "Voice access"
    }
  ]
}
```

```
]
}
```

4.1.1.2 Querying Agent Rest Information in a Specified VDN

Scenario

This interface is invoked to query all agent rest information in a specified VDN.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Internal Encapsulation Interface

BMS interfaces:

- /ccbms/ws/monitor/teamrestinfo
- /ccbms/ws/monitor/teamrestagentinfo
- /ccbms/ws/monitor/agentstatusinfobyagentids

Request URL

https://ip:port/rest/cmsapp/v1/openapi/vdn/queryagentreststate

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the request message body

No.	Parameter	Value Type	Mandatory	Description
1	cclId	Integer	Yes	ID of a call center. The value ranges from 1 to 65535.
2	vdn	Integer	Yes	ID of a VDN. The value ranges from 1 to 5000.
3	limit	Integer	Yes	Number of query records on each page. The value ranges from 1 to 1000.
4	offset	Integer	Yes	Parameter for determining the start page of the query. Value of offset = Value of limit x (Value of page – 1). The value of offset is greater than or equal to 0. page indicates the start page of the query.

Response Description

Table 3 Parameters in the response message body

No.	Parameter	Value Type	Description
1	result	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
2	resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
3	resultDatas	Object	Object in the returned result after a successful query. For details about the parameters of this object, see Table 4 . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
4	resultDesc	Object	Object in the returned result after a successful query. For details about the parameters of this object, see Table 4 .

Table 3 Parameters in the response message body

No.	Parameter	Value Type	Description
5	count	Integer	Total number of agents in the VDN.

Table 4 Parameters of resultDesc/resultDatas

No.	Parameter	Value Type	Description
2.1	onRestAgentNum	Integer	Number of resting agents.
2.2	allowRestAgentNum	Integer	Number of agents who are allowed to have a rest but still at work.
2.3	waitRestAgentNum	Integer	Number of agents who apply for a rest.
2.4	details	Array	Agent rest details. For details about the parameters of the object in this array, see Table 5 .

NOTE:

The response result is the agent rest data obtained based on the pagination parameters **limit** and **offset** in the request.

For example, if both **limit** and **offset** are set to **10**, the 10 agents on the second page (10 agents per page) are obtained, and then rest statistics on the 10 agents are collected.

Table 5 Agent rest details

No.	Parameter	Value Type	Description
2.4.1	agentId	Integer	Agent ID.
2.4.2	cantRestCause	Integer	Reason why agents are not allowed to have a rest. This parameter is valid only for the agents who wait for a rest. For details about the parameters of this object, see Table 6 .
2.4.3	vdnId	Integer	ID of a VDN.
2.4.4	cclId	Integer	ID of a call center.
2.4.5	beginTime	Long	Start time. This parameter indicates the rest start time for resting agents, and the start time of applying for or approving a rest for agents not resting. The time format is timestamp.
2.4.6	applyRestLength	Integer	Requested rest duration, in seconds.
2.4.7	name	String	Agent name.
2.4.8	reqRestDura	Long	Rest duration, that is, the rest duration of an agent who is taking a rest, in seconds.
2.4.9	skillIds	String	Skill queue to which an agent waiting for a rest signs in. The value is returned only when the value of cantRestCause is 4 .

Table 6 Reason for not allowing rest

Value of the Reason for Not Allowing Rest	Description
0	The agent is waiting for approval.
1	The agent is in busy state.
2	The number of agents who can take a rest in the VDN reaches the maximum.
3	The agent does not answer any inbound calls, because the agent does not sign in to any skill queues.
4	The number of resting agents in a skill queue the agent signs in reaches the maximum.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "ccld":1,
  "vdn":170,
  "offset":0,
  "limit":3
}
```

- Response parameters

```
{
  "result": "0",
  "resultCode": "0100000",
  "resultDatas": {
    "onRestAgentNum": 2,
    "allowRestAgentNum": 0,
    "details": [
      {
        "agentId": 1034,
        "cantRestCause": 2,
        "vdnId": 170,
        "ccld": 1,
        "beginTime": 1602830642000,
        "applyRestLength": 600,
```

```
    "name": "zhang1130user"
  },
  {
    "agentId": 1039,
    "cantRestCause": 0,
    "vdnId": 170,
    "ccId": 1,
    "beginTime": 1602830613000,
    "applyRestLength": 600,
    "name": "HW285601039",
    "reqRestDura": 238
  },
  {
    "agentId": 1099,
    "cantRestCause": 0,
    "vdnId": 170,
    "ccId": 1,
    "beginTime": 1602830638000,
    "applyRestLength": 600,
    "name": "HW285601099",
    "reqRestDura": 213
  }
],
"waitRestAgentNum": 1
},
"resultDesc": {
  "onRestAgentNum": 2,
  "allowRestAgentNum": 0,
  "details": [
    {
      "agentId": 1034,
      "cantRestCause": 2,
      "vdnId": 170,
      "ccId": 1,
      "beginTime": 1602830642000,
      "applyRestLength": 600,
      "name": "zhang1130user"
    },
    {
      "agentId": 1039,
      "cantRestCause": 0,
      "vdnId": 170,
      "ccId": 1,
```



```

        "beginTime": 1602830613000,
        "applyRestLength": 600,
        "name": "HW285601039",
        "reqRestDura": 238
    },
    {
        "agentId": 1099,
        "cantRestCause": 0,
        "vdnId": 170,
        "ccId": 1,
        "beginTime": 1602830638000,
        "applyRestLength": 600,
        "name": "HW285601099",
        "reqRestDura": 213
    }
],
    "waitRestAgentNum": 1
},
    "count": 30
}

```

4.1.1.3 Querying Agent Phone Numbers in a Specified VDN

Scenario

This interface is used to query all agent phone numbers in a specified VDN.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Internal Encapsulation Interface

BMS interface: /ccbms/ws/monitor/siteagentinfoobyagentids

Request URL

https://ip:port/rest/cmsapp/v1/openapi/vdn/allagentphones

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the request message body

No.	Parameter	Value Type	Mandatory	Description
1	cclId	Integer	Yes	ID of a call center. The value ranges from 1 to 65535.
2	vdn	Integer	Yes	ID of a VDN. The value ranges from 1 to 5000.
3	limit	Integer	Yes	Number of query records on each page. The value ranges from 1 to 1000.
4	offset	Integer	Yes	Parameter for determining the start page of the query. Value of offset = Value of limit x (Value of page – 1). The value of offset is greater than or equal to 0. page indicates the start page of the query.

Response Description

Table 3 Parameters in the response message body

No.	Parameter	Value Type	Description
1	result	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
1	resultCode	String	Query result. The options are as follows: 0100000 : success

Table 3 Parameters in the response message body

No.	Parameter	Value Type	Description
			Other values: failure For details about the failure causes, see Error Code Reference .
3	resultDatas	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
2	resultDesc	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 .
3	count	Integer	Total number of agents in the VDN.

Table 4 Parameters of resultDesc/resultDatas

No.	Parameter	Value Type	Description
2.1	agentId	Integer	Agent ID.
2.2	phone	String	Agent phone number.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "cclId":1,
  "vdn":170,
  "offset":0,
  "limit":4
}
```

- Response parameters

```
{
  "result": "0",
  "resultCode": "0100000",
  "count": 30,
  "resultDatas": [
```

```
{
  "agentId": 55002,
  "phone": 620001
},
{
  "agentId": 833,
  "phone": 620002
},
{
  "agentId": 834,
  "phone": 620003
},
{
  "agentId": 835,
  "phone": 620004
}
],
"resultDesc": [
  {
    "agentId": 55002,
    "phone": 620001
  },
  {
    "agentId": 833,
    "phone": 620002
  },
  {
    "agentId": 834,
    "phone": 620003
  },
  {
    "agentId": 835,
    "phone": 620004
  }
]
}
```

4.1.1.4 Querying Information About All Agents in a Specified VDN

Scenario

This interface is invoked to query information about all agents in a specified VDN.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Internal Encapsulation Interface

BMS interface: /ccbms/ws/monitor/siteagentinfoobyagentids

Request URL

https://ip:port/rest/cmsapp/v1/openapi/vdn/totalagentabstractex3

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the request message body

No.	Parameter	Value Type	Mandatory	Description
1	cclid	Integer	Yes	ID of a call center. The value ranges from 1 to 65535.
2	vdn	Integer	Yes	ID of a VDN. The value ranges from 1 to 5000.
3	limit	Integer	Yes	Number of query records on each page. The value ranges from 1 to 1000.
4	offset	Integer	Yes	Parameter for determining the start page of the query. Value of offset = Value of limit x (Value of page – 1). The value of offset is greater than or equal to 0.

Table 2 Parameters in the request message body

No.	Parameter	Value Type	Mandatory	Description
				page indicates the start page of the query.

Response Description

Table 3 Parameters in the response message body

No.	Parameter	Value Type	Description
1	result	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
1	resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
3	resultDatas	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
2	resultDesc	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 .
3	count	Integer	Total number of agents in the VDN.

Table 4 Parameters of resultDesc/resultDatas

No.	Parameter	Value Type	Description
2.1	agentId	Integer	Agent ID.
2.2	currentState	Integer	Current status of an agent. For details, see CTI Platform Agent Status Code Reference .
2.3	currentStateTime	Long	Duration of the current state, in seconds.

Table 4 Parameters of resultDesc/resultDatas

No.	Parameter	Value Type	Description
2.4	locationId	Integer	Home distributed access node. This parameter is a CTI reserved field and has no actual meaning.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "ccId":1,
  "vId":170,
  "offset":0,
  "limit":2
}
```

- Response parameters

```
{
  "result": "0",
  "resultCode": "0100000",
  "count": 30,
  "resultDatas": [
    {
      "agentId": 55002,
      "locationId": 65535,
      "currentTime": 153282,
      "currentState": 0
    },
    {
      "agentId": 833,
      "locationId": 65535,
      "currentTime": 153293,
      "currentState": 0
    }
  ],
  "resultDesc": [
    {
      "agentId": 55002,
```

```
    "locationId": 65535,  
    "currentStateTime": 153282,  
    "currentState": 0  
  },  
  {  
    "agentId": 833,  
    "locationId": 65535,  
    "currentStateTime": 153293,  
    "currentState": 0  
  }  
]  
}
```

4.1.1.5 Querying Information About the Current Status of All Agents in a Specified VDN

Scenario

This interface is invoked to query information about the current status of all agents in a specified VDN.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Internal Encapsulation Interface

BMS interface: /ccbms/ws/monitor/siteagentinfoobyagentids

Request URL

`https://ip:port/rest/cmsapp/v1/openapi/vdn/totalagentdynamicinfo`

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the request message body

No.	Parameter	Value Type	Mandatory	Description
1	cclId	Integer	Yes	ID of a call center. The value ranges from 1 to 65535.
2	vdn	Integer	Yes	ID of a VDN. The value ranges from 1 to 5000.
3	limit	Integer	Yes	Number of query records on each page. The value ranges from 1 to 100.
4	offset	Integer	Yes	Parameter for determining the start page of the query. Value of offset = Value of limit x (Value of page – 1). The value of offset is greater than or equal to 0. page indicates the start page of the query.

Response Description

Table 3 Parameters in the response message body

No.	Parameter	Value Type	Description
1	result	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
2	resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .

Table 3 Parameters in the response message body

No.	Parameter	Value Type	Description
3	resultDatas	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
4	resultDesc	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 .
5	count	Integer	Total number of agents in the VDN.

Table 4 Parameters of resultDesc/resultDatas

No.	Parameter	Value Type	Description
2.1	agentId	Integer	Agent ID.
2.2	state	Integer	Agent status. For details, see CTI Platform Agent Status Code Reference .

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "ccld":1,
  "vdn":170,
  "offset":0,
  "limit":4
}
```

- Response parameters

```
{
  "result": "0",
  "resultCode": "0100000",
  "count": 30,
  "resultDatas": [
    {
      "agentId": 55002,
      "state": 0
    }
  ]
}
```

```
    },
    {
      "agentId": 833,
      "state": 0
    },
    {
      "agentId": 834,
      "state": 0
    },
    {
      "agentId": 835,
      "state": 0
    }
  ],
  "resultDesc": [
    {
      "agentId": 55002,
      "state": 0
    },
    {
      "agentId": 833,
      "state": 0
    },
    {
      "agentId": 834,
      "state": 0
    },
    {
      "agentId": 835,
      "state": 0
    }
  ]
}
```

4.1.1.6 Querying Information About Static Configurations of All Agents in a Specified VDN

Scenario

This interface is invoked to query information about static configurations of all agents in a specified VDN.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Request URL

`https://ip:port/rest/cmsapp/v1/openapi/vdn/queryagentstaticconfinfo`

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the request message body

No.	Parameter	Value Type	Mandatory	Description
1	cclId	Integer	Yes	ID of a call center. The value ranges from 1 to 65535.
2	vdn	Integer	Yes	ID of a VDN. The value ranges from 1 to 5000.
3	offset	Integer	Yes	Pagination offset. The value is greater than or equal to 0.
4	limit	Integer	Yes	Pagination limit. The value ranges from 1 to 100.

Response Description

Table 3 Parameters in the response message body

No.	Parameter	Value Type	Description
1	result	String	Query result. The options are as follows: 0 : success

Table 3 Parameters in the response message body

No.	Parameter	Value Type	Description
			Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
2	resultCode	String	Query result. The options are as follows: 0100000: success Other values: failure For details about the failure causes, see Error Code Reference .
3	resultDatas	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4. NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
4	resultDesc	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 .
5	count	Integer	Total number of agents in the VDN.

Table 4 Parameters of resultDesc/resultDatas

No.	Parameter	Value Type	Description
2.1	agentId	Integer	Agent ID.
2.2	groupId	Integer	ID of the group to which an agent belongs. Currently, the value of this field is -1. Group: To facilitate the management of agents, the agents of a VDN are usually grouped. Agents in a group form an agent group. Currently, the AICC does not support the return value -1 of this field, which indicates that the agents do not belong to any agent group.
2.3	rights	String	Permission value of an agent. The value 0 indicates that the agent has no permission, and 1 indicates that the agent has permission. Table 5 describes the role permission.
2.4	name	String	Agent name.
2.5	realFlag	Integer	Whether an agent is a network agent. The value 0 indicates network agent, and 1 indicates common agent. Currently, all agents in the AICC are common agents.

Table 5 Agent permission value

Bit	Description
1	Showing busy/idle.
2	Muting/Unmuting calls.
3	Holding calls/Canceling to hold calls.
4	Connecting held calls.
5	Making common outbound calls.
6	Making internal calls.
7	Transferring calls internally.
8	Answering transferred inbound calls.
9	Making three-party calls.
10	Transferring out calls.
11	Transferring calls from the agent service to automatic service.
12	Rejecting inbound calls.
13	Listening/Listening and inserting/Whispering.
14	Intercepting calls.
15	Forcibly signing out/Forcibly showing busy/Forcibly showing idle.
16	Inspecting recordings/Inspecting voice playing.
17	Monitoring inspections.
18	Appraisal administrator.
19	Ordinary appraisal personnel.
20	Alarm console.
21	Report handler.
22	Reserved value.
23	Monitoring console.
24	Configuration administrator.
25	Report administrator.
26	Making national outbound toll calls.
27	Making international outbound toll calls.

Table 5 Agent permission value

Bit	Description
28	Making outbound calls using access codes of the call center.
29	Performing inspections in fullscreen mode.
30	Picking up calls for the agents in the same agent group.
31	Reserved value.
32	Configuring agents.
33	Configuring skills.
34	Not setting the outbound call.
35	Configuring the audit.
36–128	Reserved value.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "ccId":1,
  "vdn":170,
  "offset":0,
  "limit":100
}
```

- Response parameters

[illegible]

[illegible]

4.1.1.7 Querying the IVR Status Details of a Specified VDN

Scenario

This interface is invoked to query the IVR status details of a specified VDN. (CTI 3.6 does not support this interface.)

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Internal Encapsulation Interface

BMS interface: /ccbms/ws/monitor/queryivrdevicestatus

Request URL

https://ip:port/rest/cmsapp/v1/openapi/vdn/queryivrstatusinfo

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the request message body

No.	Parameter	Value Type	Mandatory	Description
1	ccld	Integer	Yes	ID of a call center. The value ranges from 1 to 65535 .
2	vdn	Integer	Yes	ID of a VDN. The value ranges from 1 to 5000.
3	flowId	Integer	Yes	ID of an IVR flow. Enter the ID of an existing IVR flow. The value ranges from 1 to 1500. (To obtain the IVR flow ID, log in to the web configuration console as a VDN administrator and choose Web Configuration Console > VDN Configuration > Flow > Flow .)

Response Description

Table 3 Parameters in the response message body

No.	Parameter	Value Type	Description
1	result	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
2	resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
3	resultDatas	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
4	resultDesc	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 .

Table 4 Parameters of resultDesc/resultDatas

No.	Parameter	Value Type	Description
2.1	callId	String	Call ID.
2.2	customerLevel	Integer	Level of the customer making a call. The sequence number ranges from 1 to 254. A larger value indicates the importance of a customer.
2.3	requestType	Integer	Call feature. For details about this object, see Table 1 .

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{  
  "ccld":1,  
  "vdn":170,  
  "flowld":1  
}
```

- Response parameters

```
{  
  "result": "0",  
  "resultCode": "0100000",  
  "resultDatas": [  
    {  
      "callId": "1453724606-576",  
      "customerLevel": 1,  
      "requestType": 0  
    }  
  ],  
  "resultDesc": [  
    {  
      "callId": "1453724606-576",  
      "customerLevel": 1,  
      "requestType": 0  
    }  
  ]  
}
```

4.1.1.8 Querying Information About IVR Flow Access Codes in a Specified VCC

Scenario

This interface is invoked to query information about IVR flow access codes in a specified VCC.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Internal Encapsulation Interface

BMS interface: /ccbms/ws/monitor/queryivrInfo

Request URL

`https://ip:port/rest/cmsapp/v1/openapi/vdn/queryivrprocessinfo`

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameter in the request body

No.	Parameter	Value Type	Mandatory	Description
1	cclid	Integer	Yes	ID of a call center. The value ranges from 1 to 65535.
2	vdn	Integer	Yes	ID of a VCC. The value ranges from 1 to 5000.

Response Description

Table 3 Parameter in the response body

No.	Parameter	Value Type	Description
1	result	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
2	resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure

Table 3 Parameter in the response body

No.	Parameter	Value Type	Description
			For details about the failure causes, see Error Code Reference .
3	resultDatas	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
3	resultDesc	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 .

Table 4 Parameters of resultDesc/resultDatas

No.	Parameter	Value Type	Description
2.1	id	Integer	IVR flow ID.
2.2	serviceNo	Integer	Business type ID. NOTE: This parameter has no actual meaning. Do not use it.
2.3	realFlag	Integer	Virtual or real queue. The options are 1 (real queue) and 2 (virtual queue).
2.4	accessNo	String	Access code.
2.5	description	String	Access code function description.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "ccld":1,
  "vdn":170
}
```

- Response parameters

```
{
  "result": "0",
  "resultCode": "0100000",
}
```

```

"resultDatas": [
  {
    "id": 1,
    "serviceNo": 1,
    "realFlag": 1,
    "accessNo": "85000",
    "description": "Exception handling flow"
  },
  {
    "id": 2,
    "serviceNo": 1,
    "realFlag": 1,
    "accessNo": "85001",
    "description": "Main flow"
  }
],
"resultDesc": [
  {
    "id": 1,
    "serviceNo": 1,
    "realFlag": 1,
    "accessNo": "85000",
    "description": "Exception handling flow"
  },
  {
    "id": 2,
    "serviceNo": 1,
    "realFlag": 1,
    "accessNo": "85001",
    "description": "Main flow"
  }
]
}

```

4.1.1.9 Querying Information about Each Skill Queue in a Specified VCC

Scenario

This interface is invoked to query information about each skill queue in a specified VCC.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Request URL

`https://ip:port/rest/cmsapp/v1/openapi/configures/skills`

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameter in the request body

No.	Parameter	Value Type	Mandatory	Description
1	cclid	Integer	Yes	ID of a call center. The value ranges from 1 to 65535.
2	vdn	Integer	Yes	ID of a VCC. The value ranges from 1 to 5000.

Response Description

Table 3 Parameter in the response body

No.	Parameter	Value Type	Description
1	result	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.12 and earlier versions.

Table 3 Parameter in the response body

No.	Parameter	Value Type	Description
2	resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
3	resultDatas	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 . NOTE: This field is reserved for compatibility with interface response parameters in 8.12 and earlier versions.
4	resultDesc	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 .

Table 4 Parameters of resultDesc/resultDatas

No.	Parameter	Value Type	Description
2.1	id	Integer	Skill queue ID.
2.2	mediaType	Integer	Media type. For details, see Table 1 .
2.3	submediaType	String	Submedia type. The options are as follows: -1 : no submedia type. For details about other submedia types, see Description of Call Submedia Types .
2.4	name	String	Skill queue description.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "queryParam":{
    "ccld":1,
    "vdn":1
  }
}
```



```
}
```

- Response parameters

```
{
  "result": "0",
  "resultCode": "0100000",
  "resultDatas": [
    {
      "id": 1,
      "mediaType": 1,
      "name": "Text chat 1"
    },
    {
      "id": 10,
      "mediaType": 1,
      "name": "Text chat 2"
    }
  ],
  "resultDesc": [
    {
      "id": 1,
      "mediaType": 1,
      "name": "Text chat 1"
    },
    {
      "id": 10,
      "mediaType": 1,
      "name": "Text chat 2"
    }
  ]
}
```

4.1.1.10 Querying All Rest Reason Codes in a Specific VDN

Scenario

This interface is invoked to query all rest reason codes in a specific VDN.

For details about how to configure the rest reason code, see "Configuring Rest Reason Codes" in *(For Engineer) ICD VxxxRxxxCxx Product Documentation 01 (ICD)*.

Interface Method

The request method must be set to GET. This interface supports only the GET method.

Internal Encapsulation Interface

BMS interface: /ccbms/ws/monitor/restreason

Request URL

https://ip:port/rest/cmsapp/v1/openapi/agentgroup/restreasons/{ccld}/{vdn}

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

No request body exists.

Table 2 Path parameters

No.	Parameter	Value Type	Mandatory	Description
1	ccld	Integer	Yes	ID of a call center. The value ranges from 1 to 65535.
2	vdn	Integer	Yes	ID of a VDN. The value ranges from 1 to 5000.

Response Description

Table 3 Parameters in the message body

No.	Parameter	Value Type	Description
1	result	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference .

Table 3 Parameters in the message body

No.	Parameter	Value Type	Description
			NOTE: This field is reserved for compatibility with interface response parameters in 8.12 and earlier versions.
2	resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
3	resultDatas	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 . NOTE: This field is reserved for compatibility with interface response parameters in 8.12 and earlier versions.
4	resultDesc	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 .

Table 4 Parameters of resultDesc/resultDatas

No.	Parameter	Value Type	Description
2.1	restReasonId	Integer	Rest cause code.
2.2	restReason	String	Description of a rest cause.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

None

- Response parameters

```
{
  "result": "0",
  "resultCode": "0100000",
  "resultDatas": [
    {
      "restReasonId": 100,
```

```
        "restReason": "Uncomfortable"
    }
],
"resultDesc": [
    {
        "restReasonId": 100,
        "restReason": "Uncomfortable"
    }
]
}
```

4.1.1.11 Querying Information About Calls of a Specified Access Code

Description

This interface is invoked to query information about calls of a specified access code. (CTI 3.6 does not support this interface.)

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Internal Encapsulation Interfaces

BMS interface: /ccbms/ws/monitor/querycallstatbyinno

Request URL

https://ip:port/rest/cmsapp/v1/openapi/vdn/queryaccesscodecallinfo
In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.
In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed to application/json; charset=UTF-8 .

Table 1 Parameters in the request header

No.	Parameter	Type	Mandatory	Description
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the request message body

No.	Parameter	Type	Mandatory	Description
1	cclId	Integer	Yes	ID of a call center. The value ranges from 1 to 65535.
2	vdn	Integer	Yes	ID of a VDN. The value ranges from 1 to 5000.
3	innolds	Array	Yes	Array of access code IDs on the WAS. The maximum length of the array is 500.

Response Message

Table 3 Parameters in the response message body

No.	Parameter	Type	Description
1	result	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
2	resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
3	resultDatas	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
4	resultDesc	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 .

Table 4 Parameters of resultDesc/resultDatas

No.	Parameter	Type	Description
2.1	innold	Integer	Access code ID.
2.2	allCallCount	Long	Total number of online customers using the access code.
2.3	ivrCallCount	Long	Total number of online customers using the access code in the IVR.
2.4	agentCallCount	Long	Total number of online customers who are handled by agents and use the access code.
2.5	queueCallCount	Long	Total number of online customers who are in a queue and use the access code.
2.6	longestWaitTime	Long	Maximum call waiting duration among the calls using the access code, in seconds.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "ccId":1,
  "vdn":170,
  "innolds":[1,2]
}
```

- Response parameters

```
{
  "result": "0",
  "resultCode": "0100000",
  "resultDatas": [
    {
      "innold": 1,
      "allCallCount": 1,
      "ivrCallCount": 0,
      "agentCallCount": 1,
      "queueCallCount": 0,
      "longestWaitTime": 0
    },
    {
      "innold": 2,
```

```

        "allCallCount": 1,
        "ivrCallCount": 1,
        "agentCallCount": 0,
        "queueCallCount": 0,
        "longestWaitTime": 0
    }
],
"resultDesc": [
    {
        "innold": 1,
        "allCallCount": 1,
        "ivrCallCount": 0,
        "agentCallCount": 1,
        "queueCallCount": 0,
        "longestWaitTime": 0
    },
    {
        "innold": 2,
        "allCallCount": 1,
        "ivrCallCount": 1,
        "agentCallCount": 0,
        "queueCallCount": 0,
        "longestWaitTime": 0
    }
]
}

```

4.1.1.12 Querying Call Information in a Specified VDN

Scenario

This interface is invoked to query call information in a specified VDN based on the calling number, called number, and time segment.

This interface is invoked to query only inbound call information in a specified VDN. The retention period of the queried data is one day.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Request URL

`https://ip:port/rest/cmsapp/v1/openapi/vdn/querycalls`

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the request message body

Parameter	Type	Mandatory	Description
cclid	Integer	Yes	ID of a call center. The value ranges from 1 to 65535 .
vdn	Integer	Yes	ID of a VDN. The value ranges from 1 to 5000 .
callerNo	String	No	Calling number. The value can be empty, indicating that all calling numbers are queried. The value consists of digits and must comply with the number rules.
calleeNo	String	No	Called number. The value can be empty, indicating that all called numbers are queried. The value consists of digits and must comply with the number rules.
beginDate	Long	Yes	Start time of a query period. The value is a timestamp in milliseconds. The time cannot be earlier than 00:00:00 of the previous day. The data source is time zone 0. During query, the time zone needs to be converted based on the specific time zone. For example, GMT+8 is 8 hours ahead the GMT time.
endDate	Long	Yes	End time of a query period. The value is a timestamp in milliseconds. The time cannot be later than 23:59:59 on the current day. The data source is time zone 0. During query, the time zone needs to be converted based on the specific time zone. For example, GMT+8 is 8 hours ahead the GMT time.
limit	Integer	No	Number of query records on each page. The maximum value is 100 .

Table 2 Parameters in the request message body

Parameter	Type	Mandatory	Description
offset	Integer	No	offset is used to determine the start page of the query. Value of offset = Value of limit x (Value of page – 1). The value of offset is greater than or equal to 0. The default value is 0. page indicates the start page to be queried.

Response Description

Table 3 Parameters in the response message body

Parameter	Type	Description
resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
resultDesc	Object	Object in the returned result after a successful query. For details about the parameters of this object, see Table 4 .

Table 4 Parameters of resultDesc

Parameter	Type	Description
data	Array <callinDetial>	Object array in the returned result. For details about the parameters of this object, see Table 5 .
count	Integer	Number of data records that can be queried.

Table 5 Parameters of data

Parameter	Type	Description
logDate	Long	Time when data is imported to the database. The value is a timestamp in milliseconds.
callerNo	String	Calling number.
calleeNo	String	Called number.
callId	String	Call ID.
callType	Integer	Call type. For details, see Description of Call Types .
waitBegin	Long	Waiting start time in millisecond-level timestamp.
ackBegin	Long	Response start time in millisecond-level timestamp.
callBegin	Long	Start time of a call in millisecond-level timestamp.

Table 5 Parameters of data

Parameter	Type	Description
callEnd	Long	End time of a call in millisecond-level timestamp.
vdn	Integer	ID of a VDN.
leaveReason	Integer	Reason why a call is disconnected from a device. For details, see Table 1 .
mediaType	Integer	Media type. For details, see Table 1 .
subMediaType	Integer	Submedia type. For details, see Table 3 .

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "ccId":1,
  "vdn":170,
  "callerNo": "12345",
  "calleeNo": "23131",
  "beginDate": "1451111111",
  "endDate": "1452222222",
  "limit": 10,
  "offset": 0
}
```

- Response parameters

```
{
  "resultCode":"0100000",
  "resultDesc":{"
    "data":[
      {
        "logDate": "1615717765000", //Statistics collection time
        "callerNo": "1212", //Calling number
        "calleeNo": "12121212",//Called number
        "callId": "167789709-33", //Call ID
        "callType": "1",//Call type
        "waitBegin": "1615717765000",//Start time of call waiting
        "ackBegin": "1615717765000",//Time when a call is answered
      }
    ]
  }
```

```
        "callBegin": "1615717765000", //Start time of a call
    "callEnd": "1615717765000", //End time of a call
        "vdn": 170, //VDN ID
        "leaveReason": "1", //Reason why a call is disconnected from a device
        "mediaType": 53, //Media type
        "subMediaType": 1 //Submedia type
    }
],
count: 5
}
}
```

4.1.1.13 Querying Call Information in a Specified VDN (V2)

Scenario

This interface is invoked to query call information in a specified VDN based on the calling number, called number, and time segment.

In the V1 version, this interface provides only information about inbound calls. In the V2 version, the **isCallIn** parameter is added to query information about inbound or outbound calls. The retention period of the queried data is one day.

Interface Method

This interface supports only the POST method.

URI

https://ip:port/rest/cmsapp/v2/openapi/vdn/querycalls

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

Parameter	Mandatory	Type	Default Value	Description
Content-Type	Yes	String	None	The value is fixed at application/json; charset=UTF-8 .

Table 1 Parameters in the request header

Parameter	Mandatory	Type	Default Value	Description
Authorization	Yes	String	None	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the request body

Parameter	Mandatory	Type	Default Value	Description
cclId	Yes	Integer	None	ID of a call center. The value ranges from 1 to 65535.
vdn	Yes	Integer	None	ID of a VDN. The value ranges from 1 to 5000.
callerNo	No	String	None	Calling number. The value can be empty, indicating that all calling numbers are queried. The value consists of digits and must comply with the number rules.
calleeNo	No	String	None	Called number. The value can be empty, indicating that all called numbers are queried. The value consists of digits and must comply with the number rules.
beginDate	Yes	Long	None	Start time of a query period. The value is a timestamp in milliseconds. The time cannot be earlier than 00:00:00 of the previous day.
endDate	Yes	Long	None	End time of a query period. The value is a timestamp in milliseconds. The time cannot be later than 23:59:59 on the current day.
isCallIn	Yes	String	None	If the value is true , inbound call information is queried. If the value is false , outbound call information is queried.
limit	No	Integer	None	Limit of records in a query result. The maximum value is 100 .
offset	No	Integer	None	offset is used to determine the start page of the query. Value of offset = Value of limit x (Value of page – 1). The value of offset is greater than or equal to 0. The default value is 0 . page indicates the start page to be queried.

Response Description

Table 3 Parameters in the response message body

Parameter	Type	Description
resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
resultDesc	Object	Object in the returned result after a successful query.
data	Array <callinDetial>	Object array in the returned result. For details about the parameters of this object, see Table 4 .
count	Integer	Number of data records that can be queried.

Table 4 Parameters of data

Parameter	Type	Description
logDate	Long	Time when data is imported to the database. The value is a timestamp in milliseconds.
callerNo	String	Calling number.
calleeNo	String	Called number.
callId	String	Call ID.
callType	Integer	Call type. For details, see Description of Call Types .
waitBegin	Long	Waiting start time in millisecond-level timestamp.
ackBegin	Long	Response start time in millisecond-level timestamp.
callBegin	Long	Start time of a call in millisecond-level timestamp.
callEnd	Long	End time of a call in millisecond-level timestamp.
vdn	Integer	ID of a VDN.
leaveReason	Integer	Reason why a call is disconnected from a device. For details, see Table 1 .
mediaType	Integer	Media type. For details, see Table 1 .
subMediaType	Integer	Submedia type. For details, see Table 3 .

Error Codes

For details, see [Error Code Reference](#) .

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "ccId":1,
  "vdn":170,
  "callerNo": "12345",
  "calleeNo": "23131",
  "beginDate": "1451111111",
  "endDate": "1452222222",
  "isCallIn":"true"
  "limit": 10,
  "offset": 0
}
```

- Response header

```
HTTP/1.1 200 OK
Content-Type: application/json;charset=UTF-8
Date: Mon, 02 Jul 2018 02:43:03 GMT
```

- Response parameters

```
{
  "resultCode":"0100000",
  "resultDesc":{
    "data":[
      {
        "logDate": "1615717765000", //Statistics collection time
        "callerNo": "1212", //Calling number
        "calleeNo": "12121212",//Called number
        "callId": "167789709-33", //Call ID
        "callType": "1",//Call type
        "waitBegin": "1615717765000",//Start time of call waiting
        "ackBegin": "1615717765000",//Time when a call is answered
        "callBegin": "1615717765000",//Start time of a call
        "callEnd": "1615717765000",//End time of a call
        "vdn": 170, //VDN ID
        "leaveReason": "1", //Reason why a call is disconnected from a device
        "mediaType":53, //Media type
        "subMediaType":1 //Submedia type
      }
    ],
  },
}
```

```
count: 5
}
}
```

4.1.1.14 Querying the Call track Data in a Specific VDN

Description

This interface is invoked to query call track data in a specified VDN based on the call ID.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Request URL

`https://ip:port/rest/cmsapp/v1/openapi/vdn/querycalltracedetail`

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed to application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the request message body

No.	Parameter	Type	Mandatory	Description
1	cclid	Integer	Yes	ID of a call center. The value ranges from 1 to 65535.
2	vdn	Integer	Yes	ID of a VDN. The value ranges from 1 to 5000.
3	callId	String	Yes	Call ID.
4	logDate	Long	Yes	Query time, which is used to parse the month and query the specified table. The value is a timestamp in milliseconds.

Response Message

Table 3 Parameters in the response message body

No.	Parameter	Type	Description
1	resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
2	resultDesc	Array <tbilllog>	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 .

Table 4 Parameters of resultDesc

No.	Parameter	Type	Description
2.1	callId	String	Call ID.
2.2	mediaType	Integer	Media type. For details, see Table 1 .
2.3	callIdNum	Integer	CDR sequence number. Sequence number of multiple CDRs generated for a call. The value starts from 1 and indicates the first CDR. The value -1 indicates the last CDR.
2.4	callerNo	String	Calling number.
2.5	calleeNo	String	Called number.
2.6	waitBegin	Long	Waiting start time in millisecond-level timestamp.
2.7	ackBegin	Long	Response start time in millisecond-level timestamp.
2.8	callBegin	Long	Start time of a call in millisecond-level timestamp.
2.9	callEnd	Long	End time of a call in millisecond-level timestamp.
2.10	subCcNo	Integer	ID of a call center in a network call center.
2.11	deviceType	Integer	Type of a device that processes calls. For details, see Table 1 .
2.12	deviceNo	String	ID of the device that processes the current call. For an agent device, the device ID is the agent ID. For a queue or virtual device, the device ID is the queue ID configured on the configuration console or the ID of the network shared device corresponding to the virtual device. For other device types, the value is an internal ID.
2.13	deviceIn	String	Device description.
2.14	callType	Integer	Call type. For details, see Description of Call Media Types .

Table 4 Parameters of resultDesc

No.	Parameter	Type	Description
2.15	releaseCause	Integer	Call release cause code. For details, see Cause Code .
2.16	EnterReason	Integer	Reason why a call accesses a device. For details, see Table 1 .
2.17	LeaveReason	Integer	Reason why a call is disconnected from a device. For details, see Table 1 .
2.18	CurrentSkillID	String	ID of the skill queue that processes the current call.
2.19	subMediaType	Integer	Submedia type. For details, see Table 3 .

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "ccId":1,
  "vdn":170,
  "callId": "1678978542-92",
  "logDate": "1615717765000"
}
```

- Response parameters

```
{
  "resultCode":"0100000",
  "resultDesc":
  [{
    "callId": "1678978542-92",
    "mediaType": "1",
    "subMediaType": "1",
    "callIdNum": "-1",
    "callerNo": "80000888",
    "calleeNo": "624192",
    "waitBegin": "1615717765000",//Start time of call waiting
    "ackBegin": "1615717765000",//Time when a call is answered
    "callBegin": "1615717765000",//Start time of a call
    "callEnd": "1615717765000",//End time of a call
    "subCcNo": "0",
    "deviceType": "1",
```

```

    "deviceNo": "1",
    "deviceIn": "1",
    "callType": "1",
    "releaseCause": "1", //Valid only when callIdNum is set to -1
    "EnterReason": "11",
    "LeaveReason": "11",
    "CurrentSkillID": "0", //ID of the skill queue that is processing a call
    "mediaType": 53, //Media type
    "subMediaType": 1 //Submedia type
  }
}
```

4.1.1.15 Querying Information About Static Configurations of All Agents in a Specified VDN (Not Supported by CTI 3.6)

Scenario

This interface is invoked to query static configuration information about all agents in a specified VDN. Pagination query is not supported. (CTI 3.6 does not support this interface.)

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Internal Encapsulation Interface

BMS interface: /ccbms/ws/monitor/querytotalagentinfo

Request URL

https://ip:port/rest/cmsapp/v1/openapi/vdn/querytotalagentinfo

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the request message body

No.	Parameter	Value Type	Mandatory	Description
1	cclid	Integer	Yes	ID of a call center. The value ranges from 1 to 65535.
2	vdn	Integer	Yes	ID of a VDN. The value ranges from 1 to 5000.

Response Description

Table 3 Parameters in the response message body

No.	Parameter	Value Type	Description
1	result	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
2	resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
3	resultDatas	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
4	resultDesc	Array	Object array in the returned result after a successful query.

Table 3 Parameters in the response message body

No.	Parameter	Value Type	Description
			For details about the parameters of the object in this array, see Table 4 .

Table 4 Parameters of resultDesc/resultDatas

No.	Parameter	Value Type	Description
2.1	agentId	Integer	Agent ID.
2.2	groupId	Integer	Agent group ID. Group: To facilitate the management of agents, the agents of a VDN are usually grouped. Agents in a group form an agent group.
2.3	rights	String	Permission value of an agent. The value 0 indicates that the agent has no permission, and 1 indicates that the agent has permission. Table 5 describes the role permission.
2.4	name	String	Agent name.
2.5	realFlag	Integer	Whether an agent is a network agent. The value 0 indicates network agent, and 1 indicates common agent. Currently, all agents in the AICC are common agents.

Table 5 Agent permission value

Bit	Description
1	Showing busy/idle.
2	Muting/Unmuting calls.
3	Holding/Unholding calls.
4	Connecting held calls.
5	Common outbound call.
6	Internal call.
7	Transferring calls internally.
8	Answering transferred inbound calls.
9	Making three-party calls.
10	Transferring out calls.
11	Transferring calls from the agent service to automatic service.
12	Rejecting inbound calls.
13	Listening/Listening and inserting/Whispering.

Table 5 Agent permission value

Bit	Description
14	Intercepting calls.
15	Forcibly signing out/Forcibly showing busy/Forcibly showing idle.
16	Inspecting recordings/Inspecting voice playing.
17	Monitoring inspections.
18	Appraisal administrator.
19	Ordinary appraisal personnel.
20	Alarm console.
21	Report handler.
22	Reserved value.
23	Monitoring console.
24	Configuration administrator.
25	Report administrator.
26	Making national outbound toll calls.
27	Making international outbound toll calls.
28	Making outbound calls using access codes of the call center.
29	Performing inspections in fullscreen mode.
30	Picking up calls for the agents in the same agent group.
31	Reserved value.
32	Configuring agents.
33~256	Reserved value.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "ccld":1,
```



```
    "name": "HW285601087",
    "realFlag": 1
  }
]
```

4.1.1.16 Querying Information About All Agents in a Specified VDN (Not Supported by CTI 3.6)

Scenario

This interface is invoked to query information about all agents in a specified VDN. Pagination query is not supported. (CTI 3.6 does not support this interface.)

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Internal Encapsulation Interface

BMS interface: /ccbms/ws/monitor/totalagentdynamicinfo

Request URL

https://ip:port/rest/cmsapp/v1/openapi/vdn/totalagentstate

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the request message body

No.	Parameter	Value Type	Mandatory	Description
1	cclid	Integer	Yes	ID of a call center. The value ranges from 1 to 65535.
2	vdn	Integer	Yes	ID of a VDN. The value ranges from 1 to 5000.

Response Description

Table 3 Parameters in the response message body

No.	Parameter	Value Type	Description
1	result	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
1	resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
3	resultDatas	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
2	resultDesc	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 .

Table 4 Parameters of resultDesc/resultDatas

No.	Parameter	Value Type	Description
2.1	agentId	Integer	Agent ID.
2.2	state	Integer	Agent status. For details, see CTI Platform Agent Status Code Reference .

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "ccld":1,
  "vdn":170
}
```

- Response parameters

```
{
  "result": "0",
  "resultCode": "0100000",
  "resultDatas": [
    {
      "agentId": 55002,
      "state": 0
    },
    {
      "agentId": 833,
      "state": 0
    }
  ],
  "resultDesc": [
    {
      "agentId": 55002,
      "state": 0
    },
    {
      "agentId": 833,
      "state": 0
    }
  ]
}
```

4.1.1.17 Obtaining Real-Time Monitoring Indicators by VDN

Description

This interface is invoked to obtain real-time monitoring indicators by VDN. (CTI 3.6 does not support this interface.)

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Internal Encapsulation Interfaces

BMS interfaces:

- /ccbms/ws/monitor/rindexSkill
- /ccbms/ws/monitor/queryivrInfo
- /ccbms/ws/monitor/queryivrdevicestatus
- /ccbms/ws/monitor/teamagentstatusinfobyvdn

Request URL

https://ip:port/rest/cmsapp/v1/openapi/rindex/vdn

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed to application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the request message body

No.	Parameter	Type	Mandatory	Description
1	ccId	Integer	No	ID of a call center. In the integration environment, the default value is the value of Call Center ID on the Configuration Center > System Management > CMS System Configuration page after a system administrator signs in to the AICC. If this parameter is empty, the data whose sys_key is default_cc_id and sys_type is cms_system_config in the t_cms_sysparam database table is used.

Table 2 Parameters in the request message body

No.	Parameter	Type	Mandatory	Description
				In the independent deployment environment, the default value is the value of Default Call Center ID on the Configuration Center > System Management > Parameter Settings page after a system administrator signs in to the AICC.
2	vdnIds	Array	Yes	Array of VDN IDs. The maximum length of the array is 100. The VDN ID is of the integer type. The value of vdn ranges from 1 to 5000.
3	indexIds	Array	Yes	Array of VDN real-time monitoring indicators. The indicator is of the string type. For details, see Table 2 .

Response Message

Table 3 Parameters in the response message body

No.	Parameter	Type	Description
1	result	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
2	resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
3	resultDatas	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
4	resultDesc	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 .

Table 4 Parameters of resultDesc/resultDatas

No.	Parameter	Type	Description
2.1	id	String	ID of a VDN.
2.2	idxs	List< JSONObject>	Group of indicators. For details about the parameters of this object, see Table 5 .

Table 5 Parameters of idx

No.	Parameter	Type	Description
2.2.1	id	String	Indicator ID.
2.2.2	val	Object	Indicator result.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "ccId":1,
  "vdnIds":[170,185],
  "indexIds":["IDX_COUNT_REAL_VDN_CONNECT","IDX_COUNT_REAL_VDN_CALL"]
}
```

- Response parameters

```
{
  "result": "0",
  "resultCode": "0100000",
  "resultDatas": [
    {
      "idxs": [
        {
          "val": 0,
          "id": "IDX_COUNT_REAL_VDN_CONNECT"
        },
        {
          "val": 0,
          "id": "IDX_COUNT_REAL_VDN_CALL"
        }
      ]
    }
  ],
}
```

```
"id": "170"
},
{
  "idxs": [
    {
      "val": 0,
      "id": "IDX_COUNT_REAL_VDN_CONNECT"
    },
    {
      "val": 0,
      "id": "IDX_COUNT_REAL_VDN_CALL"
    }
  ],
  "id": "185"
}
],
"resultDesc": [
  {
    "idxs": [
      {
        "val": 0,
        "id": "IDX_COUNT_REAL_VDN_CONNECT"
      },
      {
        "val": 0,
        "id": "IDX_COUNT_REAL_VDN_CALL"
      }
    ],
    "id": "170"
  },
  {
    "idxs": [
      {
        "val": 0,
        "id": "IDX_COUNT_REAL_VDN_CONNECT"
      },
      {
        "val": 0,
        "id": "IDX_COUNT_REAL_VDN_CALL"
      }
    ],
    "id": "185"
  }
]
```

```
]
}
```

4.1.2 Real-Time Interfaces for Skill Queues

4.1.2.1 Querying Signed-in Agents in Skill Queues in Batches

Scenario

This interface is invoked to query signed-in agents in skill queues in batches.



If the number of agents exceeds 10,000, the performance deteriorates.

Interface Method

The request method must be set to **POST**. This interface supports only the POST method.

Internal Encapsulation Interface

BMS interface: /ccbms/ws/monitor/agentskillstatusinfo

Request URL

https://ip:port/rest/cmsapp/v1/openapi/realindex/agent/agentsinskill

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the request body

No.	Parameter	Value Type	Mandatory	Description
1	cclId	Integer	Yes	ID of a call center. The value ranges from 1 to 65535.
2	vdn	Integer	Yes	ID of a VCC. The value ranges from 1 to 5000.
3	skillIds	Array	Yes	Object array of skill queue ID information. The array contains a maximum of 1,000 characters. The skill queue ID ranges from 1 to 10000.

Response Description

Table 3 Parameters in the response body

No.	Parameter	Value Type	Description
1	result	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.12 and earlier versions.
2	resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
3	resultDatas	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 . NOTE: This field is reserved for compatibility with interface response parameters in 8.12 and earlier versions.
4	resultDesc	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 .

Table 4 Parameters of resultDesc/resultDatas

No.	Parameter	Value Type	Description
2.1	skillId	Integer	Skill queue ID.
2.2	agentIds	Array	Array of IDs of the agents who sign in to the skill queue.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "queryParam":
  {
    "ccId":1,
    "vdn":1,
    "skillIds":[1,2]
  }
}
```

- Response parameters

```
{
  "result": "0",
  "resultCode": "0100000",
  "resultDatas": [
    {
      "skillId": 1,
      "agentIds": [
        291
      ]
    },
    {
      "skillId": 2,
      "agentIds": []
    }
  ],
  "resultDesc": [
    {
      "skillId": 1,
      "agentIds": [
```



```
291
    ]
  },
  {
    "skillId": 2,
    "agentIds": []
  }
]
```

4.1.2.2 Querying Extended Information About Skill Queues in Batches

Scenario

This interface is invoked to query extended information about skill queues in batches. (CTI 3.6 does not support this interface.)

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Internal Encapsulation Interface

BMS interface: /ccbms/ws/monitor/querytotalskillstatusex3

Request URL

https://*ip.port*/rest/cmsapp/v1/openapi/skill/totalskillstatusex3

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the request body

No.	Parameter	Value Type	Mandatory	Description
1	ccld	Integer	Yes	ID of a call center. The value ranges from 1 to 65535.
2	vdn	Integer	Yes	ID of a VCC. The value ranges from 1 to 5000.
3	skillIds	Array	Yes	Object array of skill queue ID information. The array contains a maximum of 1,000 characters. The skill queue ID ranges from 1 to 10000.

Response Description

Table 3 Parameters in the response body

No.	Parameter	Value Type	Description
1	result	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
2	resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
3	resultDatas	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 . NOTE:

Table 3 Parameters in the response body

No.	Parameter	Value Type	Description
			This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
4	resultDesc	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 .

Table 4 Parameters of resultDesc/resultDatas

No.	Parameter	Value Type	Description
2.1	skillId	Integer	Skill queue ID.
2.2	queueSize	Integer	Number of calls in waiting queue.
2.3	queueTime	Long	Maximum waiting duration of a call in the queue, in seconds.
2.4	loggedOnAgents	Integer	Number of agents who currently sign in to the skill queue.
2.5	availAgents	Integer	Number of agents available in a skill queue.
2.6	skillDescrip	String	Skill description.
2.7	callId	String	ID of the call that waits for the longest time.
2.8	maxQueueSize	Integer	Maximum number of calls that can wait in a queue. For details, see "Configuring Queue Processing and Completion Rate Parameters" in <i>(For Engineer) ICD VxxxRxxxCxx Product Documentation 01 (ICD)</i> .

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "ccId":1,
  "vdn":170,
  "skillIds":[1,2]
}
```

- Response parameters

```
{
```

```
"result": "0",
"resultCode": "0100000",
"resultDatas": [
  {
    "skillId": 1,
    "queueSize": 0,
    "loggedOnAgents": 0,
    "queueTime": 0,
    "availAgents": 0,
    "skillDescrip": "defaultVedioSkill",
    "callId": "4294967295-4294967295",
    "maxQueueSize": 100
  },
  {
    "skillId": 2,
    "queueSize": 0,
    "loggedOnAgents": 1,
    "queueTime": 0,
    "availAgents": 0,
    "skillDescrip": "defaultAudioSkill",
    "callId": "4294967295-4294967295",
    "maxQueueSize": 100
  }
],
"resultDesc": [
  {
    "skillId": 1,
    "queueSize": 0,
    "loggedOnAgents": 0,
    "queueTime": 0,
    "availAgents": 0,
    "skillDescrip": "defaultVedioSkill",
    "callId": "4294967295-4294967295",
    "maxQueueSize": 100
  },
  {
    "skillId": 2,
    "queueSize": 0,
    "loggedOnAgents": 1,
    "queueTime": 0,
    "availAgents": 0,
    "skillDescrip": "defaultAudioSkill",
    "callId": "4294967295-4294967295",
```

```
        "maxQueueSize": 100
    }
]
}
```

4.1.2.3 Querying the Call Statistics of Skill Queues in the Current 5 Minutes in Batches

Scenario

This interface is invoked to query the call statistics of skill queues in the current 5 minutes in batches. (CTI 3.6 does not support this interface.)

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Internal Encapsulation Interface

BMS interface: /ccbms/ws/monitor/querycurrentcallstatinfo

Request URL

https://*ip*.*port*/rest/cmsapp/v1/openapi/skill/currentcallstateinfo
In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.
In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the request body

No.	Parameter	Value Type	Mandatory	Description
1	cclid	Integer	Yes	ID of a call center. The value ranges from 1 to 65535.
2	vdn	Integer	Yes	ID of a VCC. The value ranges from 1 to 5000.
3	skillIds	Array	Yes	Object array of skill queue ID information. The array contains a maximum of 1,000 characters. The skill queue ID ranges from 1 to 10000.

Response Description

Table 3 Parameters in the response body

No.	Parameter	Value Type	Description
1	result	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
2	resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
3	resultDatas	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
4	resultDesc	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 .

Table 4 Parameters of resultDesc/resultDatas

No.	Parameter	Value Type	Description
2.1	id	Integer	Skill queue ID.
2.2	startTime	Long	Statistics start time. The value is a timestamp in milliseconds.
2.3	allCallCount	Long	Total number of inbound calls in the current 5 minutes.
2.4	cnndCallCount	Long	Total number of connected calls in the current 5 minutes.
2.5	abanCallCount	Long	Total number of lost calls in the current 5 minutes.
2.6	callEndCount	Long	Total number of finished calls in the current 5 minutes.
2.7	talkTimeOnlvr	Long	Call duration in the IVR system, in seconds. (When a suspended call is transferred to an agent, the suspension time is counted in the call duration.)
2.8	cnndIn20sCallCount	Long	Total number of calls that are connected in 20 seconds in the current 5 minutes.
2.9	clearUpCallCount	Long	Total number of calls that have been wrapped up in the current 5 minutes.
2.10	clearUpTime	Long	Total wrap-up duration in the current 5 minutes, in seconds.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "ccld":1,
  "vdn":170,
  "skilllds":[1,2]
}
```

- Response parameters

```
{
  "result": "0",
  "resultCode": "0100000",
  "resultDatas": [
    {
      "id": 1,
```

```
    "startTime": 1607390702000,
    "allCallCount": 0,
    "cnndCallCount": 0,
    "abanCallCount": 0,
    "callEndCount": 0,
    "talkTimeOnlvr": 0,
    "cnndIn20sCallCount": 0,
    "clearUpCallCount": 0,
    "clearUpTime": 0
  },
  {
    "id": 2,
    "startTime": 1607390702000,
    "allCallCount": 0,
    "cnndCallCount": 0,
    "abanCallCount": 0,
    "callEndCount": 0,
    "talkTimeOnlvr": 0,
    "cnndIn20sCallCount": 0,
    "clearUpCallCount": 0,
    "clearUpTime": 0
  }
],
"resultDesc": [
  {
    "id": 1,
    "startTime": 1607390702000,
    "allCallCount": 0,
    "cnndCallCount": 0,
    "abanCallCount": 0,
    "callEndCount": 0,
    "talkTimeOnlvr": 0,
    "cnndIn20sCallCount": 0,
    "clearUpCallCount": 0,
    "clearUpTime": 0
  },
  {
    "id": 2,
    "startTime": 1607390702000,
    "allCallCount": 0,
    "cnndCallCount": 0,
    "abanCallCount": 0,
    "callEndCount": 0,
```



```

    "talkTimeOnIvr": 0,
    "cnnedIn20sCallCount": 0,
    "clearUpCallCount": 0,
    "clearUpTime": 0
  }
]
}

```

4.1.2.4 Querying Information About the Status of a Specified Skill Queue

Scenario

This interface is invoked to query information about the status of a specified skill queue. (CTI 3.6 does not support this interface.)

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Internal Encapsulation Interface

BMS interface: /ccbms/ws/monitor/querybatchmultiskillqueue

Request URL

`https://ip.port/rest/cmsapp/v1/openapi/skill/queuedevicestatus`

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the request body

No.	Parameter	Value Type	Mandatory	Description
1	ccId	Integer	Yes	ID of a call center. The value ranges from 1 to 65535.
2	vdn	Integer	Yes	ID of a VCC. The value ranges from 1 to 5000.
3	skillId	Integer	Yes	Skill queue ID. The value ranges from 1 to 10000.

Response Description

Table 3 Parameters in the response body

No.	Parameter	Value Type	Description
1	result	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
2	resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
3	resultDatas	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
4	resultDesc	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 .

Table 4 Parameters of resultDesc/resultDatas

No.	Parameter	Value Type	Description
2.1	callId	String	Call ID.

Table 4 Parameters of resultDesc/resultDatas

No.	Parameter	Value Type	Description
2.2	customerLevel	Integer	Customer level. The sequence number ranges from 1 to 254. A larger value indicates the importance of a customer.
2.3	requestType	Integer	Call feature. For details, see Table 1 .

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "ccId":1,
  "vdn":170,
  "skillId":2
}
```

- Response parameters

```
{
  "result": "0",
  "resultCode": "0100000",
  "resultDatas": [
    {
      "callId": "1604285072000-840",
      "requestType": 0,
      "customerLevel": 0
    }
  ],
  "resultDesc": [
    {
      "callId": "1604285072000-840",
      "requestType": 0,
      "customerLevel": 0
    }
  ]
}
```

4.1.2.5 Querying Information About Skill Queue Configurations in Batches

Scenario

This interface is invoked to query information about skill queue configurations in batches.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Internal Encapsulation Interface

BMS interface: /ccbms/ws/monitor/skillstatusinfo

Request URL

`https://ip.port/rest/cmsapp/v1/openapi/skill/skillcfinfo`

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the request body

No.	Parameter	Value Type	Mandatory	Description
1	cclid	Integer	Yes	ID of a call center. The value ranges from 1 to 65535.
2	vdn	Integer	Yes	ID of a VCC. The value ranges from 1 to 5000.

Table 2 Parameters in the request body

No.	Parameter	Value Type	Mandatory	Description
3	skillIds	Array	Yes	Object array of skill queue ID information. The array contains a maximum of 1,000 characters. The skill queue ID ranges from 1 to 10000.

Response Description

Table 3 Parameters in the response body

No.	Parameter	Value Type	Description
1	result	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
2	resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
3	resultDatas	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
2	resultDesc	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 .

Table 4 Parameters of resultDesc/resultDatas

No.	Parameter	Value Type	Description
2.1	skillId	Integer	Skill queue ID.
2.2	maxCallNum	Integer	Maximum number of queuing calls.
2.3	maxHoldCallNum	Integer	Maximum number of held calls.
2.4	maxWaitTime	Long	Maximum waiting duration, in seconds.

Table 4 Parameters of resultDesc/resultDatas

No.	Parameter	Value Type	Description
2.5	maxAnswerWaitTime	Long	Maximum ring duration, in seconds.
2.6	maxTalkTime	Long	Maximum call duration, in seconds.
2.7	maxRestAgentNum	Integer	Maximum number of agents that are allowed to have a rest.
2.8	maxAgentWrapupTime	Integer	Maximum agent wrap-up duration, in seconds.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "ccId":1,
  "vdn":170,
  "skillIds":[1,2]
}
```

- Response parameters

```
{
  "result": "0",
  "resultCode": "0100000",
  "resultDatas": [
    {
      "skillId": 1,
      "maxHoldCallNum": 5,
      "maxTalkTime": 86400,
      "maxAgentWrapupTime": 5,
      "maxWaitTime": 60,
      "maxRestAgentNum": 2,
      "maxCallNum": 100,
      "maxAnswerWaitTime": 30
    },
    {
      "skillId": 2,
      "maxHoldCallNum": 5,
      "maxTalkTime": 86400,
      "maxAgentWrapupTime": 5,
```

```

        "maxWaitTime": 60,
        "maxRestAgentNum": 2,
        "maxCallNum": 100,
        "maxAnswerWaitTime": 30
    }
],
"resultDesc": [
    {
        "skillId": 1,
        "maxHoldCallNum": 5,
        "maxTalkTime": 86400,
        "maxAgentWrapupTime": 5,
        "maxWaitTime": 60,
        "maxRestAgentNum": 2,
        "maxCallNum": 100,
        "maxAnswerWaitTime": 30
    },
    {
        "skillId": 2,
        "maxHoldCallNum": 5,
        "maxTalkTime": 86400,
        "maxAgentWrapupTime": 5,
        "maxWaitTime": 60,
        "maxRestAgentNum": 2,
        "maxCallNum": 100,
        "maxAnswerWaitTime": 30
    }
]
}

```

4.1.2.6 Querying the Statistics Details of Skill Queues in Batches

Scenario

This interface is invoked to query the statistics details of skill queues in batches. (CTI 3.6 does not support this interface.)

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Internal Encapsulation Interfaces

BMS interfaces:

- /ccbms/ws/monitor/querycurrentcallstatinfo
- /ccbms/ws/monitor/queryStatInfoOfEverySkill

Request URL

`https://ip.port/rest/cmsapp/v1/openapi/skill/stateinfo`

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the request body

No.	Parameter	Value Type	Mandatory	Description
1	cclId	Integer	Yes	ID of a call center. The value ranges from 1 to 65535 .
2	vdn	Integer	Yes	ID of a VCC. The value ranges from 1 to 5000.
3	skillIds	Array	Yes	Object array of skill queue ID information. The array contains a maximum of 1,000 characters. The skill queue ID ranges from 1 to 10000.

Response Description

Table 3 Parameters in the response body

No.	Parameter	Value Type	Description
1	result	String	Query result. The options are as follows: 0: success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.12 and earlier versions.
2	resultCode	String	Query result. The options are as follows: 0100000: success Other values: failure For details about the failure causes, see Error Code Reference .
3	resultDatas	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4. NOTE: This field is reserved for compatibility with interface response parameters in 8.12 and earlier versions.
4	resultDesc	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4.

Table 4 Parameters of resultDesc/resultDatas

No.	Parameter	Value Type	Description
2.1	skillId	Integer	Skill queue ID.
2.2	agentLoginNums	Long	Number of signed-in agents.
2.3	agentTalkingNums	Long	Number of agents in talking state.
2.4	agentIdleNums	Long	Number of idle agents.
2.5	agentAvailableNums	Long	Number of available agents. The value may be different from the number of idle agents when an agent can have one-to-many conversations.
2.6	agentSetBusyNums	Long	Number of agents who indicate busy.
2.7	agentRestNums	Long	Number of agents who take a rest.
2.8	agentMaxIdleTimeIn5	Long	Maximum idle duration of agents in 5 minutes.

Table 4 Parameters of resultDesc/resultDatas

No.	Parameter	Value Type	Description
2.9	agentWorkNums	Long	Number of agents in after-call processing state, including agents in wrap-up state.
2.10	agentStudyNums	Long	Number of agents who are in the Study state.
2.11	agentOccupyNums	Long	Number of agents in the occupied state.
2.12	agentPreOccupyNums	Long	Number of preoccupied agents.
2.13	totalCallNums	Long	Total number of processed calls, including the calls that are being processed.
2.14	processingCallNums	Long	Number of calls that are being processed.
2.15	maxCallWaitTimesIn5	Long	Maximum time that a call can wait in 5 minutes.
2.16	callWaitNums	Long	Number of queuing calls (number of calls that are waiting).
2.17	evenCallTimeIn5	Long	Average call duration in 5 minutes.
2.18	evenWaitTimeIn5	Long	Average call waiting duration in 5 minutes.
2.19	allCallCount	Long	Total number of inbound calls in the current 5 minutes.
2.20	cnnedIn20sCallCount	Long	Total number of calls that are connected in 20 seconds in the current 5 minutes.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "queryParam":
  {
    "ccld":1,
    "vdn":1,
    "skillIds":[1,2]
  }
}
```

- Response parameters

```
{
```

```
"result": "0",
"resultCode": "0100000",
"resultDatas": [
  {
    "skillId": 1,
    "agentLoginNums": 1,
    "agentTalkingNums": 0,
    "agentIdleNums": 0,
    "agentAvailableNums": 0,
    "agentSetBusyNums": 1,
    "agentRestNums": 0,
    "agentMaxIdleTimeIn5": 0,
    "agentWorkNums": 0,
    "agentStudyNums": 0,
    "agentOccupyNums": 0,
    "agentPreOccupyNums": 0,
    "totalCallNums": 0,
    "processingCallNums": 0,
    "maxCallWaitTimesIn5": 0,
    "callWaitNums": 0,
    "evenCallTimeIn5": 0,
    "evenWaitTimeIn5": 0,
    "allCallCount": 0,
    "cnnedIn20sCallCount": 0
  },
  {
    "skillId": 2,
    "agentLoginNums": 0,
    "agentTalkingNums": 0,
    "agentIdleNums": 0,
    "agentAvailableNums": 0,
    "agentSetBusyNums": 0,
    "agentRestNums": 0,
    "agentMaxIdleTimeIn5": 0,
    "agentWorkNums": 0,
    "agentStudyNums": 0,
    "agentOccupyNums": 0,
    "agentPreOccupyNums": 0,
    "totalCallNums": 0,
    "processingCallNums": 0,
    "maxCallWaitTimesIn5": 0,
    "callWaitNums": 0,
    "evenCallTimeIn5": 0,
```

```
        "evenWaitTimeIn5": 0,
        "allCallCount": 0,
        "cnndIn20sCallCount": 0
    }
],
"resultDesc": [
    {
        "skillId": 1,
        "agentLoginNums": 1,
        "agentTalkingNums": 0,
        "agentIdleNums": 0,
        "agentAvailableNums": 0,
        "agentSetBusyNums": 1,
        "agentRestNums": 0,
        "agentMaxIdleTimeIn5": 0,
        "agentWorkNums": 0,
        "agentStudyNums": 0,
        "agentOccupyNums": 0,
        "agentPreOccupyNums": 0,
        "totalCallNums": 0,
        "processingCallNums": 0,
        "maxCallWaitTimesIn5": 0,
        "callWaitNums": 0,
        "evenCallTimeIn5": 0,
        "evenWaitTimeIn5": 0,
        "allCallCount": 0,
        "cnndIn20sCallCount": 0
    },
    {
        "skillId": 2,
        "agentLoginNums": 0,
        "agentTalkingNums": 0,
        "agentIdleNums": 0,
        "agentAvailableNums": 0,
        "agentSetBusyNums": 0,
        "agentRestNums": 0,
        "agentMaxIdleTimeIn5": 0,
        "agentWorkNums": 0,
        "agentStudyNums": 0,
        "agentOccupyNums": 0,
        "agentPreOccupyNums": 0,
        "totalCallNums": 0,
        "processingCallNums": 0,
```

```

    "maxCallWaitTimesIn5": 0,
    "callWaitNums": 0,
    "evenCallTimeIn5": 0,
    "evenWaitTimeIn5": 0,
    "allCallCount": 0,
    "cnndIn20sCallCount": 0
  }
]
}
```

4.1.2.7 Querying All Calls Queuing in a Skill Queue in Batches

Scenario

This interface is invoked to query all calls queuing in a skill queue in batches. (This interface cannot be used to query multimedia queue information. CTI 3.6 does not support this interface.)

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Internal Encapsulation Interfaces

BMS interfaces:

- /ccbms/ws/monitor/querybatchmultiskillqueue
- /ccbms/ws/monitor/skillcallqueueinfos

Request URL

`https://ip.port/rest/cmsapp/v1/openapi/skill/batchmultiskill`

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the request body

No.	Parameter	Value Type	Mandatory	Description
1	cclId	Integer	Yes	ID of a call center. The value ranges from 1 to 65535.
2	vdn	Integer	Yes	ID of a VCC. The value ranges from 1 to 5000.
3	skillIds	Array	Yes	Object array of skill queue ID information. The array contains a maximum of 1,000 characters. The skill queue ID ranges from 1 to 10000.

Response Description

Table 3 Parameters in the response body

No.	Parameter	Value Type	Description
1	result	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
2	resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
3	resultDatas	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 . NOTE:

Table 3 Parameters in the response body

No.	Parameter	Value Type	Description
			This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
2	resultDesc	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 .

Table 4 Parameters of resultDesc/resultDatas

No.	Parameter	Value Type	Description
2.1	skillId	Integer	Skill queue ID. The value ranges from 1 to 10000.
2.2	callPropertyExs	Array	Array of calls queuing in the skill queue. For details about the parameters of the object in this array, see Table 5 .

Table 5 Parameters of callPropertyExs

No.	Parameter	Value Type	Description
2.2.1	callId	String	Call ID.
2.2.2	customerLevel	Integer	Level of the customer making a call. The sequence number ranges from 1 to 254. A larger value indicates the importance of a customer.
2.2.3	requestType	Integer	Call feature. For details, see Table 1 .
2.2.4	calling	Array	Calling number.
2.2.5	called	Array	Called number.
2.2.6	mediaType	String	Call media type. Multimedia calls cannot be queried. For details, see Table 1 .
2.2.7	orgiCalled	String	Original called number. If the called number changes before the call accesses the system, this parameter records the number of the first called party.
2.2.8	callInTime	Integer	Call arrival time (millisecond-level timestamp).
2.2.9	callPriority	Long	Call priority. For details, see "Configuring Skill Groups" in <i>(For Engineer) ICD VxxxRxxxCxx Product Documentation 01 (ICD)</i> .
2.2.10	waitTime	Long	Duration that a call waits in the current queue.
2.2.11	callQueueFlag	Long	Whether the current queue is the main queue of a call. 1 : yes 0 : no

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "ccId":1,
  "vdn":170,
  "skillIds":[1,2]
}
```

- Response parameters

```
{
  "result": "0",
  "resultCode": "0100000",
  "resultDatas": [
    {
      "skillId": 1,
      "callPropertyExs": []
    },
    {
      "skillId": 2,
      "callPropertyExs": [
        {
          "requestType": 0,
          "callPriority": 0,
          "waitTime": 0,
          "callQueueFlag": 1,
          "callId": "1607392439000-379",
          "customerLevel": 0,
          "calling": "1233",
          "called": "34565",
          "mediaType": "MEDIA_TYPE_PHONE",
          "orgiCalled": "234545",
          "callInTime": 1223455666
        }
      ]
    }
  ],
  "resultDesc": [
    {
```



```

    "skillId": 1,
    "callPropertyExs": []
  },
  {
    "skillId": 2,
    "callPropertyExs": [
      {
        "requestType": 0,
        "callPriority": 0,
        "waitTime": 0,
        "callQueueFlag": 1,
        "callId": "1607392439000-379",
        "customerLevel": 0,
        "calling": "1233",
        "called": "34565",
        "mediaType": "MEDIA_TYPE_PHONE",
        "orgiCalled": "234545",
        "callInTime": 1223455666
      }
    ]
  }
]
}

```

4.1.2.8 Querying the Number of Calls that Each Queue Answers from Other Queues in Batches When Calls are Queuing in Multiple Skill Queues

Scenario

This interface is invoked to query the number of calls that each queue answers from other queues in batches when calls are queuing in multiple skill queues.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Internal Encapsulation Interface

BMS interface: /ccbms/ws/monitor/calloverloadstatus

Request URL

`https://ip.port/rest/cmsapp/v1/openapi/skill/batchskilloverload`

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the request body

No.	Parameter	Value Type	Mandatory	Description
1	cclid	Integer	Yes	ID of a call center. The value ranges from 1 to 65535.
2	vdn	Integer	Yes	ID of a VCC. The value ranges from 1 to 5000.
3	skillIds	Array	Yes	Object array of skill queue ID information. The array contains a maximum of 1,000 characters. The skill queue ID ranges from 1 to 10000.

Response Description

Table 3 Parameters in the response body

No.	Parameter	Value Type	Description
1	result	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference . NOTE:

Table 3 Parameters in the response body

No.	Parameter	Value Type	Description
			This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
2	resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
3	resultDatas	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
4	resultDesc	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 .

Table 4 Parameters of resultDesc/resultDatas

No.	Parameter	Value Type	Description
2.1	skillId	Integer	Skill queue ID. The value ranges from 1 to 10000.
2.2	overloadDetails	Array	Details about the number of calls in other queues. For details about the parameters of the object in this array, see Table 5 .

Table 5 Parameter description of overloadDetails

No.	Parameter	Value Type	Description
2.2.1	queueId	Integer	Skill queue ID.
2.2.2	callNum	Long	Number of answered calls.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "ccId":1,
```

```
"vdn":170,  
"skillIds":[1,2]  
}
```

- Response parameters

```
{  
  "result": "0",  
  "resultCode": "0100000",  
  "resultDatas": [  
    {  
      "skillId": 1,  
      "overloadDetails": [  
        {  
          "queueId": 2,  
          "callNum": 1  
        }  
      ]  
    },  
    {  
      "skillId": 2,  
      "overloadDetails": []  
    }  
  ],  
  "resultDesc": [  
    {  
      "skillId": 1,  
      "overloadDetails": [  
        {  
          "queueId": 2,  
          "callNum": 1  
        }  
      ]  
    },  
    {  
      "skillId": 2,  
      "overloadDetails": []  
    }  
  ]  
}
```

4.1.2.9 Querying the Number of Agents in Each State in a Skill Group

Scenario

This interface is invoked to query the number of agents in each state in a skill group.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Internal Encapsulation Interface

BMS interface: /ccbms/ws/monitor/agentstatusinfobyskillids

Request URL

`https://ip.port/rest/cmsapp/v1/openapi/skill/agentstatusbyskill`

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the request body

No.	Parameter	Value Type	Mandatory	Description
1	cclid	Integer	Yes	ID of a call center. The value ranges from 1 to 65535.
2	vdn	Integer	Yes	ID of a VCC. The value ranges from 1 to 5000.

Table 2 Parameters in the request body

No.	Parameter	Value Type	Mandatory	Description
3	skillIds	Array	Yes	Object array of skill queue ID information. The array contains a maximum of 1,000 characters. The skill queue ID ranges from 1 to 10000.
4	isSkillAllMatch	String	No	Whether the agent skill queue is fully matched. The value is of the Boolean type. The default value is false . The options are as follows: true : yes. The statistics are collected only when the skill queue to which an agent logs in completely matches the queried skill queue. false : no. The statistics are collected if the skill queue to be queried is contained in the skill queues to which the agent logs in.

Response Description

Table 3 Parameters in the response body

No.	Parameter	Value Type	Description
1	result	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
2	resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
3	resultDatas	Object	Object in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
4	resultDesc	Object	Object in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 .

Table 4 Parameters of resultDesc/resultDatas

No.	Parameter	Value Type	Description
2.1	workSubStateBegin	Integer	Number of agents in wrap-up state.
2.2	lateAdjust	Integer	Number of agents in wrap-up state.
2.3	usable	Integer	Number of available agents.
2.4	rest	Integer	Number of agents in rest state.
2.5	studying	Integer	Number of agents in study state.
2.6	setBusy	Integer	Number of agents in busy state.
2.7	occupy	Integer	Number of agents in occupied state.
2.8	idle	Integer	Number of agents in idle state.
2.9	preoccupy	Integer	Number of agents in preoccupied state.
2.10	answering	Integer	Number of agents in answering state.
2.11	talking	Integer	Number of agents in talking state.
2.12	loginEd	Integer	Number of signed-in agents.
2.13	notlogged	int	Number of agents who have not signed in.
2.14	callOut	Integer	Number of agents who make outbound calls.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "ccId":1,
  "vdn":170,
  "skillIds":[1,2],
  "isSkillAllMatch":"true"
}
```

- Response parameters

```
{
  "result": "0",
  "resultCode": "0100000",
```

```
"resultDatas": {
  "workSubStateBegin": 0,
  "lateAdjust": 0,
  "usable": 0,
  "rest": 0,
  "studying": 0,
  "setBusy": 1,
  "occupy": 0,
  "idle": 0,
  "preoccupy": 0,
  "answering": 0,
  "talking": 0,
  "loginEd": 1,
  "notlogged": 0,
  "callOut": 1
},
"resultDesc": {
  "workSubStateBegin": 0,
  "lateAdjust": 0,
  "usable": 0,
  "rest": 0,
  "studying": 0,
  "setBusy": 1,
  "occupy": 0,
  "idle": 0,
  "preoccupy": 0,
  "answering": 0,
  "talking": 0,
  "loginEd": 1,
  "notlogged": 0,
  "callOut": 1
}
}
```

4.1.2.10 Querying Details About Signed-in Agents in a Skill Queue

Scenario

This interface is invoked to return details about signed-in agents in a skill queue based on the skill queue ID.

 NOTE

If the number of agents exceeds 10,000, the performance deteriorates.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Internal Encapsulation Interface

BMS interface: /ccbms/ws/monitor/agentstatusinfo

Request URL

https://ip:port/rest/cmsapp/v1/openapi/skill/agentbyconfigedskill

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the message body

No.	Parameter	Value Type	Mandatory	Description
1	cclid	Integer	Yes	ID of a call center. The value ranges from 1 to 65535.
2	vdn	Integer	Yes	ID of a VCC. The value ranges from 1 to 5000.
3	skillid	Integer	Yes	Skill queue ID. The value ranges from 1 to 10000.

Response Description

Table 3 Parameters in the message body

No.	Parameter	Value Type	Description
1	result	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.12 and earlier versions.
2	resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
3	resultDatas	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 . NOTE: This field is reserved for compatibility with interface response parameters in 8.12 and earlier versions.
4	resultDesc	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 .

Table 4 Parameters of resultDesc/resultDatas

No.	Parameter	Value Type	Description
2.1	workNo	Integer	Agent ID.
2.2	name	String	Agent name.
2.3	status	Integer	Agent status. For details, see Table 5 .
2.4	ctiStatus	Integer	Platform agent status. For details, see CTI Platform Agent Status Code Reference .
2.5	groupId	Integer	Agent group ID.
2.6	mediaType	Integer	Agent media type. For details, see Table 1 .
2.7	vdnId	Integer	ID of the VCC to which a call belongs.
2.8	phoneNumber	String	Agent phone number.
2.9	currentStateTime	Long	Duration of the current state, in seconds.

Table 4 Parameters of resultDesc/resultDatas

No.	Parameter	Value Type	Description
2.10	loginDate	Long	Sign-in time. The value is a timestamp in milliseconds.

Table 5 Agent status

Value	Description
0	Unknown. The value corresponds to status 9 on the CTI platform.
2	An exception occurs when the interface is invoked.
3	Showing busy. The value corresponds to status 7 on the CTI platform.
4	Idle. The value corresponds to status 1 on the CTI platform.
5	Wrap-up. The value corresponds to status 6 on the CTI platform.
7	Talking. The value corresponds to states 2, 3, 4, and 5 on the CTI platform.
8	Rest. The value corresponds to status 8 on the CTI platform.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "queryParam":
  {
    "ccId":1,
    "vdn":1,
    "skillId":1
  }
}
```

- Request parameters

```
{
  "result": "0",
  "resultCode": "0100000",
  "resultDatas": [
    {
      "workNo": 20011,
```

```

    "name": "20011",
    "status": 4,
    "ctiStatus": 1,
    "groupId": 1,
    "mediaType": 1,
    "vdnId": 1,
    "phoneNumber": "70942",
    "currentStateTime": 1505,
    "loginDate": 1501125758000
  },
  "resultDesc": [
    {
      "workNo": 20011,
      "name": "20011",
      "status": 4,
      "ctiStatus": 1,
      "groupId": 1,
      "mediaType": 1,
      "vdnId": 1,
      "phoneNumber": "70942",
      "currentStateTime": 1505,
      "loginDate": 1501125758000
    }
  ]
}

```

4.1.2.11 Querying Service Quality for Skill Queues in Batches

Scenario

This interface is used to query data of a 5-minute interval that includes the start point of the last 5 minutes.

The last 5 minutes indicate the 5 minutes before the current time. For a 5-minute interval, both the start time and end time end with 0 or 5, in minutes, for example, 00 to 05 and 05 to 10.

NOTICE

If the software parameter **P50** is set to **1** on the WAS configuration console, the **callstatinfos** field does not count the number of calls in the scenarios where no CDR is generated, for example, no agent is on duty, the queue is busy, or the queue overflows.

Involved scenarios:

1. When a call is routed to skill queue 1 and no agent is on duty in skill queue 1, the call is transferred to skill queue 2. In this scenario, only media CDRs related to skill queue 2 are generated, and no CDR is generated for skill queue 1. Calls only of skill queue 2 are counted.
2. When a call is routed to skill queue 1 and the number of queuing calls in skill queue 1 exceeds the maximum (the forwarding on busy condition is set to the device usage), the call is transferred from skill queue 1 to skill queue 2. In this scenario, only media CDRs related to skill queue 2 are generated, and no CDR is generated for skill queue 1. Calls only of skill queue 2 are counted.
3. When a call is routed to skill queue 1 and the predicted queuing duration in skill queue 1 exceeds the configured duration for transfer in busy hours (the forwarding on busy condition is set to the waiting duration), the call is transferred from skill queue 1 to skill queue 2. In this scenario, only media CDRs related to skill queue 2 are generated, and no CDR is generated for skill queue 1. Calls only of skill queue 2 are counted.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Internal Encapsulation Interface

BMS interface: /ccbms/ws/monitor/callStatInfos

Request URL

https://ip:port/rest/cmsapp/v1/openapi/skill/servicequality

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

NOTE

The current interface performance is affected. Interface flow control is added to ensure that the interface can be invoked only once per second.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the request body

No.	Parameter	Value Type	Mandatory	Description
1	ccId	Integer	Yes	ID of a call center. The value ranges from 1 to 65535 .
2	vdn	Integer	Yes	ID of a VCC. The value ranges from 1 to 5000 .
3	skillIds	Array	Yes	Object array of skill queue ID information. The array contains a maximum of 1,000 characters. The skill queue ID ranges from 1 to 10000.

Response Description

Table 3 Parameters in the response body

No.	Parameter	Value Type	Description
1	resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
2	resultDesc	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 .

Table 4 Parameters of resultDesc

No.	Parameter	Value Type	Description
2.1	skillId	Integer	Skill queue ID.
2.2	statStartTime	Long	Time when the statistics collection starts. The value is a timestamp in milliseconds.
2.3	inCallNum	Long	Total number of inbound calls in the current 5 minutes in a skill queue.
2.4	connectedNum	Long	Total number of connected calls in the current 5 minutes in a skill queue.
2.5	callLostNum	Long	Total number of lost calls in the current 5 minutes in a skill queue.
2.6	in15sCallNum	Long	Total number of calls that are connected within 15 seconds in the last 5 minutes in a skill queue.
2.7	wait15sLostNum	Long	Total number of lost calls whose queuing duration is less than 15 seconds in the current 5 minutes in a skill queue.
2.8	in20sCallNum	Long	Total number of calls that are connected within 20 seconds in the last 5 minutes in a skill queue.

Table 4 Parameters of resultDesc

No.	Parameter	Value Type	Description
2.9	wait20sLostNum	Long	Total number of lost calls whose queuing duration is less than 20 seconds in the current 5 minutes in a skill queue.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "ccId":1,
  "vdn":1,
  "skillIds":[1,2]
}
```

- Response parameters

```
{
  "resultCode": "0100000",
  "resultDesc": [
    {
      "in15sCallNum": 0,
      "skillId": 1,
      "wait15sLostNum": 0,
      "callLostNum": 0,
      "in20sCallNum": 0,
      "statStartTime": 1611714900000,
      "wait20sLostNum": 0,
      "inCallNum": 0,
      "connectedNum": 0
    }
  ]
}
```

4.1.2.12 Obtaining Real-Time Monitoring Indicators by Skill Queue

Scenario

This interface is invoked to obtain real-time monitoring indicators by skill queue.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Internal Encapsulation Interface

BMS interface: /ccbms/ws/monitor/rindexSkill

Request URL

`https://ip.port/rest/cmsapp/v1/openapi/skill/rindex`

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the message body

Parameter	Value Type	Mandatory	Description
cclId	Integer	Yes	ID of a call center. The value ranges from 1 to 65535.
vdn	Integer	Yes	ID of a VCC. The value ranges from 1 to 5000.
skillIds	Array	Yes	Object array of skill queue ID information. The array contains a maximum of 1,000 characters. The skill queue ID ranges from 1 to 10000.
indexIds	Array	Yes	Array of indicators. The maximum length of the array depends on the number of real-time monitoring indicators by skill. The maximum value is 12 currently.

Table 2 Parameters in the message body

Parameter	Value Type	Mandatory	Description
			For details about the parameters of the object, see Table 2 .

Response Description

Table 3 Parameters in the message body

Parameter	Value Type	Description
result	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
resultDatas	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
resultDesc	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 .

Table 4 Parameters of resultDesc/resultDatas

Parameter	Value Type	Description
id	Integer	Skill queue ID.
idxs	Array	Group of indicators. For details about the parameters of the object, see Table 5 .

Table 5 Parameters of idxs

Parameter	Value Type	Description
id	String	Indicator ID.
val	Object	Indicator result. For details about the parameters of the object, see Table 2 .

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "ccId":1,
  "vdn":170,
  "skillIds":[1,2],

  "indexIds":["IDX_COUNT_REAL_SKILL_QUEUE","IDX_COUNT_REAL_SKILL_HANDLE","IDX_DURATION_SKILL_MAX_QUEUE_ACK",
    "IDX_COUNT_20S_CALL_FOR_5MIN","IDX_COUNT_REAL_AGENT_SIGN_IN","IDX_COUNT_REAL_AGENT_CONNECT",
    "IDX_COUNT_REAL_AGENT_SET_IDLE","IDX_COUNT_REAL_AGENT_USABLE","IDX_COUNT_REAL_AGENT_SET_BUSY",
    "IDX_COUNT_REAL_AGENT_ACW","IDX_COUNT_REAL_AGENT_REST","IDX_COUNT_REAL_AGENT_STUDY"]
}
```

- Response parameters

```
{
  "result": "0",
  "resultCode": "0100000",
  "resultDatas": [
    {
      "id": "1",
      "idxs": [
        {
          "id": "IDX_COUNT_REAL_SKILL_QUEUE",
          "val": "0"
        },
        {
          "id": "IDX_COUNT_REAL_SKILL_HANDLE",
```

```
    "val": "0"
  },
  {
    "id": "IDX_DURATION_SKILL_MAX_QUEUE_ACK",
    "val": "0"
  },
  {
    "id": "IDX_COUNT_20S_CALL_FOR_5MIN",
    "val": "0"
  },
  {
    "id": "IDX_COUNT_REAL_AGENT_SIGN_IN",
    "val": "0"
  },
  {
    "id": "IDX_COUNT_REAL_AGENT_CONNECT",
    "val": "0"
  },
  {
    "id": "IDX_COUNT_REAL_AGENT_SET_IDLE",
    "val": "0"
  },
  {
    "id": "IDX_COUNT_REAL_AGENT_USABLE",
    "val": "0"
  },
  {
    "id": "IDX_COUNT_REAL_AGENT_SET_BUSY",
    "val": "0"
  },
  {
    "id": "IDX_COUNT_REAL_AGENT_ACW",
    "val": "0"
  },
  {
    "id": "IDX_COUNT_REAL_AGENT_REST",
    "val": "0"
  },
  {
    "id": "IDX_COUNT_REAL_AGENT_STUDY",
    "val": "0"
  }
]
```

```
,
{
  "id": "2",
  "idxs": [
    {
      "id": "IDX_COUNT_REAL_SKILL_QUEUE",
      "val": "0"
    },
    {
      "id": "IDX_COUNT_REAL_SKILL_HANDLE",
      "val": "0"
    },
    {
      "id": "IDX_DURATION_SKILL_MAX_QUEUE_ACK",
      "val": "0"
    },
    {
      "id": "IDX_COUNT_20S_CALL_FOR_5MIN",
      "val": "0"
    },
    {
      "id": "IDX_COUNT_REAL_AGENT_SIGN_IN",
      "val": "1"
    },
    {
      "id": "IDX_COUNT_REAL_AGENT_CONNECT",
      "val": "0"
    },
    {
      "id": "IDX_COUNT_REAL_AGENT_SET_IDLE",
      "val": "0"
    },
    {
      "id": "IDX_COUNT_REAL_AGENT_USABLE",
      "val": "0"
    },
    {
      "id": "IDX_COUNT_REAL_AGENT_SET_BUSY",
      "val": "1"
    },
    {
      "id": "IDX_COUNT_REAL_AGENT_ACW",
      "val": "0"
    }
  ]
}
```

```
    },
    {
      "id": "IDX_COUNT_REAL_AGENT_REST",
      "val": "0"
    },
    {
      "id": "IDX_COUNT_REAL_AGENT_STUDY",
      "val": "0"
    }
  ]
}
],
"resultDesc": [
  {
    "id": "1",
    "idxs": [
      {
        "id": "IDX_COUNT_REAL_SKILL_QUEUE",
        "val": "0"
      },
      {
        "id": "IDX_COUNT_REAL_SKILL_HANDLE",
        "val": "0"
      },
      {
        "id": "IDX_DURATION_SKILL_MAX_QUEUE_ACK",
        "val": "0"
      },
      {
        "id": "IDX_COUNT_20S_CALL_FOR_5MIN",
        "val": "0"
      },
      {
        "id": "IDX_COUNT_REAL_AGENT_SIGN_IN",
        "val": "0"
      },
      {
        "id": "IDX_COUNT_REAL_AGENT_CONNECT",
        "val": "0"
      },
      {
        "id": "IDX_COUNT_REAL_AGENT_SET_IDLE",
        "val": "0"
      }
    ]
  }
]
```

```
,
{
  "id": "IDX_COUNT_REAL_AGENT_USABLE",
  "val": "0"
},
{
  "id": "IDX_COUNT_REAL_AGENT_SET_BUSY",
  "val": "0"
},
{
  "id": "IDX_COUNT_REAL_AGENT_ACW",
  "val": "0"
},
{
  "id": "IDX_COUNT_REAL_AGENT_REST",
  "val": "0"
},
{
  "id": "IDX_COUNT_REAL_AGENT_STUDY",
  "val": "0"
}
]
},
{
  "id": "2",
  "idxs": [
    {
      "id": "IDX_COUNT_REAL_SKILL_QUEUE",
      "val": "0"
    },
    {
      "id": "IDX_COUNT_REAL_SKILL_HANDLE",
      "val": "0"
    },
    {
      "id": "IDX_DURATION_SKILL_MAX_QUEUE_ACK",
      "val": "0"
    },
    {
      "id": "IDX_COUNT_20S_CALL_FOR_5MIN",
      "val": "0"
    }
  ],
}
```

```
    "id": "IDX_COUNT_REAL_AGENT_SIGN_IN",
    "val": "1"
  },
  {
    "id": "IDX_COUNT_REAL_AGENT_CONNECT",
    "val": "0"
  },
  {
    "id": "IDX_COUNT_REAL_AGENT_SET_IDLE",
    "val": "0"
  },
  {
    "id": "IDX_COUNT_REAL_AGENT_USABLE",
    "val": "0"
  },
  {
    "id": "IDX_COUNT_REAL_AGENT_SET_BUSY",
    "val": "1"
  },
  {
    "id": "IDX_COUNT_REAL_AGENT_ACW",
    "val": "0"
  },
  {
    "id": "IDX_COUNT_REAL_AGENT_REST",
    "val": "0"
  },
  {
    "id": "IDX_COUNT_REAL_AGENT_STUDY",
    "val": "0"
  }
]
}
```

4.1.3 Real-Time Interfaces for Agents

4.1.3.1 Querying Skill Queues to Which Agents Sign In in Batches

Scenario

This interface is invoked to query skill queues to which specified agents sign in in batches.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Internal Encapsulation Interface

CTI3.6: BMS interface: /ccbms/ws/monitor/siteagentcallandskillinfoobyagentids

CTI3.8: BMS interface: /ccbms/ws/monitor/queryAgentsLoginSkills

Request URL

https://ip:port/rest/cmsapp/v1/openapi/agentgroup/siteagentcallandskillinfoobyagentids

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the message body

No.	Parameter	Value Type	Mandatory	Description
1	ccld	Integer	Yes	ID of a call center. The value ranges from 1 to 65535.
2	vdn	Integer	Yes	ID of a VCC. The value ranges from 1 to 5000.
3	agentIds	Array	Yes	Array of agent IDs. The maximum length of the array is 6000. Value range: 101–59999 and 70000–999999

Response Description

Table 3 Parameters in the message body

No.	Parameter	Value Type	Description
1	result	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
2	resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
3	resultDatas	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
3	resultDesc	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 .

Table 4 Parameters of resultDesc/resultDatas

No.	Parameter	Value Type	Description
2.1	agentId	Integer	Agent ID.
2.2	skills	Array	Skill queue array of an agent. For details about the parameters of the object, see Table 5 .

Table 5 Parameters of skills

No.	Parameter	Value Type	Description
2.2.1	skillId	Integer	Skill queue ID.
2.2.2	isLogin	Integer	Whether an agent has signed in. The value 1 indicates that the agent has signed in, and 0 indicates that the agent does not sign in.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "ccId":1,
  "vdn":170,
  "agentIds":[1031,1099]
}
```

- Response parameters

```
{
  "result": "0",
  "resultCode": "0100000",
  "resultDatas": [
    {
      "skills": [
        {
          "skillId": 2,
          "isLogin": 0
        }
      ],
      "agentId": 1031
    },
    {
      "skills": [
        {
          "skillId": 2,
          "isLogin": 0
        },
        {
          "skillId": 4,
          "isLogin": 0
        }
      ],
      "agentId": 1099
    }
  ],
  "resultDesc": [
    {
      "skills": [
        {

```

```

        "skillId": 2,
        "isLogin": 0
    },
    ],
    "agentId": 1031
},
{
    "skills": [
        {
            "skillId": 2,
            "isLogin": 0
        },
        {
            "skillId": 4,
            "isLogin": 0
        }
    ],
    "agentId": 1099
}
]
}

```

4.1.3.2 Querying Agent Information in Batches

Scenario

This interface is invoked to query information about specified agents in batches.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Internal Encapsulation Interface

CTI3.6: BMS interface: /ccbms/ws/monitor/agentstatusinfoobyagentids

CTI3.8: BMS interface: /ccbms/ws/monitor/queryAgentStatusEx3Message

Request URL

https://ip:port/rest/cmsapp/v1/openapi/agentgroup/agentstatusinfoobyagentids

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the message body

No.	Parameter	Value Type	Mandatory	Description
1	ccId	Integer	Yes	ID of a call center. The value ranges from 1 to 65535.
2	vdn	Integer	Yes	ID of a VCC. The value ranges from 1 to 5000.
3	agentIds	Array	Yes	Array of agent IDs. The maximum length of the array is 6000. Value range: 101–59999 and 70000–999999

Response Description

Table 3 Parameters in the message body

No.	Parameter	Value Type	Description
1	result	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
2	resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .

Table 3 Parameters in the message body

No.	Parameter	Value Type	Description
3	resultDatas	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
2	resultDesc	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 .

Table 4 Parameters of resultDesc/resultDatas

No.	Parameter	Value Type	Description
2.1	agentId	Integer	Agent ID.
2.2	locationId	Integer	Home distributed access node.
2.3	currentStateReason	Integer	Reason for the current state. Currently, this parameter is valid for only the rest state.
2.4	prevState	Integer	Previous status. For details, see CTI Platform Agent Status Code Reference .
2.5	prevStateTime	Long	Duration of the previous state, in seconds.
2.6	currentState	Integer	Current status. For details, see CTI Platform Agent Status Code Reference .
2.7	currentStateTime	Long	Duration of the current state, in seconds.
2.8	logonTime	Long	Sign-in time (millisecond-level timestamp). NOTE: If an agent in the query result has not signed in, the value is a future time (in 2106).
2.9	agentRelease	Integer	Number of times that an agent proactively releases calls.
2.10	currentPrivateCallNum	Integer	Number of private calls of an agent.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "ccId":1,
  "vdn":170,
  "agentIds":[1031,1099]
}
```

- Response parameters

```
{
  "result": "0",
  "resultCode": "0100000",
  "resultDatas": [
    {
      "prevState": 1,
      "agentId": 1031,
      "logonTime": 4294967295000,
      "agentRelease": 0,
      "locationId": 65535,
      "currentPrivateCallNum": 0,
      "currentStateTime": 2224439,
      "currentStateReason": 65535,
      "prevStateTime": 4294967295,
      "currentState": 0
    },
    {
      "prevState": 1,
      "agentId": 1099,
      "logonTime": 4294967295000,
      "agentRelease": 0,
      "locationId": 65535,
      "currentPrivateCallNum": 0,
      "currentStateTime": 230703,
      "currentStateReason": 258,
      "prevStateTime": 4294967295,
      "currentState": 0
    }
  ],
  "resultDesc": [
    {
      "prevState": 1,
      "agentId": 1031,
      "logonTime": 4294967295000,
      "agentRelease": 0,
```

```
    "locationId": 65535,  
    "currentPrivateCallNum": 0,  
    "currentStateTime": 2224439,  
    "currentStateReason": 65535,  
    "prevStateTime": 4294967295,  
    "currentState": 0  
  },  
  {  
    "prevState": 1,  
    "agentId": 1099,  
    "logonTime": 4294967295000,  
    "agentRelease": 0,  
    "locationId": 65535,  
    "currentPrivateCallNum": 0,  
    "currentStateTime": 230703,  
    "currentStateReason": 258,  
    "prevStateTime": 4294967295,  
    "currentState": 0  
  }  
]  
}
```

4.1.3.3 Querying the Current Call ID of a Specified Agent

Scenario

This interface is invoked to query information about the current call ID of a specified agent.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Internal Encapsulation Interface

BMS interface: /ccbms/ws/monitor/queryagentCalls

Request URL

https://ip:port/rest/cmsapp/v1/openapi/agent/queryagentcalls

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the message body

No.	Parameter	Value Type	Mandatory	Description
1	cclid	Integer	Yes	ID of a call center. The value ranges from 1 to 65535.
2	vdn	Integer	Yes	ID of a VCC. The value ranges from 1 to 5000.
3	agentId	Integer	Yes	Agent ID. Value range: 101–59999 and 70000–999999

Response Description

Table 3 Parameters in the message body

No.	Parameter	Value Type	Description
1	result	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
2	resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .

Table 3 Parameters in the message body

No.	Parameter	Value Type	Description
3	resultDatas	Array	Array of IDs of calls that are being handled by an agent in the returned result after a successful query. NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
2	resultDesc	Array	Array of IDs of calls that are being handled by an agent in the returned result after a successful query.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "ccld":1,
  "vdn":170,
  "agentId":1099
}
```

- Response parameters

```
{
  "result": "0",
  "resultCode": "0100000",
  "resultDatas": [
    "1605492949-1447"
  ],
  "resultDesc": [
    "1605492949-1447"
  ]
}
```

4.1.3.4 Querying Call Information About All Agents in Batches

Scenario

This interface is invoked to query real-time call information about all agents in batches.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Internal Encapsulation Interface

BMS interface: /ccbms/ws/monitor/queryagentCalls

Request URL

https://ip:port/rest/cmsapp/v1/openapi/agentgroup/queryagentcalls

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the message body

No.	Parameter	Value Type	Mandatory	Description
1	cclId	Integer	Yes	ID of a call center. The value ranges from 1 to 65535.
2	vdn	Integer	Yes	ID of a VCC. The value ranges from 1 to 5000.
3	agentIds	Array	Yes	Array of agent IDs. The maximum length of the array is 3000. Value range: 101–59999 and 70000–999999

Response Description

Table 3 Parameters in the message body

No.	Parameter	Value Type	Description
1	result	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
2	resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
3	resultDatas	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
4	resultDesc	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 .

Table 4 Parameters of resultDesc/resultDatas

No.	Parameter	Value Type	Description
2.1	agentId	Integer	Agent ID.
2.2	callList	Array	Call information array. For details about the parameters of the object, see Table 5 .

Table 5 Parameters of callList

No.	Parameter	Value Type	Description
2.2.1	callId	String	Call ID.
2.2.2	callStatus	Integer	Call status. For details, see Table 6 .
2.2.3	callInOut	Integer	Inbound and outbound call flags. The value 0 indicates inbound call, and 1 indicates outbound call.
2.2.4	mediaType	Integer	Media type. For details, see Table 1 .

Table 5 Parameters of callList

No.	Parameter	Value Type	Description
2.2.5	submediaType	String	Submedia type. The options are as follows: -1: no submedia type. For details about other submedia types, see Description of Call Submedia Types .

Table 6 Call status

No.	Call Type	Description
0	IDLE	Indicates that the called party is idle (no call).
1	OCCUPY	Indicates that the called party is occupied.
2	QUEUE	Indicates that the call is queuing in a skill queue.
3	PRIVATE_QUEUE	Indicates that the call is queuing in a private skill queue.
4	ALERTING	Indicates that the phone of the called party rings and waits for answer.
5	TALK	Indicates that the calling and called parties are in conversation or interaction.
6	HOLD	Indicates that the call is held.
7	SUSPEND	Indicates that the call is suspended.
8	MUTE	Indicates that the call is muted.
9	TRITALK	Indicates that a three-party call is initiated.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "ccId":1,
  "vdn":170,
  "agentIds":[1031,1099]
}
```

- Response parameters

```
{
```

```
"result": "0",
"resultCode": "0100000",
"resultDatas": [
  {
    "agentId": 1034,
    "callList": []
  },
  {
    "agentId": 1099,
    "callList": [
      {
        "callId": "1605494261-1449",
        "callStatus": 5,
        "callInOut": 0,
        "submediaType": "Email",
        "mediaType": 5
      }
    ]
  }
],
"resultDesc": [
  {
    "agentId": 1034,
    "callList": []
  },
  {
    "agentId": 1099,
    "callList": [
      {
        "callId": "1605494261-1449",
        "callStatus": 5,
        "callInOut": 0,
        "submediaType": "Email",
        "mediaType": 5
      }
    ]
  }
]
```

4.1.3.5 Querying Private Calls of a Specified Agent

Scenario

This interface is invoked to query private calls of a specified agent. To access a private skill queue, perform the following steps.

If both of the following conditions are met, the system transfers a call to the private skill queue of an agent. When an agent is idle, the agent first answers the calls in the private queue.

- The call is transferred to the agent.

Generally, a call is assigned to an agent in the following cases:

- When a call is in a special situation (such as **Skill Timeout**, **Skill Busy**, **Skill NoAgents**, or **Canceling queuing**), the call is transferred to another agent.
- A call is transferred by an agent to another agent for a certain reason.
- The agent is in talking or working state and cannot handle calls.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Internal Encapsulation Interface

BMS interface: /ccbms/ws/monitor/agentprivatecall

Request URL

https://ip:port/rest/cmsapp/v1/openapi/agent/agentprivatecall

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the message body

No.	Parameter	Value Type	Mandatory	Description
1	cclid	Integer	Yes	ID of a call center. The value ranges from 1 to 65535.
2	vdn	Integer	Yes	ID of a VCC. The value ranges from 1 to 5000.
3	agentId	Integer	Yes	Agent ID. Value range: 101–59999 and 70000–999999

Response Description

Table 3 Parameters in the message body

No.	Parameter	Value Type	Description
1	result	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
2	resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
3	resultDatas	Array	Array of IDs of private calls made by an agent in the returned result after a successful query. NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.

Table 3 Parameters in the message body

No.	Parameter	Value Type	Description
4	resultDesc	Array	Array of IDs of private calls made by an agent in the returned result after a successful query.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "ccId":1,
  "vdn":170,
  "agentId":1099
}
```

- Response parameters

```
{
  "result": "0",
  "resultCode": "0100000",
  "resultDatas": [
    "1605492949-1447"
  ],
  "resultDesc": [
    "1605492949-1447"
  ]
}
```

4.1.3.6 Querying the Number of Agents in the Agent State in batches

Scenario

This interface is invoked to query the number of agents in the agent state based on the agent ID list.

NOTE

If the number of agents exceeds 10,000, the performance deteriorates.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Internal Encapsulation Interface

BMS interface: /ccbms/ws/monitor/teamagentstatusinfo

Request URL

https://ip:port/rest/cmsapp/v1/openapi/agentgroup/teamagentstatusinfo

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the message body

No.	Parameter	Value Type	Mandatory	Description
1	ccId	Integer	Yes	ID of a call center. The value ranges from 1 to 65535.
2	vdn	Integer	Yes	ID of a VCC. The value ranges from 1 to 5000.
3	agentIds	Array	Yes	Array of agent IDs. The maximum length of the array is 6000. Value range: 101–59999 and 70000–999999

Response Description

Table 3 Parameters in the message body

No.	Parameter	Value Type	Description
1	result	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
2	resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
3	resultDatas	Object	Object in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
2	resultDesc	Object	Object in the returned result after a successful query. For details about the parameters of the object, see Table 4 .

Table 4 Parameters of resultDesc/resultDatas

No.	Parameter	Value Type	Description
2.1	loginEd	Integer	Number of signed-in agents.
2.2	talking	Integer	Number of agents in talking state.
2.3	idle	Integer	Number of agents in idle state.
2.4	usable	Integer	Number of available agents.
2.5	setBusy	Integer	Number of agents in busy state.
2.6	workSubstateBegin	Integer	Number of agents in wrap-up state.
2.7	rest	Integer	Number of agents in rest state.
2.8	studying	Integer	Number of agents in study state.
2.9	occupy	Integer	Number of agents in occupied state.
2.10	preoccupy	Integer	Number of agents in preoccupied state.

Table 4 Parameters of resultDesc/resultDatas

No.	Parameter	Value Type	Description
2.11	answering	Integer	Number of agents in answering state.
2.12	lateAdjust	Integer	Number of agents in wrap-up state.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "ccId":1,
  "vdn":170,
  "agentIds":[1031,1099]
}
```

- Response parameters

```
{
  "result": "0",
  "resultCode": "0100000",
  "resultDatas": {
    "workSubStateBegin": 0,
    "lateAdjust": 0,
    "usable": 1,
    "rest": 0,
    "setBusy": 0,
    "studying": 0,
    "occupy": 0,
    "idle": 1,
    "preoccupy": 0,
    "answering": 0,
    "talking": 0,
    "loginEd": 1
  },
  "resultDesc": {
    "workSubStateBegin": 0,
    "lateAdjust": 0,
    "usable": 1,
    "rest": 0,
```

```

    "setBusy": 0,
    "studying": 0,
    "occupy": 0,
    "idle": 1,
    "preoccupy": 0,
    "answering": 0,
    "talking": 0,
    "loginEd": 1
  }
}

```

4.1.3.7 Querying Details of a Specified Agent

Scenario

This interface is invoked to query details of a specified agent.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Internal Encapsulation Interface

BMS interface: /ccbms/ws/monitor/siteagentinfoobyagentids

Request URL

https://ip:port/rest/cmsapp/v1/openapi/agent/agentbyworkno

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the message body

No.	Parameter	Value Type	Mandatory	Description
1	cclid	Integer	Yes	ID of a call center. The value ranges from 1 to 65535.
2	vdn	Integer	Yes	ID of a VCC. The value ranges from 1 to 5000.
3	agentId	Integer	Yes	Agent ID. Value range: 101–59999 and 70000–999999

Response Description

Table 3 Parameters in the message body

No.	Parameter	Value Type	Description
1	result	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
2	resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
3	resultDatas	Object	Object in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.

Table 3 Parameters in the message body

No.	Parameter	Value Type	Description
4	resultDesc	Object	Object in the returned result after a successful query. For details about the parameters of the object, see Table 4 .

Table 4 Parameters of resultDesc/resultDatas

No.	Parameter	Value Type	Description
2.1	workNo	Integer	Agent ID.
2.2	vdnId	Integer	ID of the VCC to which a call belongs.
2.3	citStatus	Integer	Platform status of an agent. For details, see CTI Platform Agent Status Code Reference .
2.4	name	String	Agent name.
2.5	phoneNumber	String	Agent phone number.
2.6	loginDate	Long	Sign-in time (millisecond-level timestamp).
2.7	currentStateTime	Long	Duration of the current state, in seconds.
2.8	mediaType	Integer	Agent media type. For details, see Table 1 .
2.9	status	Integer	Agent status. For details, see Table 5 .

Table 5 Agent status

Value	Description
0	Unknown. The value corresponds to status 9 on the CTI platform.
1	Sign-in. The value is not used.
2	Sign-out. The value corresponds to status 0 on the CTI platform.
3	Showing busy. The value corresponds to status 7 on the CTI platform.
4	Idle. The value corresponds to status 1 on the CTI platform.
5	Wrap-up. The value corresponds to status 6 on the CTI platform.
6	Same as the idle state. The value is not used.
7	Talking. The value corresponds to states 2, 3, 4, and 5 on the CTI platform.
8	Rest. The value corresponds to status 8 on the CTI platform.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "queryParam":
  {
    "ccId":1,
    "vdn":1,
    "agentId":101
  }
}
```

- Response parameters

```
{
  "result": "0",
  "resultCode": "0100000",
  "resultDatas": {
    "workNo": 101,
    "vdnId": 1,
    "citStatus": 0,
    "name": "HW35154101",
    "phoneNumber": null,
    "loginDate": 4294967295000,
    "currentStateTime": 436879,
    "mediaType": 0,
    "status": 2
  },
  "resultDesc": {
    "workNo": 101,
    "vdnId": 1,
    "citStatus": 0,
    "name": "HW35154101",
    "phoneNumber": null,
    "loginDate": 4294967295000,
    "currentStateTime": 436879,
    "mediaType": 0,
    "status": 2
  }
}
```

4.1.3.8 Querying Agent Rest Information

Scenario

This interface is invoked to query all agent rest information.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Internal Encapsulation Interface

- CTI3.6 BMS interface: /ccbms/ws/monitor/teamrestinfo
- CTI3.6 BMS interface: /ccbms/ws/monitor/teamrestagentinfo
- CTI3.8 BMS interface: /ccbms/ws/monitor/queryAgentRestDetailInfo

Request URL

https://ip:port/rest/cmsapp/v1/openapi/agentgroup/agentrestinfos

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the message body

No.	Parameter	Value Type	Mandatory	Description
1	cclid	Integer	Yes	ID of a call center. The value ranges from 1 to 65535.
2	vdn	Integer	Yes	ID of a VDN.

Table 2 Parameters in the message body

No.	Parameter	Value Type	Mandatory	Description
				The value ranges from 1 to 5000.
3	offset	Integer	No	Parameter for determining the start page of the query. Value of offset = Value of limit x (Value of page – 1). The value of offset is greater than or equal to 0. page indicates the start page of the query. This field is valid only in CTI3.6.
4	limit	Integer	No	Number of query records on each page. The value ranges from 1 to 100. limit and offset must exist or not exist at the same time. This field is valid only in CTI3.6.

Response Description

Table 3 Parameters in the message body

No.	Parameter	Value Type	Description
1	result	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
2	resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
3	resultDatas	Object	Object in the returned result after a successful query. For details about the parameters of the object, see Table 4 . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
4	resultDesc	Object	Object in the returned result after a successful query. For details about the parameters of this object, see Table 4 .

Table 4 Parameters of resultDesc/resultDatas

No.	Parameter	Value Type	Description
2.1	allowRestNum	Integer	Number of agents who are allowed to have a rest but still at work.
2.2	onRestNum	Integer	Number of agents that are resting.
2.3	waitRestNum	Integer	Number of agents who apply for a rest.
2.4	agentsRestDetails	Array	List of details about agents who take a rest in the VDN. For details about the parameters of the object in this array, see Table 5 .

Table 5 Parameters of agentsrestdetails

No.	Parameter	Value Type	Description
2.4.1	agentName	String	Agent name.
2.4.2	workNo	Integer	Agent ID.
2.4.3	cantRestReason	Integer	Reason why agents are not allowed to have a rest. This parameter is valid only for the agents who wait for a rest. For details about the parameters of this object, see Table 6 .
2.4.4	beginTime	Long	Start time. For resting agents, this parameter indicates the rest start time. For agents who are not in rest, this parameter indicates the requested or approved rest start time. The value is in the timestamp format.
2.4.5	restDuration	Long	Requested rest duration, in seconds.

Table 6 Reason for not allowing rest

No.	Value of the Reason for Not Allowing Rest	Description
2.4.3.1	0	The agent is waiting for approval.
2.4.3.2	1	The agent is in busy state.
2.4.3.3	2	The number of agents who can take a rest in the VDN reaches the maximum.
2.4.3.4	3	The agent does not answer any inbound calls, because the agent does not sign in to any skill queues.
2.4.3.5	4	The number of resting agents in a skill queue the agent signs in reaches the maximum.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
```

Authorization: *****

- Request parameters

```
{
  "queryParam":
  {
    "cclId":1,
    "vdn":1,
    "offset":0,
    "limit":100
  }
}
```

- Response parameters

```
{
  "result": "0",
  "resultCode": "0100000",
  "resultDatas": {
    "allowRestNum": 101,
    "onRestNum": 1,
    "waitRestNum": 0,
    "agentsRestDetails": [
      {
        "agentName": "z1",
        "workNo": 55001,
        "cantRestReason": 1,
        "beginTime": 2345,
        "restDuration": 22333
      }
    ]
  },
  "resultDesc": {
    "allowRestNum": 101,
    "onRestNum": 1,
    "waitRestNum": 0,
    "agentsRestDetails": [
      {
        "agentName": "z1",
        "workNo": 55001,
        "cantRestReason": 1,
        "beginTime": 2345,
        "restDuration": 22333
      }
    ]
  }
}
```

```

    ]
  }
}
```

4.1.3.9 Checking Whether an Agent Is Processing a Call

Scenario

This interface is invoked to check whether a specified agent is processing a call.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Internal Encapsulation Interface

BMS interface: /ccbms/ws/monitor/siteagentinfoobyagentids

Request URL

https://ip:port/rest/cmsapp/v1/openapi/calldata/isprocesscall

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the message body

No.	Parameter	Value Type	Mandatory	Description
1	cclid	Integer	Yes	ID of a call center. The value ranges from 1 to 65535.

Table 2 Parameters in the message body

No.	Parameter	Value Type	Mandatory	Description
2	vdn	Integer	Yes	ID of a VCC. The value ranges from 1 to 5000.
3	agentId	Integer	Yes	Agent ID. Value range: 101–59999 and 70000–999999

Response Description

Table 3 Parameters in the message body

No.	Parameter	Value Type	Description
1	result	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
2	resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
3	resultDatas	Boolean	If the query is successful, true (yes) or false (no) is returned. NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
4	resultDesc	Boolean	If the query is successful, true (yes) or false (no) is returned.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
```

```
"queryParam":  
{  
  "ccld":1,  
  "vdn":1,  
  "agentId":101  
}  
}
```

- Response parameters

```
{  
  "result": "0",  
  "resultCode": "0100000",  
  "resultDatas": false,  
  "resultDesc": false  
}
```

4.1.3.10 Querying System Status by Agent

Scenario

This interface is invoked to query system status by agent.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Internal Encapsulation Interface

BMS interface: /ccbms/ws/monitor/teamagentstatusinfo

Request URL

https://ip:port/rest/cmsapp/v1/openapi/agentgroup/systemstate

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the message body

No.	Parameter	Value Type	Mandatory	Description
1	cclId	Integer	Yes	ID of a call center. The value ranges from 1 to 65535.
2	vdn	Integer	Yes	ID of a VDN. The value ranges from 1 to 5000.
3	offset	Integer	Yes	Parameter for determining the start page of the query. Value of offset = Value of limit x (Value of page – 1). The value of offset is greater than or equal to 0. page indicates the start page of the query.
4	limit	Integer	Yes	Number of query records on each page. The value ranges from 1 to 100.

Response Description

Table 3 Parameters in the message body

No.	Parameter	Value Type	Description
1	result	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
2	resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .

Table 3 Parameters in the message body

No.	Parameter	Value Type	Description
3	resultDatas	Object	Object in the returned result after a successful query. For details about the parameters of this object, see Table 4 . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
4	resultDesc	Object	Object in the returned result after a successful query. For details about the parameters of this object, see Table 4 .

Table 4 Parameters of resultDesc/resultDatas

No.	Parameter	Value Type	Description
2.1	busyAgent	Integer	Number of busy agents.
2.2	talkingNum	Integer	Number of agents in talking state.
2.3	idleAgent	Integer	Number of idle agents.
2.4	totalLoginAgent	Integer	Number of signed-in agents.
NOTE: The response result is the agent status data obtained based on the pagination parameters limit and offset in the request. For example, if both limit and offset are set to 10 , the 10 agents on the second page (10 agents per page) are obtained, and then status statistics on the 10 agents are collected.			

Message Example

- Message body

The following provides an example of the request message body of this interface:

```
{
  "queryParam":
  {
    "ccId":1,
    "vdn":1,
    "offset":0,
    "limit":100
  }
}
```

- The following provides an example of the response message body of this interface:


```
{
  "result": "0",
  "resultCode": "0100000",
  "resultDatas": {
    "busyAgent": 0,
    "talkingNum": 0,
    "idleAgent": 0,
    "totalLoginAgent": 0
  },
  "resultDesc": {
    "busyAgent": 0,
    "talkingNum": 0,
    "idleAgent": 0,
    "totalLoginAgent": 0
  }
}
```

4.1.3.11 Obtaining Real-Time Monitoring Indicators by Agent

Scenario

This interface is used to obtain real-time monitoring indicators by agent.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Internal Encapsulation Interface

BMS interface: /ccbms/ws/monitor/agentstatusinfobyagentids

Request URL

<https://ip:port/rest/cmsapp/v1/openapi/agentgroup/rindex>

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the message body

Parameter	Value Type	Mandatory	Description
cclid	Integer	Yes	ID of a call center. The value ranges from 1 to 65535.
vdn	Integer	Yes	ID of a VCC. The value ranges from 1 to 5000.
agentIds	Array	Yes	Array of agent IDs. The maximum length of the array is 100. Value range: 101–59999 and 70000–999999
indexIds	Array	Yes	Array of indicators. The maximum length of the array depends on the number of real-time monitoring indicators by agent. The maximum value is 4 . For details about the parameters of the object, see Table 2 .

Response Description

Table 3 Parameters in the message body

Parameter	Value Type	Description
result	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .

Table 3 Parameters in the message body

Parameter	Value Type	Description
resultDatas	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
resultDesc	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 .

Table 4 Parameters of resultDesc/resultDatas

Parameter	Value Type	Description
id	Integer	Agent ID.
idxs	Array	Group of indicators. For details about the parameters of the object, see Table 5 .

Table 5 Parameter description of idxs

Parameter	Value Type	Description
id	String	Indicator ID.
val	Object	Indicator result. For details about the parameters of the object, see Table 2 .

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "ccId":1,
  "vdn":170,
  "agentIds":[1032,1099],

  "indexIds":["IDX_AGENT_PHONE","IDX_AGENT_CURRENT_STATE","IDX_DURATION_AGENT_CURRENT_STATE","IDX_A
GENT_LOGIN_SKILLS"]
}
```

- Response parameters

```
{
  "result": "0",
  "resultCode": "0100000",
  "resultDatas": [
    {
      "idxs": [
        {
          "val": "88880103",
          "id": "IDX_AGENT_PHONE"
        },
        {
          "val": 7,
          "id": "IDX_AGENT_CURRENT_STATE"
        },
        {
          "val": 17728,
          "id": "IDX_DURATION_AGENT_CURRENT_STATE"
        },
        {
          "val": [
            3
          ],
          "id": "IDX_AGENT_LOGIN_SKILLS"
        }
      ],
      "id": 1032
    },
    {
      "idxs": [
        {
          "val": "88881609",
          "id": "IDX_AGENT_PHONE"
        },
        {
          "val": 1,
          "id": "IDX_AGENT_CURRENT_STATE"
        },
        {
          "val": 1414,
          "id": "IDX_DURATION_AGENT_CURRENT_STATE"
        }
      ],
      "id": 1032
    }
  ]
}
```

```

        {
            "val": [
                2,
                4
            ],
            "id": "IDX_AGENT_LOGIN_SKILLS"
        }
    ],
    "id": 1099
}
],
"resultDesc": [
    {
        "idxs": [
            {
                "val": "88880103",
                "id": "IDX_AGENT_PHONE"
            },
            {
                "val": 7,
                "id": "IDX_AGENT_CURRENT_STATE"
            },
            {
                "val": 17728,
                "id": "IDX_DURATION_AGENT_CURRENT_STATE"
            },
            {
                "val": [
                    3
                ],
                "id": "IDX_AGENT_LOGIN_SKILLS"
            }
        ],
        "id": 1032
    },
    {
        "idxs": [
            {
                "val": "88881609",
                "id": "IDX_AGENT_PHONE"
            },
            {
                "val": 1,

```

```

        "id": "IDX_AGENT_CURRENT_STATE"
    },
    {
        "val": 1414,
        "id": "IDX_DURATION_AGENT_CURRENT_STATE"
    },
    {
        "val": [
            2,
            4
        ],
        "id": "IDX_AGENT_LOGIN_SKILLS"
    }
],
"id": 1099
}
]
}

```

4.1.3.12 Obtaining Agent Operation Details

Scenario

This interface is invoked to query agent operation details. The retention period of the queried data is one day.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Request URL

`https://ip:port/rest/cmsapp/v1/openapi/agent/agentoprinfo`

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the message body

Parameter	Value Type	Mandatory	Description
currentAgentId	Integer	Yes	Agent ID. The value ranges from 0 to 999999.
operateType	Integer	No	Operation type. For details, see Table 1 .
beginTime	String	Yes	Statistics start time. The value must be in <i>yyyy-MM-dd HH:mm:ss</i> format. The time in the UTC time zone is used. The time range must be an integer multiple of 15 minutes, and the time span cannot be greater than 24 hours.
endTime	String	Yes	Statistics end time. The value must be in <i>yyyy-MM-dd HH:mm:ss</i> format. The time in the UTC time zone is used. The time range must be an integer multiple of 15 minutes, and the time span cannot be greater than 24 hours.
cclid	Integer	No	ID of a call center. In the integration environment, the default value is the value of Call Center ID on the Configuration Center > System Management > CMS System Configuration page after the system administrator signs in to the AICC. If this parameter is empty, the data whose sys_key is default_cc_id and sys_type is cms_system_config in the t_cms_sysparam database table is used. In the independent deployment environment, the default value is the value of Default Call Center ID on the Configuration Center > System Management > Parameter Settings page after the system administrator signs in to the AICC.

NOTE

The operation objects and operation reasons are marked "-", indicating that the agent operation table does not need to be recorded for the corresponding operation types.

Response Description

Table 3 Parameters in the message body

Parameter	Value Type	Description
result	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
resultDatas	List	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
resultDesc	List	Object list in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 .

Table 4 Parameters of resultDesc/resultDatas

Parameter	Value Type	Description
id	String	Configuration ID.
callId	String	Call ID.
phoneNo	String	Agent phone number or conference ID.
subCcNo	Integer	Subsystem ID.
currentAgentId	Integer	ID of an agent.
currentSkillId	Integer	ID of the skill queue to which a call belongs.
mediaType	Integer	Media type. For details, see Table 1 .
submediaType	String	Submedia type. The options are as follows: -1 : no submedia type. For details about other submedia types, see Description of Call Submedia Types .

Table 4 Parameters of resultDesc/resultDatas

Parameter	Value Type	Description
beginTime	String	Start time of the statistics collection operation. The UTC time zone is used.
endTime	String	End time of the statistics collection operation. The UTC time zone is used.
vdn	Integer	ID of the VCC that processes a call.
callType	Integer	Call type. For details, see Table 1 .
serviceNo	Integer	Business type ID. NOTE: This parameter has no actual meaning. Do not use it.
operateType	Integer	Operation type. For details, see Table 1 .
operateObject	Integer	Operation object. For details, see Table 1 .
operateCause	Integer	Operation reason. For details, see Table 1 .
agentType	Integer	Agent type. For details, see Table 1 .
locationId	Integer	Distributed node ID.
logonSn	String	Sign-in SN.
skillInfoSn	String	Index of the skills owned by an agent. If this field is empty, the agent does not have the task skill. For example, when an agent signs in, this field is empty.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "ccId": 1,
  "operateType": 2,
  "currentAgentId": 35001,
  "beginTime": "2024-08-02 00:00:00",
  "endTime": "2024-08-02 23:00:00"
}
```

- Response parameters

```
{
  "result": "0",
  "resultCode": "0100000",
  "resultDatas": [
    {
      "callId": "5",
      "operateCause": 0,
      "vdn": 1,
      "agentType": 11,
      "operateType": 2,
      "currentSkillId": 1,
      "mediaType": 10,
      "submediaType": "Email",
      "operateObject": 23,
      "callType": 0,
      "phoneNo": "'05276789'",
      "logonSn": "2024080214194100005209009",
      "skillInfoSn": "2024080214194300005209050",
      "locationId": 1,
      "currentAgentId": 35001,
      "subCcNo": 1,
      "serviceNo": 1,
      "beginTime": "2024-08-02 06:19:44",
      "endTime": "2024-08-02 06:20:08",
      "id": "172257960827964297019273683535"
    }
  ],
  "resultDesc": [
    {
      "callId": "5",
      "operateCause": 0,
      "vdn": 1,
      "agentType": 11,
      "operateType": 2,
      "currentSkillId": 1,
      "mediaType": 10,
      "submediaType": "Email",
      "operateObject": 23,
      "callType": 0,
      "phoneNo": "'05276789'",
      "logonSn": "2024080214194100005209009",
      "skillInfoSn": "2024080214194300005209050",
```

```

    "locationId": 1,
    "currentAgentId": 35001,
    "subCcNo": 1,
    "serviceNo": 1,
    "beginTime": "2024-08-02 06:19:44",
    "endTime": "2024-08-02 06:20:08",
    "id": "172257960827964297019273683535"
  }
]
}
```

4.1.3.13 Querying Agent Information by Agent Status in Pagination Mode

Scenario

This interface is invoked to query agent information in pagination mode by agent status.

NOTE

If the current environment supports super agent IDs but no agent has signed in, no value is returned when this interface is invoked to query agent information.

Interface Method

This interface supports only the POST method.

Internal Encapsulation Interface

BMS interface: /ccbms/ws/monitor/siteagentinfoobyagentids

URI

<https://ip:port/rest/cmsapp/v1/openapi/vdn/queryAgentByStatus>

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

Parameter	Mandatory	Value Type	Default Value	Description
Content-Type	Yes	String	None	The value is fixed at application/json; charset=UTF-8 .
Authorization	Yes	String	None	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the request body

Parameter	Mandatory	Value Type	Default Value	Description
cclid	Yes	String (1–25)	None	Call center ID. The value ranges from 1 to 65535.
vdn	Yes	String	None	ID of a virtual call center. The value ranges from 1 to 5000.
status	Optional	String	None	Agent status. For details about the value, see CTI Platform Agent Status Code Reference .
offset	Yes	Number	None	offset is used to determine the start page of the query. The value range of offset is as follows: The value of offset is equal to limit x (page – 1). The value is greater than or equal to 0. page indicates the start page to be queried.
limit	Yes	Number	None	Number of records on each page. The maximum value is 1000.

Response Description

Table 3 Parameters in the response message

Parameter	Value Type	Description
returnCode	String	Result code returned. 0: success Other values: failure
description	String	Request result description.
data	Array	Response data.

Table 4 Parameters in the response message body

Parameter	Value Type	Description
agentId	Number	Agent ID.
locationId	Number	Home distributed access node. This parameter is a CTI reserved field and has no actual meaning.
currentState	Number	Agent status. CTI Platform Agent Status Code Reference describes the parameters.
currentStateTime	Number	Duration of the current state, in seconds.

Error Codes

None

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "ccId": "1","vdn": "3","status": "0","limit": 10,"offset": 0
}
```

- Response parameters

```
{
  "returnCode": "0",
  "data": [
    {
      "agentId": 108,
      "locationId": 65535,
      "currentStateTime": 426849,
      "currentState": 0
    },
    {
      "agentId": 109,
      "locationId": 65535,
      "currentStateTime": 426849,
      "currentState": 0
    }
  ]
}
```

```
],
  "description": "success"
}
```

4.1.4 Real-Time Interfaces for Call Details

4.1.4.1 Querying Basic Information About a Call with a Specified Call ID

Scenario

This interface is invoked to query basic information about a call with a specified call ID. (CTI 3.6 does not support this interface.)
The call with the call ID must be in progress.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Internal Encapsulation Interface

BMS interface: /ccbms/ws/monitor/querybasecallinfo

Request URL

https://ip:port/rest/cmsapp/v1/openapi/calldata/querybasiccallinfo
In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.
In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the message body

Parameter	Value Type	Mandatory	Description
cclId	Integer	Yes	ID of a call center. The value ranges from 1 to 65535 .
vdn	Integer	Yes	ID of a VCC. The value ranges from 1 to 5000.
callId	String	Yes	Call ID. The value contains a maximum of 20 characters. The value is in the format of <i>Call start time-Call ID</i> . <i>Call start time</i> is a timestamp in milliseconds. An example is 1659513159000-546750 .

Response Description

Table 3 Parameters in the message body

Parameter	Value Type	Description
result	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
resultDatas	Object	Object in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
resultDesc	Object	Object in the returned result after a successful query. For details about the parameters of the object, see Table 4 .

Table 4 Parameters of resultDesc/resultDatas

Parameter	Value Type	Description
time	Date	Time when a call enters the platform. The unit is millisecond.
dsn	Integer	Call task ID.
handle	Integer	Number of times that a call task control block is reused.
server	Integer	ID of a media server that processes calls.
keyId	String	Unique ID of a call.
callingNum	String	Calling number.
calledNum	String	Called number.
mid	Integer	UAP module ID.
trunkNo	Integer	Trunk ID.
priority	Integer	User level.
deviceType	Integer	Type of the device that is being processed. For details, see Table 1 .
deviceNo	Integer	ID of the device that is processing a call. For an agent device, the device ID is the agent ID. For a queue or virtual device, the device ID is the queue ID configured on the configuration console or the ID of the network shared device corresponding to the virtual device. For other device types, the value is an internal ID.
callSkill	Integer	Skill queue description.
partyNum	Integer	Number of devices involved in a call track.
callFeature	Integer	Call feature. For details, see Table 1 .
orgiCalledNum	Integer	Original called number. If the called number changes before the call accesses the system, this parameter records the number of the first called party.
mediaType	Integer	Media type. For details, see Table 1 .
submediaType	int	Submedia type. The options are as follows: -1: no submedia type. For details about other submedia types, see Description of Call Submedia Types .
place	Integer	User location (call access place).
waitTime	Long	Waiting time in a queue.

Table 4 Parameters of resultDesc/resultDatas

Parameter	Value Type	Description
totalTalkingTimes	Long	Call duration.
logonTimes	Long	Total sign-in duration.
serviceType	Integer	Business type.
callInCause	Integer	Reason for an inbound call.
enterReason	Integer	Reason for accessing the device. For details, see Table 1 .
mediaBillType	String	Multimedia report type.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "ccId":1,
  "vdn":1,
  "callId":"1453723531-574"
}
```

- Response parameters

```
{
  "result": "0",
  "resultCode": "0100000",
  "resultDatas": {
    "time": 1606115215000,
    "dsn": 0,
    "handle": 0,
    "server": 0,
    "keyId": "",
    "callingNum": "88881364",
    "calledNum": "650550",
    "mid": 5,
    "trunkNo": 582,
    "priority": 0,
    "deviceType": 2,
```

```
"deviceNo": 1177,
"callSkill": "defaultAudioSkill",
"partyNum": 1,
"callFeature": 0,
"orgiCalledNum": "",
"mediaType": 5,
"place": 0,
"waitTime": 0,
"totalTalkingTimes": 275,
"logonTimes": 284,
"serviceType": 2,
"callInCause": 65535,
"enterReason": 0,
"submediaType": 0,
"mediaBillType": 0
},
"resultDesc": {
  "time": 1606115215000,
  "dsn": 0,
  "handle": 0,
  "server": 0,
  "keyId": "",
  "callingNum": "88881364",
  "calledNum": "650550",
  "mid": 5,
  "trunkNo": 582,
  "priority": 0,
  "deviceType": 2,
  "deviceNo": 1177,
  "callSkill": "defaultAudioSkill",
  "partyNum": 1,
  "callFeature": 0,
  "orgiCalledNum": "",
  "mediaType": 5,
  "place": 0,
  "waitTime": 0,
  "totalTalkingTimes": 275,
  "logonTimes": 284,
  "serviceType": 2,
  "callInCause": 65535,
  "enterReason": 0,
  "submediaType": 0,
  "mediaBillType": 0
}
```

```
}  
}
```

4.1.4.2 Querying Private Call Details of a Specified Agent

Scenario

The call with the call ID must be in progress.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Internal Encapsulation Interface

BMS interface: /ccbms/ws/monitor/callinfo

Request URL

https://ip:port/rest/cmsapp/v1/openapi/calldata/querydetailcallinfo

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the message body

Parameter	Value Type	Mandatory	Description
ccld	Integer	Yes	ID of a call center. The value ranges from 1 to 65535.
vdn	Integer	Yes	ID of a VCC. The value ranges from 1 to 5000.

Table 2 Parameters in the message body

Parameter	Value Type	Mandatory	Description
callId	String	Yes	Call ID, which is the private call ID of a specified agent. The value contains a maximum of 20 characters. For details about how to obtain the value, see Querying Private Calls of a Specified Agent .

Response Description

Table 3 Parameters in the message body

Parameter	Value Type	Description
result	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
resultDatas	Object	Object in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
resultDesc	Object	Object in the returned result after a successful query. For details about the parameters of the object, see Table 4 .

Table 4 Parameters of resultDesc/resultDatas

Parameter	Value Type	Description
cb	Long	Internal ID.
time	Long	Time when a call enters the platform.
dsn	Integer	Call task ID.

Table 4 Parameters of resultDesc/resultDatas

Parameter	Value Type	Description
handle	Integer	Number of times that a call task control block is reused.
server	Integer	ID of a media server that processes calls.
keyId	String	Business ID of a call.
callingNum	String	Calling number.
calledNum	String	Called number.
mid	Integer	Module ID.
trunkNo	Integer	Trunk ID.
priority	Integer	User level.
deviceType	Integer	Device type. For details, see Table 1 .
deviceNo	Integer	Device ID. For an agent device, the device ID is the agent ID. For a queue or virtual device, the device ID is the queue ID configured on the configuration console or the ID of the network shared device corresponding to the virtual device. For other device types, the value is an internal ID.
callData	String	Call data.
callDataLen	Integer	Valid length of call data.
callSkill	String	Call skill description.
partyNum	Integer	Number of devices involved in a call track.
callFeature	String	Call feature. For details, see Table 1 .
orgiCalledNum	String	Original called number. If the called number changes before the call accesses the system, this parameter records the number of the first called party.
mediaType	String	Call media type. For details, see Table 1 .
submediaType	int	Submedia type. The options are as follows: -1: no submedia type. For details about other submedia types, see Description of Call Submedia Types .
place	Integer	User location (call access place).
waitTime	Long	Queuing time, in seconds.

Table 4 Parameters of resultDesc/resultDatas

Parameter	Value Type	Description
totalTalkingTimes	Long	Call duration, in seconds.
logonTimes	Long	Total sign-in duration, in seconds.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "ccId":1,
  "vdn":1,
  "callId":"1453723531-574"
}
```

- Response parameters

```
{
  "result": "0",
  "resultDatas": {
    "trunkNo": 582,
    "deviceType": 2,
    "deviceNo": 1177,
    "dsn": 49,
    "callData": "",
    "totalTalkingTimes": 116,
    "cb": 1,
    "keyId": "",
    "callingNum": "88881363",
    "calledNum": "650550",
    "orgiCalledNum": "",
    "callDataLen": 0,
    "callSkill": "defaultAudioSkill",
    "partyNum": 2,
    "callFeature": 0,
    "place": 0,
    "waitTime": 0,
    "logonTimes": 127,
```

```
"handle": 0,
"time": 1606120429,
"mediaType": "MEDIA_TYPE_PHONE",
"submediaType": 0,
"server": 0,
"mid": 5,
"priority": 0
},
"resultDesc": {
  "trunkNo": 582,
  "deviceType": 2,
  "deviceNo": 1177,
  "dsn": 49,
  "callData": "",
  "totalTalkingTimes": 116,
  "cb": 1,
  "keyId": "",
  "callingNum": "88881363",
  "calledNum": "650550",
  "orgiCalledNum": "",
  "callDataLen": 0,
  "callSkill": "defaultAudioSkill",
  "partyNum": 2,
  "callFeature": 0,
  "place": 0,
  "waitTime": 0,
  "logonTimes": 127,
  "handle": 0,
  "time": 1606120429,
  "mediaType": "MEDIA_TYPE_PHONE",
  "submediaType": 0,
  "server": 0,
  "mid": 5,
  "priority": 0
}
}
```

4.1.4.3 Querying CDRs of a Specified Call ID

Scenario

This interface is invoked to query CDRs by call ID. The retention period of CDRs is one day. The time zone is that of the tenant to which the CDRs belong.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Request URL

`https://ip:port/rest/cmsapp/v1/openapi/calldata/querycalldetailbyid`

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the message body

No.	Parameter	Type	Mandatory	Description
1	cclId	Integer	Yes	ID of a call center. The value ranges from 1 to 65535.
2	isCallin	String	Yes	Whether a call is an inbound call. The options are as follows: true : inbound call false : outbound call
3	callId	String	Yes	Call ID.

Response Description

Table 3 Parameters in the message body

No.	Parameter	Type	Description
1	result	String	Query result. The options are as follows:

Table 3 Parameters in the message body

No.	Parameter	Type	Description
			0: success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
2	resultCode	String	Query result. The options are as follows: 0100000: success Other values: failure For details about the failure causes, see Error Code Reference .
3	resultDatas	Object	Object in the returned result after a successful query. For details about the parameters of this object, see Table 4 . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
2	resultDesc	Object	Object in the returned result after a successful query. For details about the parameters of this object, see Table 4 .

Table 4 Parameters of resultDesc/resultDatas

No.	Parameter	Type	Description
2.1	vdn	Integer	ID of a VDN.
2.2	callId	String	Call ID.
2.3	callerNo	String	Calling number.
2.4	calleeNo	String	Called number.
2.5	callType	Integer	Call type. For details, see Description of Call Types .
2.6	agentId	Integer	Agent ID.
2.7	skillId	Integer	Skill queue ID.
2.8	anonyCall	Integer	Anonymous call identifier. (This parameter is valid only for inbound calls.)
2.9	queueWaitTime	Integer	Queue waiting duration, in seconds.
2.10	ivrCallTime	Integer	IVR call duration, in seconds.
2.11	ackTime	Integer	Ring duration, in seconds.
2.12	callTime	Integer	Call duration, in seconds.

Table 4 Parameters of resultDesc/resultDatas

No.	Parameter	Type	Description
2.13	releaseCause	Integer	Call release cause code. For details, see Cause Code .
2.14	beginTime	String	Call start time. (This parameter is valid only for predicted and preempted outbound calls.) The format is <i>yyyy-MM-dd HH:mm:ss</i> .
2.15	endTime	String	Call end time. (This parameter is valid only for predicted and preempted outbound calls.) The format is <i>yyyy-MM-dd HH:mm:ss</i> .
2.16	result	String	Call result. (This parameter is valid only for predicted and preempted outbound calls.) 0000 : The OBS successfully makes an outbound call before the call is transferred to the automatic flow. 0900 : The OBS successfully makes an outbound call before the call is transferred to an agent. 0101 : The outbound call fails. -1 : No number can be called in the current task. 0601 : The call is returned.
2.17	resultCode	String	Call result code. (This parameter is valid only for predicted and preempted outbound calls.) This parameter indicates an external result code corresponding to the failure cause code. The external result code can be customized. The failure cause code is associated with the external result code.
2.18	busiResult	String	Outbound call business result code. (This parameter is valid only for predicted and preempted outbound calls.) For details about how to configure the iSales frontend page, see the operation guide.
2.19	busiSubStatus	String	Outbound call business result sub-status code. (This parameter is valid only for predicted and preempted outbound calls.) For details about how to configure the iSales frontend page, see the operation guide.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "queryParam":
  {
    "ccld":1,
```

```
"isCallin": "true",  
"callId": "1000000000-0001"  
}  
}
```

- Response parameters

```
{  
  "result": "0",  
  "resultCode": "0100000",  
  "resultDatas": {  
    "vdn": 1,  
    "callId": "1000000000-0001",  
    "callerNo": "10000",  
    "calleeNo": "10001",  
    "callType": 0,  
    "agentId ": 100,  
    "skillId": 1,  
    "anonyCall": 0,  
    "queueWaitTime": 10,  
    "ivrCallTime": 0,  
    "ackTime": 5,  
    "callTime": 30,  
    "releaseCause": 0  
  },  
  "resultDesc": {  
    "vdn": 1,  
    "callId": "1000000000-0001",  
    "callerNo": "10000",  
    "calleeNo": "10001",  
    "callType": 0,  
    "agentId ": 100,  
    "skillId": 1,  
    "anonyCall": 0,  
    "queueWaitTime": 10,  
    "ivrCallTime": 0,  
    "ackTime": 5,  
    "callTime": 30,  
    "releaseCause": 0  
  }  
}
```

4.2 Historical Data Query

4.2.1 Obtaining Historical VCC Monitoring Indicators

Scenario

Obtain historical monitoring indicators by VCC. By default, data within 5 minutes is obtained from the settlement table, and the real-time cache data is not queried. To query the real-time cache data, change the value of **history_monitor_indicator/history_monitor_indicator_agent** field in **t_cms_sysparam** in the database to **true**. Since real-time cache data settlement by the interface has performance bottleneck, contact the maintenance personnel to obtain the real-time cache data volume of an interface before enabling this function.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Request URL

https://ip:port/rest/cmsapp/v1/openapi/hisindex/vdn

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the request body

No.	Parameter	Value Type	Mandatory	Description
1	cclid	Integer	No	ID of a call center. In the integration environment, the default value is the value of Call Center ID on the Configuration Center > System Management

Table 2 Parameters in the request body

No.	Parameter	Value Type	Mandatory	Description
				> CMS System Configuration page after the system administrator signs in to the AICC. If this parameter is empty, the data whose sys_key is default_cc_id and sys_type is cms_system_config in the t_cms_sysparam database table is used. In the independent deployment environment, the default value is the value of Default Call Center ID on the Configuration Center > System Management > Parameter Settings page after the system administrator signs in to the AICC.
2	vdnIds	Array	Yes	Array of VCC IDs. The maximum length of the array is 100. The VCC ID is an integer.
3	indexIds	Array	Yes	Array of indicators. The indicator is of the string type. For details, see Table 1 .
4	timeGrain	String	Yes/No	This parameter is mandatory when Table 1 contains the timeGrain parameter. This parameter can be used to collect statistics on historical monitoring indicators by minute. The options are 15MIN, 30MIN, and 60MIN. If this parameter is not passed, statistics on the current day in the time zone where the tenant is located are collected.
5	anonyCall	Integer	No	Anonymous call identifier. This parameter is available only for inbound calls. The value 1 indicates anonymous calls, and the value 0 indicates non-anonymous calls. If this parameter is not passed, all anonymous calls and non-anonymous calls are queried.
6	calleeNo	String	No	Called number.
7	mediaType	Integer	No	Call media type. For details, see Table 1 .
8	subMediaType	Integer	No	Submedia type. For details, see Table 3 .
9	enableOffset	String	No	The options are Y and N . If Y is transferred, the data of the previous day is queried. If N is transferred, the data of the current day is queried. If no value is transferred, the default value N is used.

Table 3 Submedia types

No.	Description
1	Web.
2	WhatsApp.
3	LINE.
4	WeChat.

Table 3 Submedia types

No.	Description
5	Facebook Messenger.
6	Twitter.
8	Email.
9	SMS message.

Response Description

Table 4 Parameters in the response body

No.	Parameter	Value Type	Description
1	result	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
2	resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
3	resultDatas	List <JSON object>	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 5 . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
4	resultDesc	List <JSON object>	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 5 .

Table 5 Parameters of resultDesc/resultDatas

No.	Parameter	Value Type	Description
2.1	id	String	ID of a VCC.
2.2	idxs	List <JSON object>	Group of indicators. For details about the parameters of the object, see Table 6 .

Table 6 Parameters of idx

No.	Parameter	Value Type	Description
2.1.1	id	String	Indicator ID.
2.1.2	val	Object	Indicator result collected by minute. The UTC time zone is used. The value is in the format of "<i>Time</i>":"<i>Indicator result</i>", indicating the indicator result collected in a time range starting from <i>Time</i>. The time range is determined by the timeGrain parameter in the request. For example, "2020-11-11 09:15:00.0": "1" indicates that the indicator result is 1 from 09:15 to 09:30. In this example, the value of timeGrain in the request is 15. If the indicator result is 0, the current dataset is not displayed. For example, when you query data of 09:15 to 11:15, if the indicator result at 10:15 is 0 and the indicator results at other time points are greater than 0, the following information is returned: <pre>{ "2020-11-11 09:15:00.0": "2", "2020-11-11 11:15:00.0": "1" }</pre>

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "ccId":1,
  "vdnIds":[29,30],
  "indexIds":["IDX_COUNT_TOTAL_CALL","IDX_COUNT_CONNECTED_CALL"],
  "timeGrain":"15MIN",
  "anonyCall":1,
  "calleeNo":"640001",
  "mediaType":53,
  "subMediaType":3
}
```

- Response parameters

```
{
  "result": "0",
  "resultCode": "0100000",
  "resultDatas": [
    {
```

```

      "idxs": [
        {
          "val": {
            "2020-11-11 09:15:00.0": "2",
            "2020-11-11 10:15:00.0": "1"
          },
          "id": "IDX_COUNT_TOTAL_CALL"
        },
        {
          "val": {
            "2020-11-11 09:15:00.0": "2",
            "2020-11-11 10:15:00.0": "1"
          },
          "id": "IDX_COUNT_CONNECTED_CALL"
        }
      ],
      "id": "29"
    },
    {
      "idxs": [
        {
          "val": {
            "2020-11-11 09:15:00.0": "1",
            "2020-11-11 10:15:00.0": "1"
          },
          "id": "IDX_COUNT_TOTAL_CALL"
        },
        {
          "val": {
            "2020-11-11 09:15:00.0": "1",
            "2020-11-11 10:15:00.0": "1"
          },
          "id": "IDX_COUNT_CONNECTED_CALL"
        }
      ],
      "id": "30"
    }
  ],
  "resultDesc": [
    {
      "idxs": [
        {
          "val": {

```



```

        "2020-11-11 09:15:00.0": "2",
        "2020-11-11 10:15:00.0": "1"
    },
    "id": "IDX_COUNT_TOTAL_CALL"
},
{
    "val": {
        "2020-11-11 09:15:00.0": "2",
        "2020-11-11 10:15:00.0": "1"
    },
    "id": "IDX_COUNT_CONNECTED_CALL"
}
],
"id": "29"
},
{
    "idxs": [
        {
            "val": {
                "2020-11-11 09:15:00.0": "1",
                "2020-11-11 10:15:00.0": "1"
            },
            "id": "IDX_COUNT_TOTAL_CALL"
        },
        {
            "val": {
                "2020-11-11 09:15:00.0": "1",
                "2020-11-11 10:15:00.0": "1"
            },
            "id": "IDX_COUNT_CONNECTED_CALL"
        }
    ],
    "id": "30"
}
]
}

```

4.2.2 Querying Information About Missed Calls in a Specified VCC

Scenario

This interface is invoked to query information about all missed calls that are connected to agents but not answered by agents in a specified VCC.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Request URL

`https://ip:port/rest/cmsapp/v1/callin/queryMissedCalls`

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the request body

No.	Parameter	Value Type	Mandatory	Description
1	vdn	Integer	Yes	ID of a VCC. The value ranges from 1 to 5000.
2	offset	Number	Yes	Parameter for determining the start page of the query. Value of offset = Value of limit x (Value of page - 1)
3	limit	Number	Yes	Number of queried records. A maximum of 5000 records can be queried at a time.
4	beginTime	String	No	Start time for the query (zero-zone time). The format is <i>yyyy-MM-dd HH:mm:ss</i> .
5	endTime	String	No	End time for the query (zero-zone time). The format is <i>yyyy-MM-dd HH:mm:ss</i> .

Response Description

Table 3 Response parameters

Parameter	Value Type	Mandatory	Description
returnCode	String	Yes	Query result. The options are as follows: 0 : success Other values: failure
total	Number	Yes	Total number of missed calls.
missedCalls	MissedCall[]	Yes	List of details about missed calls. For details, see Table 4 .
returnDesc	String	No	Query result description.

Table 4 Parameters of MissedCall

Parameter	Value Type	Mandatory	Description
callId	String	Yes	Call ID.
callerNo	String	Yes	Calling number.
calleeNo	String	Yes	Called number.
waitBegin	Datetime	Yes	Call waiting start time.
callEnd	Datetime	Yes	Call end time.
mediaType	Integer	No	Media type. For details, see Table 1 .
subMediaType	Integer	No	Submedia type. For details, see Table 3 .

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "vdn":159,
  "offset":0,
  "limit":3,
  "beginTime":"2022-11-17 00:13:43",
  "endTime":"2022-11-17 23:44:13"
}
```

- Response parameters

```
{
```

```

"returnCode": "0",
"total": 2,
"missedCalls": [
  {
    "callId": "1668048253-16797218",
    "callerNo": "1659960139858526939",
    "calleeNo": "6300111010",
    "waitBegin": "2022-11-17 23:44:13",
    "callEnd": "2022-11-17 23:44:25",
    "mediaType": 53,
    "subMediaType": 1
  },
  {
    "callId": "1668050023-1859",
    "callerNo": "88880990",
    "calleeNo": "31045665",
    "waitBegin": "2022-11-17 00:13:43",
    "callEnd": "2022-11-17 00:13:47",
    "mediaType": 53,
    "subMediaType": 1
  }
],
"returnDesc": "query success"
}

```

4.2.3 Obtaining Historical IVR Monitoring Indicators

Scenario

Obtain historical monitoring indicators by IVR. By default, data within 5 minutes is obtained from the settlement table, and the real-time cache data is not queried. To query the real-time cache data, change the value of **history_monitor_indicator/history_monitor_indicator_agent** field in **t_cms_sysparam** in the database to **true**. Since real-time cache data settlement by the interface has performance bottleneck, contact the maintenance personnel to obtain the real-time cache data volume of an interface before enabling this function.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Request URL

https://ip:port/rest/cmsapp/v1/openapi/hisindex/ivr

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the request body

Parameter	Value Type	Mandatory	Description
cclid	Integer	No	ID of a call center. In the integration environment, the default value is the value of Call Center ID on the Configuration Center > System Management > CMS System Configuration page after the system administrator signs in to the AICC. If this parameter is empty, the data whose sys_key is default_cc_id and sys_type is cms_system_config in the t_cms_sysparam database table is used. In the independent deployment environment, the default value is the value of Default Call Center ID on the Configuration Center > System Management > Parameter Settings page after the system administrator signs in to the AICC.
vdn	Integer	Yes	ID of a VCC.
serviceNos	Array	Yes	Array of business type IDs. The maximum length of the array is 500. The business type is of the integer type. In public cloud mode, the value is 1 by default. To change the value, contact the system administrator. In independent deployment mode, log in to the WAS as a VCC administrator, choose Web Configuration Console > VDN * Configuration > Flow > Flow, and check the value of Flow Service Type corresponding to the IVR flow. For example, 1 Voice indicates that the business type code corresponding to the IVR flow is 1. NOTE: This parameter has no actual meaning. Do not use it.

Table 2 Parameters in the request body

Parameter	Value Type	Mandatory	Description
indexIds	Array	Yes	Array of indicators. The indicator is of the string type. For details, see Table 1 .
timeGrain	String	Yes/No	This parameter is mandatory when Table 1 contains the timeGrain parameter. This parameter can be used to collect statistics on historical monitoring indicators by minute. The options are 15MIN , 30MIN , and 60MIN . If this parameter is not passed, statistics on the current day in the time zone where the tenant is located are collected.
mediaType	Integer	No	Call media type. For details, see Table 1 .
subMediaType	Integer	No	Submedia type. For details, see Table 3 .
enableOffset	String	No	The options are Y and N . If Y is transferred, the data of the previous day is queried. If N is transferred, the data of the current day is queried. If no value is transferred, the default value N is used.

Response Description

Table 3 Parameters in the response body

Parameter	Value Type	Description
result	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
resultDatas	List <JSON object>	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.

Table 3 Parameters in the response body

Parameter	Value Type	Description
resultDesc	List <JSON object>	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 .

Table 4 Parameters of resultDesc/resultDatas

Parameter	Value Type	Description
id	String	Business type ID.
idxs	List <JSON object>	Group of indicators. For details about the parameters of the object, see Table 5 .

Table 5 Parameters of idx

Parameter	Value Type	Description
id	String	Indicator ID.
val	Object	Indicator result collected by minute. The UTC time zone is used. The value is in the format of " <i>Time</i> ":" <i>Indicator result</i> ", indicating the indicator result collected in a time range starting from <i>Time</i> . The time range is determined by the timeGrain parameter in the request. For example, " 2020-11-11 09:15:00.0 ": " 1 " indicates that the indicator result is 1 from 09:15 to 09:30. In this example, the value of timeGrain in the request is 15 .

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "ccld":1,
  "vdn":170,
  "timeGrain":"15MIN",
  "serviceNos":[3,2],
  "indexIds":["IDX_COUNT_IVR_INBOUND_CALL"],
  "mediaType":53,
  "subMediaType":3
}
```

- Response parameters

```
{
```

```

"result": "0",
"resultCode": "0100000",
"resultDatas": [
  {
    "idxs": [
      {
        "val": {
          "2020-11-11 09:15:00.0": "1",
          "2020-11-11 10:15:00.0": "1"
        },
        "id": "IDX_COUNT_IVR_INBOUND_CALL"
      }
    ],
    "id": "3"
  },
  {
    "idxs": [
      {
        "val": {
          "2020-11-11 09:15:00.0": "1",
          "2020-11-11 10:15:00.0": "1"
        },
        "id": "IDX_COUNT_IVR_INBOUND_CALL"
      }
    ],
    "id": "3"
  }
]
}

```

4.2.4 Obtaining Historical Monitoring Indicators by Skill Queue

Scenario

Obtain historical monitoring indicators by skill queue. By default, data within 5 minutes is obtained from the settlement table, and the real-time cache data is not queried. To query the real-time cache data, change the value of **history_monitor_indicator/history_monitor_indicator_agent** field in **t_cms_sysparam** in the database to **true**. Since real-time cache data settlement by the interface has performance bottleneck,

contact the maintenance personnel to obtain the real-time cache data volume of an interface before enabling this function.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Request URL

`https://ip:port/rest/cmsapp/v1/openapi/hisindex/skill`

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the request body

Parameter	Value Type	Mandatory	Description
cclid	Integer	No	Call center ID. The value ranges from 1 to 65535. In the integration environment, the default value is the value of Call Center ID on the Configuration Center > System Management > CMS System Configuration page after the system administrator signs in to the AICC. If this parameter is empty, the data whose sys_key is default_cc_id and sys_type is cms_system_config in the t_cms_sysparam database table is used. In the independent deployment environment, the default value is the value of Default Call Center ID on the Configuration Center > System Management > Parameter Settings page after the system administrator signs in to the AICC.
vdn	Integer	Yes	ID of a VCC. The value ranges from 1 to 5000.
timeGrain	String	True/False	This parameter is mandatory when Table 1 contains the timeGrain parameter.

Table 2 Parameters in the request body

Parameter	Value Type	Mandatory	Description
			This parameter can be used to collect statistics on historical monitoring indicators by minute. The options are 15MIN , 30MIN , and 60MIN . If this parameter is not passed, statistics on the current day in the time zone where the tenant is located are collected.
skillIds	List <Integer>	Yes	Array of skill queue IDs. The array contains a maximum of 1,000 characters. The skill queue is of the integer type.
indexIds	List <String>	Yes	Array of indicators. The indicator is of the string type. The maximum length of the array depends on the number of historical monitoring indicators by skill. For details, see Table 1 .
mediaType	Integer	No	Call media type. For details, see Table 1 .
subMediaType	Integer	No	Submedia type. For details, see Table 3 .
enableOffset	String	No	The options are Y and N . If Y is transferred, the data of the previous day is queried. If N is transferred, the data of the current day is queried. If no value is transferred, the default value N is used.

Response Description

Table 3 Parameters in the response body

Parameter	Value Type	Description
result	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
resultDatas	List <JSON object>	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 . NOTE:

Table 3 Parameters in the response body

Parameter	Value Type	Description
		This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
resultDesc	List <JSON object>	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 .

Table 4 Parameters of resultDesc/resultDatas

Parameter	Value Type	Description
id	String	Skill queue ID.
idxs	List <JSON object>	Group of indicators. For details about the parameters of the object, see Table 1 .

Table 5 Parameters of idx

Parameter	Value Type	Description
id	String	Indicator ID.
val	Object	Indicator result collected by minute. The UTC time zone is used. The value is in the format of " <i>Time</i> ":" <i>Indicator result</i> ", indicating the indicator result collected in a time range starting from <i>Time</i> . The time range is determined by the timeGrain parameter in the request. For example, " 2020-11-11 09:15:00.0 ": " 1 " indicates that the indicator result is 1 from 09:15 to 09:30. In this example, the value of timeGrain in the request is 15 .

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "cclId":1,
  "vdn":1,
  "skillIds":[11,24],
  "timeGrain":"15MIN",
  "indexIds":["IDX_COUNT_SKILL_REQUEST","IDX_COUNT_SKILL_CONNECTED"],
  "mediaType":53,
  "subMediaType":3
}
```

- Response parameters

```
{
  "result": "0",
  "resultCode": "0100000",
  "resultDatas": [
    {
      "idxs": [
        {
          "val": {
            "2020-08-24 10:30:00.0": "15",
            "2020-08-24 10:45:00.0": "19"
          },
          "id": "IDX_COUNT_SKILL_REQUEST"
        },
        {
          "val": {
            "2020-08-24 10:30:00.0": "6",
            "2020-08-24 10:45:00.0": "4"
          },
          "id": "IDX_COUNT_SKILL_CONNECTED"
        }
      ],
      "id": "12"
    },
    {
      "idxs": [
        {
          "val": {
            "2020-08-24 10:00:00.0": "16",
            "2020-08-24 10:15:00.0": "14"
          },
          "id": "IDX_COUNT_SKILL_REQUEST"
        },
        {
          "val": {
            "2020-08-24 10:00:00.0": "5",
            "2020-08-24 10:15:00.0": "4"
          },
          "id": "IDX_COUNT_SKILL_CONNECTED"
        }
      ],
      "id": "13"
    }
  ]
}
```

```
    }
  ],
  "resultDesc": [
    {
      "idxs": [
        {
          "val": {
            "2020-08-24 10:30:00.0": "15",
            "2020-08-24 10:45:00.0": "19"
          },
          "id": "IDX_COUNT_SKILL_REQUEST"
        },
        {
          "val": {
            "2020-08-24 10:30:00.0": "6",
            "2020-08-24 10:45:00.0": "4"
          },
          "id": "IDX_COUNT_SKILL_CONNECTED"
        }
      ],
      "id": "12"
    },
    {
      "idxs": [
        {
          "val": {
            "2020-08-24 10:00:00.0": "16",
            "2020-08-24 10:15:00.0": "14"
          },
          "id": "IDX_COUNT_SKILL_REQUEST"
        },
        {
          "val": {
            "2020-08-24 10:00:00.0": "5",
            "2020-08-24 10:15:00.0": "4"
          },
          "id": "IDX_COUNT_SKILL_CONNECTED"
        }
      ],
      "id": "13"
    }
  ]
}
```

}

4.2.5 Obtaining Historical Monitoring Indicators by Agent

Scenario

Obtain historical monitoring indicators by agent. By default, data within 5 minutes is obtained from the settlement table, and the real-time cache data is not queried. To query the real-time cache data, change the value of **history_monitor_indicator/history_monitor_indicator_agent** field in **t_cms_sysparam** in the database to **true**. Since real-time cache data settlement by the interface has performance bottleneck, contact the maintenance personnel to obtain the real-time cache data volume of an interface before enabling this function.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Request URL

`https://ip:port/rest/cmsapp/v1/openapi/hisindex/agent`

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the request body

Parameter	Value Type	Mandatory	Description
ccld	Integer	No	Call center ID. The value ranges from 1 to 65535. In the integration environment, the default value is the value of Call Center ID on the Configuration Center > System Management > CMS

Table 2 Parameters in the request body

Parameter	Value Type	Mandatory	Description
			System Configuration page after the system administrator signs in to the AICC. If this parameter is empty, the data whose sys_key is default_cc_id and sys_type is cms_system_config in the t_cms_sysparam database table is used. In the independent deployment environment, the default value is the value of Default Call Center ID on the Configuration Center > System Management > Parameter Settings page after the system administrator signs in to the AICC.
vdn	Integer	Yes	ID of a VCC. The value ranges from 1 to 5000.
agentIds	List <Integer>	Yes	Array of agent IDs. The array contains a maximum of 100 characters. The agent ID is an integer ranging from 0 to 999999.
timeGrain	String	Yes/No	This parameter is mandatory when Table 1 contains the timeGrain parameter. This parameter can be used to collect statistics on historical monitoring indicators by minute. The options are 15MIN, 30MIN, and 60MIN. If this parameter is not passed, statistics on the current day in the time zone where the tenant is located are collected.
indexIds	List <String>	Yes	Array of indicators. The indicator is of the string type. The maximum length of the array depends on the number of historical agent monitoring indicators. For details, see Table 1 .
mediaType	Integer	No	Call media type. For details, see Table 1 .
subMediaType	Integer	No	Submedia type. For details, see Table 3 .
enableOffset	String	No	The options are Y and N . If Y is transferred, the data of the previous day is queried. If N is transferred, the data of the current day is queried. If no value is transferred, the default value N is used.

Response Description

Table 3 Parameters in the response body

Parameter	Value Type	Description
result	String	Query result. The options are as follows: 0: success Other values: failure For details about the failure causes, see Error Code Reference . NOTE:

Table 3 Parameters in the response body

Parameter	Value Type	Description
		This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
resultDatas	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions. If the return value is empty, the current agent has not signed in.
resultDesc	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 . NOTE: If the return value is empty, the current agent has not signed in.

Table 4 Parameters of resultDesc/resultDatas

Parameter	Value Type	Description
id	String	Agent ID.
idxs	Array	Group of indicators. For details about the parameters of the object, see Table 5 .

Table 5 Parameters of idx

Parameter	Value Type	Description
id	String	Indicator ID.
val	Object	Indicator result. The UTC time zone is used.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters


```
{
  "ccId":1,
  "vdn":22,
  "agentIds":[295],
  "timeGrain":"15MIN",

  "indexIds":["IDX_COUNT_AGENT_COLLATION","IDX_COUNT_AGENT_SET_BUSY","IDX_DURATION_AGENT_CALL_IN"],
  "mediaType":53,
  "subMediaType":3
}
```

- Response parameters

```
{
  "result": "0",
  "resultCode": "0100000",
  "resultDatas": [
    {
      "idxs": [
        {
          "val": {
            "2020-08-24 10:30:00.0": "15",
            "2020-08-24 10:45:00.0": "19"
          },
          "id": "IDX_COUNT_AGENT_COLLATION"
        },
        {
          "val": {
            "2020-08-24 10:30:00.0": "15",
            "2020-08-24 10:45:00.0": "19"
          },
          "id": "IDX_DURATION_AGENT_AVG_COLLATION"
        },
        {
          "val": {
            "2020-08-24 10:30:00.0": "15",
            "2020-08-24 10:45:00.0": "19"
          },
          "id": "IDX_COUNT_AGENT_SET_BUSY"
        }
      ],
      "id": "11"
    },
    {
```

```

      "idxs": [
        {
          "val": {
            "2020-08-24 10:30:00.0": "15",
            "2020-08-24 10:45:00.0": "19"
          },
          "id": "IDX_COUNT_AGENT_COLLATION"
        },
        {
          "val": {
            "2020-08-24 10:30:00.0": "15",
            "2020-08-24 10:45:00.0": "19"
          },
          "id": "IDX_DURATION_AGENT_AVG_COLLATION"
        },
        {
          "val": {
            "2020-08-24 10:30:00.0": "15",
            "2020-08-24 10:45:00.0": "19"
          },
          "id": "IDX_COUNT_AGENT_SET_BUSY"
        }
      ],
      "id": "12"
    }
  ],
  "resultDesc": [
    {
      "idxs": [
        {
          "val": {
            "2020-08-24 10:30:00.0": "15",
            "2020-08-24 10:45:00.0": "19"
          },
          "id": "IDX_COUNT_AGENT_COLLATION"
        },
        {
          "val": {
            "2020-08-24 10:30:00.0": "15",
            "2020-08-24 10:45:00.0": "19"
          },
          "id": "IDX_DURATION_AGENT_AVG_COLLATION"
        }
      ],

```

```
{
  "val": {
    "2020-08-24 10:30:00.0": "15",
    "2020-08-24 10:45:00.0": "19"
  },
  "id": "IDX_COUNT_AGENT_SET_BUSY"
}
],
"id": "11"
},
{
  "idxs": [
    {
      "val": {
        "2020-08-24 10:30:00.0": "15",
        "2020-08-24 10:45:00.0": "19"
      },
      "id": "IDX_COUNT_AGENT_COLLATION"
    },
    {
      "val": {
        "2020-08-24 10:30:00.0": "15",
        "2020-08-24 10:45:00.0": "19"
      },
      "id": "IDX_DURATION_AGENT_AVG_COLLATION"
    },
    {
      "val": {
        "2020-08-24 10:30:00.0": "15",
        "2020-08-24 10:45:00.0": "19"
      },
      "id": "IDX_COUNT_AGENT_SET_BUSY"
    }
  ],
  "id": "12"
}
]
```

4.2.6 Obtaining Historical Monitoring Indicators in a Specified Time Range by Skill

Scenario

This interface is invoked to obtain historical monitoring indicators in specified time range by skill.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Request URL

`https://ip:port/rest/cmsapp/v1/openapi/hisindex/skillforday`

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the message body

Parameter	Value Type	Mandatory	Description
cclId	Integer	No	ID of a call center. In the integration environment, the default value is the value of Call Center ID on the Configuration Center > System Management > CMS System Configuration page after the system administrator signs in to the AICC. If this parameter is empty, the data whose sys_key is default_cc_id and sys_type is cms_system_config in the t_cms_sysparam database table is used. In the independent deployment environment, the default value is the value of Default Call Center ID on the Configuration Center > System Management > Parameter Settings page after the system administrator signs in to the AICC.

Table 2 Parameters in the message body

Parameter	Value Type	Mandatory	Description
vdn	Integer	Yes	ID of a VCC.
skillIds	List<Integer>	Yes	Array of skill queue IDs. The array contains a maximum of 1,000 characters. The skill queue is of the integer type.
indexIds	List<String>	Yes	Indicator code array. The indicator is of the string type. The maximum length of the array depends on the number of historical monitoring indicators by skill. For details about the indicators, see Table 1 and Table 3 .
beginDayTime	String	Yes	Start date and time, which is the time in the time zone where the tenant is located.
endDayTime	String	Yes	End date and time, which is the time in the time zone where the tenant is located.
mediaType	Integer	No	Call media type. For details, see Table 1 .
subMediaType	Integer	No	Submedia type. For details, see Table 3 .

Response Description

Table 3 Parameters in the message body

Parameter	Value Type	Description
result	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
resultDatas	List <JSON object>	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.

Table 3 Parameters in the message body

Parameter	Value Type	Description
resultDesc	List <JSON object>	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 .

Table 4 Parameters of resultDesc/resultDatas

Parameter	Value Type	Description
id	String	Skill queue ID.
idxs	List <JSON object>	Group of indicators. For details about the parameters of the object, see Table 5 .

Table 5 Parameters of idxs

Parameter	Value Type	Description
id	String	Indicator name.
val	object	Indicator result.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "ccId":1,
  "vdn":4,
  "skillIds":[11,12],
  "indexIds":["IDX_DURATION_SKILL_AVG_ABANDON_ACK","IDX_COUNT_SKILL_REQUEST"],
  "beginDayTime":"2020-08-21 00:00:00",
  "endDayTime":"2020-08-22 00:00:00",
  "mediaType":53,
  "subMediaType":3
}
```

- Response parameters

```
{
  "result": "0",
  "resultCode": "0100000",
  "resultDatas": [
```

```
{
  "idxs": [
    {
      "val": "3",
      "id": "IDX_COUNT_SKILL_REQUEST"
    },
    {
      "val": "35",
      "id": "IDX_DURATION_SKILL_AVG_ABANDON_ACK"
    }
  ],
  "id": "11"
},
{
  "idxs": [
    {
      "val": "5",
      "id": "IDX_COUNT_SKILL_REQUEST"
    },
    {
      "val": "15",
      "id": "IDX_DURATION_SKILL_AVG_ABANDON_ACK"
    }
  ],
  "id": "12"
}
],
"resultDesc": [
  {
    "idxs": [
      {
        "val": "3",
        "id": "IDX_COUNT_SKILL_REQUEST"
      },
      {
        "val": "35",
        "id": "IDX_DURATION_SKILL_AVG_ABANDON_ACK"
      }
    ],
    "id": "11"
  },
  {
    "idxs": [
```

```

    {
      "val": "5",
      "id": "IDX_COUNT_SKILL_REQUEST"
    },
    {
      "val": "15",
      "id": "IDX_DURATION_SKILL_AVG_ABANDON_ACK"
    }
  ],
  "id": "12"
}
]
}
```

4.2.7 Obtaining Historical Monitoring Indicators in a Specified Month Range by Skill

Scenario

This interface is invoked to obtain historical monitoring indicators in specified month by skill.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Request URL

https://ip:port/rest/cmsapp/v1/openapi/hisindex/skillformonth

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the message body

Parameter	Value Type	Mandatory	Description
cclid	Integer	No	ID of a call center. In the integration environment, the default value is the value of Call Center ID on the Configuration Center > System Management > CMS System Configuration page after the system administrator signs in to the AICC. If this parameter is empty, the data whose sys_key is default_cc_id and sys_type is cms_system_config in the t_cms_sysparam database table is used. In the independent deployment environment, the default value is the value of Default Call Center ID on the Configuration Center > System Management > Parameter Settings page after the system administrator signs in to the AICC.
vdn	Integer	Yes	ID of a VCC.
skillIds	List<Integer>	Yes	Array of skill queue IDs. The array contains a maximum of 1,000 characters. The skill queue is of the integer type.
indexIds	List<String>	Yes	Array of indicators. The indicator is of the string type. The maximum length of the array depends on the number of historical monitoring indicators by skill. For details about the indicators, see Table 1 and Table 3.
beginMonthTime	String	Yes	Start month, which is in the time zone of the tenant.
endMonthTime	String	Yes	End month, which is in the time zone of the tenant.
mediaType	Integer	No	Call media type. For details, see Table 1 .
subMediaType	Integer	No	Submedia type. For details, see Table 3 .

Response Description

Table 3 Parameters in the message body

Parameter	Value Type	Description
result	String	Query result. The options are as follows: 0: success

Table 3 Parameters in the message body

Parameter	Value Type	Description
		Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
resultDatas	List <JSON object>	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
resultDesc	List <JSON object>	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 .

Table 4 Parameters of resultDesc/resultDatas

Parameter	Value Type	Description
id	String	Skill queue ID.
idxs	List <JSON object>	Group of indicators. For details about the parameters of the object, see Table 5 .

Table 5 Parameters of idxs

Parameter	Value Type	Description
id	String	Indicator name.
val	Object	Indicator result.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
```

```

"ccId":1,
"vdn":4,
"skillIds":[11,12],
"indexIds":["IDX_DURATION_SKILL_AVG_ABANDON_ACK","IDX_COUNT_SKILL_REQUEST"],
"beginMonthTime":"202007",
"endMonthTime":"202008",
"mediaType":53,
"subMediaType":3
}

```

- Response parameters

```

{
  "result": "0",
  "resultCode": "0100000",
  "resultDatas": [
    {
      "idxs": [
        {
          "val": "3",
          "id": "IDX_COUNT_SKILL_REQUEST"
        },
        {
          "val": "35",
          "id": "IDX_DURATION_SKILL_AVG_ABANDON_ACK"
        }
      ],
      "id": "11"
    },
    {
      "idxs": [
        {
          "val": "10",
          "id": "IDX_COUNT_SKILL_REQUEST"
        },
        {
          "val": "35",
          "id": "IDX_DURATION_SKILL_AVG_ABANDON_ACK"
        }
      ],
      "id": "12"
    }
  ],
  "resultDesc": [

```

```
{
  "idxs": [
    {
      "val": "3",
      "id": "IDX_COUNT_SKILL_REQUEST"
    },
    {
      "val": "35",
      "id": "IDX_DURATION_SKILL_AVG_ABANDON_ACK"
    }
  ],
  "id": "11"
},
{
  "idxs": [
    {
      "val": "10",
      "id": "IDX_COUNT_SKILL_REQUEST"
    },
    {
      "val": "35",
      "id": "IDX_DURATION_SKILL_AVG_ABANDON_ACK"
    }
  ],
  "id": "12"
}
]
```

4.2.8 Obtaining Historical Monitoring Indicators in a Specified Time Range by Agent

Scenario

This interface is invoked to obtain historical monitoring indicators in specified time range by agent.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Request URL

`https://ip:port/rest/cmsapp/v1/openapi/hisindex/agentforday`

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the message body

Parameter	Value Type	Mandatory	Description
cclId	Integer	No	ID of a call center. In the integration environment, the default value is the value of Call Center ID on the Configuration Center > System Management > CMS System Configuration page after the system administrator signs in to the AICC. If this parameter is empty, the data whose sys_key is default_cc_id and sys_type is cms_system_config in the t_cms_sysparam database table is used. In the independent deployment environment, the default value is the value of Default Call Center ID on the Configuration Center > System Management > Parameter Settings page after the system administrator signs in to the AICC.
vdn	Integer	Yes	ID of a VCC.
agentIds	List <Integer>	Yes	Array of agent queue IDs. The array contains a maximum of 1,000 characters. The agent queue ID is an integer ranging from 0 to 999999.
indexIds	List <String>	Yes	Indicator code array. The indicator is of the string type. The maximum length of the array depends on the number of historical agent monitoring indicators. For details about the indicators, see Table 1 and Table 3 .
beginDayTime	String	Yes	Start date and time, which is in the time zone of the tenant.
endDayTime	String	Yes	End date and time, which is in the time zone of the tenant.
mediaType	Integer	No	Call media type. For details, see Table 1 .

Table 2 Parameters in the message body

Parameter	Value Type	Mandatory	Description
subMediaType	Integer	No	Submedia type. For details, see Table 3 .

Response Description

Table 3 Parameters in the message body

Parameter	Value Type	Description
result	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
resultDatas	List <JSON object>	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
resultDesc	List <JSON object>	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 .

Table 4 Parameters of resultDesc/resultDatas

Parameter	Value Type	Description
id	String	Agent ID.
idxs	List <JSON object>	Group of indicators. For details about the parameters of the object, see Table 5 .

Table 5 Parameters of idxs

Parameter	Value Type	Description
id	String	Indicator name.
val	Object	Indicator result.

Message Example

- Request header

Content-Type: application/json;charset=UTF-8

Authorization: *****

- Request parameters

```
{
  "ccId":1,
  "vdn":4,
  "agentIds":[11,12],
  "indexIds":["IDX_DURATION_AGENT_REST","IDX_COUNT_AGENT_REQUEST"],
  "beginDayTime":"2020-08-21 00:00:00",
  "endDayTime":"2020-08-22 00:00:00",
  "mediaType":53,
  "subMediaType":3
}
```

- Response parameters

```
{
  "result": "0",
  "resultCode": "0100000",
  "resultDatas": [
    {
      "idxs": [
        {
          "val": "3",
          "id": "IDX_COUNT_AGENT_REQUEST"
        },
        {
          "val": "35",
          "id": "IDX_DURATION_AGENT_REST"
        }
      ],
      "id": "11"
    },
  ],
}
```

```
{
  "idxs": [
    {
      "val": "5",
      "id": "IDX_COUNT_AGENT_REQUEST"
    },
    {
      "val": "15",
      "id": "IDX_DURATION_AGENT_REST"
    }
  ],
  "id": "12"
},
{
  "resultDesc": [
    {
      "idxs": [
        {
          "val": "3",
          "id": "IDX_COUNT_AGENT_REQUEST"
        },
        {
          "val": "35",
          "id": "IDX_DURATION_AGENT_REST"
        }
      ],
      "id": "11"
    },
    {
      "idxs": [
        {
          "val": "5",
          "id": "IDX_COUNT_AGENT_REQUEST"
        },
        {
          "val": "15",
          "id": "IDX_DURATION_AGENT_REST"
        }
      ],
      "id": "12"
    }
  ]
}
```



```
}

```

4.2.9 Obtaining Historical Monitoring Indicators in a Specified Month by Agent

Scenario

This interface is invoked to obtain historical monitoring indicators in a specified month by agent.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Request URL

https://ip:port/rest/cmsapp/v1/openapi/hisindex/agentformonth

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the message body

Parameter	Value Type	Mandatory	Description
cclid	Integer	No	ID of a call center. In the integration environment, the default value is the value of Call Center ID on the Configuration Center > System Management > CMS System Configuration page after the system administrator signs in to the AICC. If this parameter is empty, the data whose sys_key is default_cc_id and sys_type is cms_system_config in the t_cms_sysparam database table is used.

Table 2 Parameters in the message body

Parameter	Value Type	Mandatory	Description
			In the independent deployment environment, the default value is the value of Default Call Center ID on the Configuration Center > System Management > Parameter Settings page after the system administrator signs in to the AICC.
vdn	Integer	Yes	ID of a VCC.
agentIds	List <Integer>	Yes	Array of agent IDs. The array contains a maximum of 1,000 characters. The agent queue ID is an integer ranging from 0 to 999999.
indexIds	List <String>	Yes	Array of indicators. The indicator is of the string type. The maximum length of the array depends on the number of historical agent monitoring indicators. For details about the indicators, see Table 1 and Table 3 .
beginMonthTime	String	Yes	Start month, which is in the time zone of the tenant.
endMonthTime	String	Yes	End month, which is in the time zone of the tenant.
mediaType	Integer	No	Call media type. For details, see Table 1 .
subMediaType	Integer	No	Submedia type. For details, see Table 3 .

Response Description

Table 3 Parameters in the message body

Parameter	Value Type	Description
result	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
resultDatas	List <JSON object>	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 .

Table 3 Parameters in the message body

Parameter	Value Type	Description
		NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
resultDesc	List <JSON object>	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 .

Table 4 Parameters of resultDesc/resultDatas

Parameter	Value Type	Description
id	String	Agent ID.
idxs	List <JSON object>	Group of indicators. For details about the parameters of the object, see Table 5 .

Table 5 Parameters of idxs

Parameter	Value Type	Description
id	String	Indicator name.
val	Object	Indicator result.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "ccId":1,
  "vdn":4,
  "agentIds":[11,12],
  "indexIds":["IDX_DURATION_AGENT_REST","IDX_COUNT_AGENT_REQUEST"],
  "beginMonthTime":"202007",
  "endMonthTime":"202008",
  "mediaType":53,
  "subMediaType":3
}
```

- Response parameters

```
{
```

```
"result": "0",
"resultCode": "0100000",
"resultDatas": [
  {
    "idxs": [
      {
        "val": "3",
        "id": "IDX_COUNT_AGENT_REQUEST"
      },
      {
        "val": "35",
        "id": "IDX_DURATION_AGENT_REST"
      }
    ],
    "id": "11"
  },
  {
    "idxs": [
      {
        "val": "10",
        "id": "IDX_COUNT_AGENT_REQUEST"
      },
      {
        "val": "35",
        "id": "IDX_DURATION_AGENT_REST"
      }
    ],
    "id": "12"
  }
],
"resultDesc": [
  {
    "idxs": [
      {
        "val": "3",
        "id": "IDX_COUNT_AGENT_REQUEST"
      },
      {
        "val": "35",
        "id": "IDX_DURATION_AGENT_REST"
      }
    ],
    "id": "11"
  }
]
```

```

    },
    {
      "idxs": [
        {
          "val": "10",
          "id": "IDX_COUNT_AGENT_REQUEST"
        },
        {
          "val": "35",
          "id": "IDX_DURATION_AGENT_REST"
        }
      ],
      "id": "12"
    }
  ]
}

```

4.2.10 Collecting Statistics on Result Sets in Daily Settlement Table for Inbound Calls

Scenario

This interface is invoked to collect statistics on result sets in the daily settlement table for inbound calls based on conditions.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Request URL

`https://ip:port/rest/cmsapp/v1/callinday/querymanualindexesbycondition`

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the message body

Parameter	Value Type	Mandatory	Description
calleeNos	Array	Yes	Array of called numbers. Statistics are collected by each number in the array. A called number can contain only digits. The array contains a maximum of 1,000 characters.
beginLogDay	String	Yes	Start call time for statistics collection. The value must be in the <i>yyyy-MM-dd</i> format and in the time zone of the tenant.
endLogDay	String	Yes	End call time for statistics collection. The value must be in the <i>yyyy-MM-dd</i> format and in the time zone of the tenant.
callTypes	Array	Yes	Call type array of the integer type. Statistics are collected based on each type in the array. The call type is only the digit. The array contains a maximum of 1,000 characters.
skillIds	Array	No	Skill queue IDs. An array of integers. Statistics are collected by type in the array. The array contains a maximum of 1,000 characters.
subCcNo	Integer	Either this parameter or tenantId must be set.	Subsystem number.
vdn	Integer	Either this parameter or tenantId must be set.	ID of a VCC.
tenantId	String	Either this parameter or subCcNo (or vdn) must be set.	Tenant ID.
ccId	Integer	No	ID of a call center. In the integration environment, the default value is the value of Call Center ID on the Configuration Center > System Management > CMS System Configuration page after the system administrator signs in to the AICC. If this parameter is empty, the data whose

Table 2 Parameters in the message body

Parameter	Value Type	Mandatory	Description
			sys_key is default_cc_id and sys_type is cms_system_config in the t_cms_sysparam database table is used. In the independent deployment environment, the default value is the value of Default Call Center ID on the Configuration Center > System Management > Parameter Settings page after the system administrator signs in to the AICC.

Response Description

Table 3 Parameters in the message body

Parameter	Value Type	Description
result	String	Query result. The options are as follows: 0: success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
resultCode	String	Query result. The options are as follows: 0100000: success Other values: failure For details about the failure causes, see Error Code Reference .
resultDatas	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4. NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
resultDesc	Array	Object in the returned result after a successful query. For details about the parameters of the object, see Table 4.

Table 4 Parameters of resultDesc/resultDatas

Parameter	Value Type	Description
beginLogDay	String	Start call time, which is the same as the input parameter.

Table 4 Parameters of resultDesc/resultDatas

Parameter	Value Type	Description
endLogDay	String	End call time, which is the same as the input parameter.
calleeNo	String	Called number.
callType	Integer	Call type.
vdn	Integer	ID of a VCC.
serviceNo	Integer	Business type. NOTE: This parameter has no actual meaning. Do not use it.
orgCcNo	Integer	Subsystem number of the original call center.
subCcNo	Integer	Subsystem number.
serCcNo	Integer	Subsystem number of the service party.
currentAgentId	Integer	Agent ID. Agent ID in the last CDR of a call, which is not recommended for indicator settlement.
currentSkillId	Integer	Skill queue ID. Skill queue ID in the last CDR of a call, which is not recommended for indicator settlement.
failQueueWaitNum	Integer	Number of waits for lost calls.
failQueueWaitTime	Integer	Waiting duration of lost calls, in seconds.
waitAnsNum	Integer	Number of calls that are waiting for answers.
waitAnsTime	Integer	Ring duration, in seconds.
succWaitAnsTime	Integer	Duration of waiting for the answer in a successful call, in seconds.
noAckNum	Integer	Number of calls that are not answered.
succQueueWaitTime	Integer	Queuing duration of a successful call, in seconds.
succQueueWaitNum	Integer	Number of queuing times for a successful call.
queOverTolvrNum	Integer	Number of hang-up calls transferred from a queue to the IVR.
queOverTolvrTime	Integer	Total hang-up duration for transferring calls from a queue to the IVR, in seconds.
hangUpNum	Integer	Number of manual-to-IVR calls.
hangUpTime	Integer	Total duration of manual-to-IVR calls.

Table 4 Parameters of resultDesc/resultDatas

Parameter	Value Type	Description
hangUpReleaseNum	Integer	Number of manual-to-IVR calls that are released.
inOccupyNum	Integer	Number of manual requests.
inCallSuccNum	Integer	Number of manual calls.
inCallTime	Integer	Manual call duration, in seconds.
maxQueueWaitTime	Integer	Maximum queuing duration, in seconds.
minQueueWaitTime	Integer	Minimum queuing duration, in seconds.
maxWaitAnsTime	Integer	Maximum ring duration, in seconds.
minWaitAnsTime	Integer	Minimum ring duration, in seconds.
maxCallTime	Integer	Maximum call duration, in seconds.
minCallTime	Integer	Minimum call duration, in seconds.
succAckTimeLt3	Integer	Number of successful calls whose ring duration is shorter than 3 seconds.
callTimeLt3	Integer	Number of successful calls whose call duration is shorter than 3 seconds.
failTimeLt3	Integer	Number of failed calls whose waiting duration is shorter than 3 seconds.
succAckTimeLt5	Integer	Number of successful calls whose ring duration is shorter than 5 seconds.
callTimeLt5	Integer	Number of successful calls whose call duration is shorter than 5 seconds.
failTimeLt5	Integer	Number of failed calls whose waiting duration is shorter than 5 seconds.
succAckTimeLt10	Integer	Number of successful calls whose ring duration is shorter than 10 seconds.
callTimeLt10	Integer	Number of successful calls whose call duration is shorter than 10 seconds.
failTimeLt10	Integer	Number of failed calls whose waiting duration is shorter than 10 seconds.
succAckTimeLt30	Integer	Number of successful calls whose ring duration is shorter than 30 seconds.
callTimeLt30	Integer	Number of successful calls whose call duration is shorter than 30 seconds.
failTimeLt30	Integer	Number of failed calls whose waiting duration is shorter than 30 seconds.
succAckTimeLt60	Integer	Number of successful calls whose ring duration is shorter than 60 seconds.
callTimeLt60	Integer	Number of successful calls whose call duration is shorter than 60 seconds.
failTimeLt60	Integer	Number of failed calls whose waiting duration is shorter than 60 seconds.
succAckTimeGt10	Integer	Number of successful calls whose ring duration is longer than 10 seconds.

Table 4 Parameters of resultDesc/resultDatas

Parameter	Value Type	Description
callTimeGt10	Integer	Number of successful calls whose call duration is longer than 10 seconds.
failTimeGt10	Integer	Number of failed calls whose waiting duration is longer than 10 seconds.
succAckTimeGt60	Integer	Number of successful calls whose ring duration is longer than 60 seconds.
callTimeGt60	Integer	Number of successful calls whose call duration is longer than 60 seconds.
failTimeGt60	Integer	Number of failed calls whose waiting duration is longer than 60 seconds.
minIvrCallTime	Integer	Minimum IVR call duration, in seconds.
agentToQueueNum	Integer	Number of manual-to-queue calls.
inVdnTime	Integer	Total duration of VCC occupation by inbound calls, in seconds.
succWaitAnsNum	Integer	Number of successful call responses.
ivrOccupyNum	Integer	Number of IVR inbound call requests.
autoUserAbandonNum	Integer	Number of calls that are hung up by users during IVR voice playing.
autoAnsTime	Integer	System-level IVR inbound call ringing duration, in seconds.
ivrOnlySuccNum	Integer	Number of successful calls that are connected using only the IVR.
autoCallSuccNum	Integer	Number of system-level IVR calls.
maxIvrCallTime	Integer	Maximum IVR call duration, in seconds.
ivrOnlyTime	Integer	Total duration of inbound calls that are connected using only the IVR, in seconds.
netCallSuccNum	Integer	Number of successful inbound calls through network devices.
ivrOnlyNum	Integer	Number of inbound calls that are connected using only the IVR.
autoCallNum	Integer	Number of system-level IVR inbound calls.
failWaitAnsTime	Integer	Ring duration for lost calls, in seconds.
queueWaitTime	Integer	Queuing duration, in seconds.
ivrCallTime	Integer	IVR inbound call duration, in seconds.
netCallNum	Integer	Number of inbound calls through network devices.
ivrCallSuccNum	Integer	Number of IVR inbound calls.
agentToAgentNum	Integer	Number of calls that are transferred from the manual service to the manual service.

Table 4 Parameters of resultDesc/resultDatas

Parameter	Value Type	Description
autoTransferNum	Integer	Number of system-level IVR inbound calls that are transferred out.
inCallCostMinute	Integer	Charging duration of a manual call, in minutes.
inCallAllCostMinute	Integer	Total charging duration of each call, in minutes, including the charging duration of manual calls and IVR calls.
reserved_col1- reserved_col20	Integer	Reserved field. The service meaning is determined based on the configuration in the t_cms_custom_indicator_config table. By default, the reserved field 3 indicates the number of inbound calls whose waiting duration for answer is less than or equal to 15 seconds.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "queryParam": {
    "calleeNos": ["10000"],
    "beginLogDay": "2020-01-01",
    "endLogDay": "2020-01-01",
    "callTypes": [0],
    "skillIds": [1],
    "subCcNo": 1,
    "vdn": 3,
    "ccId": 1  }
}
```

- Response parameters

```
{
  "result": "0",
  "resultCode": "0100000",
  "resultDatas": [
    {
      "callTimeGt10": 6,
      "vdn": 152,
      "minIvrCallTime": 0,
      "queOverToIvrNum": 0,

```

```
"failTimeGt10": 0,  
"inCallTime": 6584,  
"agentToQueueNum": 0,  
"inVdnTime": 0,  
"callType": 6,  
"maxQueueWaitTime": 0,  
"succWaitAnsNum": 0,  
"ivrOccupyNum": 0,  
"beginLogDay": "2020-08-24",  
"autoUserAbandonNum": 0,  
"callTimeLt60": 2,  
"maxCallTime": 5702,  
"failTimeLt60": 0,  
"hangUpTime": 0,  
"failTimeLt10": 0,  
"succAckTimeGt10": 0,  
"minWaitAnsTime": 2,  
"autoAnsTime": 0,  
"ivrOnlySuccNum": 0,  
"autoCallSuccNum": 0,  
"hangUpNum": 0,  
"callTimeLt10": 1,  
"currentSkillId": -1,  
"failQueueWaitNum": 0,  
"minQueueWaitTime": 0,  
"succQueueWaitNum": 0,  
"inOccupyNum": 7,  
"maxIvrCallTime": 0,  
"currentAgentId": 1133,  
"ivrOnlyTime": 0,  
"subCcNo": 1,  
"succAckTimeLt30": 7,  
"succAckTimeGt60": 0,  
"serviceNo": 65535,  
"maxWaitAnsTime": 3,  
"netCallSuccNum": 0,  
"failTimeLt5": 0,  
"ivrOnlyNum": 0,  
"minCallTime": 5,  
"failTimeLt3": 0,  
"inCallSuccNum": 7,  
"queOverToIvrTime": 0,  
"autoCallNum": 0,
```

```

    "failWaitAnsTime": 0,
    "succAckTimeLt60": 7,
    "queueWaitTime": 0,
    "hangUpReleaseNum": 0,
    "endLogDay": "2020-08-25",
    "ivrCallTime": 0,
    "orgCcNo": 1,
    "calleeNo": "1134",
    "failTimeLt30": 0,
    "failTimeGt60": 0,
    "succWaitAnsTime": 20,
    "callTimeGt60": 5,
    "callTimeLt5": 0,
    "netCallNum": 0,
    "callTimeLt30": 1,
    "callTimeLt3": 0,
    "serCcNo": 1,
    "succAckTimeLt5": 7,
    "ivrCallSuccNum": 0,
    "succQueueWaitTime": 0,
    "agentToAgentNum": 0,
    "failQueueWaitTime": 0,
    "succAckTimeLt10": 7,
    "noAckNum": 0,
    "waitAnsNum": 7,
    "waitAnsTime": 20,
    "succAckTimeLt3": 1,
    "autoTransferNum": 0,
    "inCallCostMinute": 10,
    "reserved_col1": 0,
    "inCallAllCostMinute": 1,
    .....
  },
  {.....}
],
"resultDesc": [
  {
    "callTimeGt10": 6,
    "vdn": 152,
    "minIvrCallTime": 0,
    "queOverToIvrNum": 0,
    "failTimeGt10": 0,
    "inCallTime": 6584,

```

```
"agentToQueueNum": 0,
"inVdnTime": 0,
"callType": 6,
"maxQueueWaitTime": 0,
"succWaitAnsNum": 0,
"ivrOccupyNum": 0,
"beginLogDay": "2020-08-24",
"autoUserAbandonNum": 0,
"callTimeLt60": 2,
"maxCallTime": 5702,
"failTimeLt60": 0,
"hangUpTime": 0,
"failTimeLt10": 0,
"succAckTimeGt10": 0,
"minWaitAnsTime": 2,
"autoAnsTime": 0,
"ivrOnlySuccNum": 0,
"autoCallSuccNum": 0,
"hangUpNum": 0,
"callTimeLt10": 1,
"currentSkillId": -1,
"failQueueWaitNum": 0,
"minQueueWaitTime": 0,
"succQueueWaitNum": 0,
"inOccupyNum": 7,
"maxIvrCallTime": 0,
"currentAgentId": 1133,
"ivrOnlyTime": 0,
"subCcNo": 1,
"succAckTimeLt30": 7,
"succAckTimeGt60": 0,
"serviceNo": 65535,
"maxWaitAnsTime": 3,
"netCallSuccNum": 0,
"failTimeLt5": 0,
"ivrOnlyNum": 0,
"minCallTime": 5,
"failTimeLt3": 0,
"inCallSuccNum": 7,
"queOverToIvrTime": 0,
"autoCallNum": 0,
"failWaitAnsTime": 0,
"succAckTimeLt60": 7,
```

```

    "queueWaitTime": 0,
    "hangUpReleaseNum": 0,
    "endLogDay": "2020-08-25",
    "ivrCallTime": 0,
    "orgCcNo": 1,
    "calleeNo": "1134",
    "failTimeLt30": 0,
    "failTimeGt60": 0,
    "succWaitAnsTime": 20,
    "callTimeGt60": 5,
    "callTimeLt5": 0,
    "netCallNum": 0,
    "callTimeLt30": 1,
    "callTimeLt3": 0,
    "serCcNo": 1,
    "succAckTimeLt5": 7,
    "ivrCallSuccNum": 0,
    "succQueueWaitTime": 0,
    "agentToAgentNum": 0,
    "failQueueWaitTime": 0,
    "succAckTimeLt10": 7,
    "noAckNum": 0,
    "waitAnsNum": 7,
    "waitAnsTime": 20,
    "succAckTimeLt3": 1,
    "autoTransferNum": 0,
    "inCallCostMinute": 10,
    "reserved_col1": 0,
    "inCallAllCostMinute": 1,
    .....
  },
  {.....}
]
}

```

4.2.11 Collecting Statistics on Result Sets in Daily Settlement Table for Outbound Calls

Interface Function

This interface is invoked to collect statistics on result sets in the daily settlement table for outbound calls based on conditions.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Request URL

`https://ip:port/rest/cmsapp/v1/calloutday/querymanualindexesbycondition`

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the message body

Parameter	Value Type	Mandatory	Description
callerNos	Array	Yes	Array of calling numbers. Statistics are collected by each number in the array. A calling number can contain only digits. The array contains a maximum of 1,000 characters.
beginLogDay	String	Yes	Start call time for statistics collection. The value must be in the <i>yyyy-MM-dd</i> format and in the time zone of the tenant.
endLogDay	String	Yes	End call time for statistics collection. The value must be in the <i>yyyy-MM-dd</i> format and in the time zone of the tenant.
callTypes	Array	Yes	Call type array of the integer type. Statistics are collected based on each type in the array. The call type is only the digit. The array contains a maximum of 1,000 characters.
skillIds	Array	No	Skill queue IDs. An array of integers. Statistics are collected by type in the array. The array contains a maximum of 1,000 characters.
subCcNo	Integer	Either this parameter or	Subsystem number.

Table 2 Parameters in the message body

Parameter	Value Type	Mandatory	Description
		tenantId must be set.	
vdn	Integer	Either this parameter or tenantId must be set.	ID of a VCC.
tenantId	String	Either this parameter or subCcNo (or vdn) must be set.	Tenant ID.
ccId	Integer	No	ID of a call center. In the integration environment, the default value is the value of Call Center ID on the Configuration Center > System Management > CMS System Configuration page after the system administrator signs in to the AICC. If this parameter is empty, the data whose sys_key is default_cc_id and sys_type is cms_system_config in the t_cms_sysparam database table is used. In the independent deployment environment, the default value is the value of Default Call Center ID on the Configuration Center > System Management > Parameter Settings page after the system administrator signs in to the AICC.

Response Description

Table 3 Parameters in the message body

Parameter	Value Type	Description
result	String	Query result. The options are as follows: 0: success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
resultCode	String	Query result. The options are as follows: 0100000: success Other values: failure For details about the failure causes, see Error Code Reference .

Table 3 Parameters in the message body

Parameter	Value Type	Description
resultDatas	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
resultDesc	Array	Object in the returned result after a successful query. For details about the parameters of the object, see Table 4 .

Table 4 Parameters of resultDesc/resultDatas

Parameter	Value Type	Description
beginLogDay	String	Start call time, which is the same as the input parameter.
endLogDay	String	End call time, which is the same as the input parameter.
callerNo	String	Calling number.
callType	Integer	Call type.
vdn	Integer	ID of a VCC.
serviceNo	Integer	Business type. NOTE: This parameter has no actual meaning. Do not use it.
orgCcNo	Integer	Subsystem number of the original call center.
subCcNo	Integer	Subsystem number.
serCcNo	Integer	Subsystem number of the service party.
currentAgentId	Integer	Outbound agent ID. Outbound agent ID in the last CDR of a call, which is not recommended for indicator settlement.
currentSkillId	Integer	Skill queue ID. Skill queue ID in the last CDR of a call, which is not recommended for indicator settlement.
obsServiceNo	String	ID of an outbound campaign.
failQueueWaitNum	Integer	Number of waits for lost calls.
failQueueWaitTime	Integer	Waiting duration of lost calls, in seconds.
outOccupyNum	Integer	Number of manual outbound calls.

Table 4 Parameters of resultDesc/resultDatas

Parameter	Value Type	Description
outSuccWaitAnsNum	Integer	Number of manual outbound calls that are answered.
outCallSuccNum	Integer	Number of successful manual outbound calls.
outWaitAnsTime	Integer	Ring duration for outbound calls, in seconds.
outCallTime	Integer	Outbound call duration, in seconds.
maxWaitAnsTime	Integer	Maximum ring duration, in seconds.
minWaitAnsTime	Integer	Minimum ring duration, in seconds.
maxCallTime	Integer	Maximum call duration, in seconds.
minCallTime	Integer	Minimum call duration, in seconds.
succAckTimeLt3	Integer	Number of successful calls whose ring duration is shorter than 3 seconds.
callTimeLt3	Integer	Number of successful calls whose call duration is shorter than 3 seconds.
failTimeLt3	Integer	Number of failed calls whose waiting duration is shorter than 3 seconds.
succAckTimeLt5	Integer	Number of successful calls whose ring duration is shorter than 5 seconds.
callTimeLt5	Integer	Number of successful calls whose call duration is shorter than 5 seconds.
failTimeLt5	Integer	Number of failed calls whose waiting duration is shorter than 5 seconds.
succAckTimeLt10	Integer	Number of successful calls whose ring duration is shorter than 10 seconds.
callTimeLt10	Integer	Number of successful calls whose call duration is shorter than 10 seconds.
failTimeLt10	Integer	Number of failed calls whose waiting duration is shorter than 10 seconds.
succAckTimeLt30	Integer	Number of successful calls whose ring duration is shorter than 30 seconds.
callTimeLt30	Integer	Number of successful calls whose call duration is shorter than 30 seconds.
failTimeLt30	Integer	Number of failed calls whose waiting duration is shorter than 30 seconds.

Table 4 Parameters of resultDesc/resultDatas

Parameter	Value Type	Description
succAckTimeLt60	Integer	Number of successful calls whose ring duration is shorter than 60 seconds.
callTimeLt60	Integer	Number of successful calls whose call duration is shorter than 60 seconds.
failTimeLt60	Integer	Number of failed calls whose waiting duration is shorter than 60 seconds.
succAckTimeGt10	Integer	Number of successful calls whose ring duration is longer than 10 seconds.
callTimeGt10	Integer	Number of successful calls whose call duration is longer than 10 seconds.
failTimeGt10	Integer	Number of failed calls whose waiting duration is longer than 10 seconds.
succAckTimeGt60	Integer	Number of successful calls whose ring duration is longer than 60 seconds.
callTimeGt60	Integer	Number of successful calls whose call duration is longer than 60 seconds.
failTimeGt60	Integer	Number of failed calls whose waiting duration is longer than 60 seconds.
agentHangUpLt3	Integer	Number of calls released by calling parties in less than 3 seconds.
preAgentSuccNum	Integer	Number of predicted outbound calls that are forwarded to agents.
preViewedTransferNum	Integer	Number of previewed outbound calls that are transferred by an agent.
hangUpTime	Integer	Total duration of manual-to-IVR calls.
preOccupiedSuccAckTime	Integer	Waiting duration before a preempted outbound call is connected, in seconds.
autoAnsTime	Integer	System-level IVR outbound call ringing duration, in seconds.
autoCallSuccNum	Integer	Number of system-level IVR outbound calls.
preOccupiedSuccNum	Integer	Waiting duration before a preempted outbound call is connected, in seconds.
hangUpNum	Integer	Number of manual-to-IVR calls.
preAgentNum	Integer	Number of predicted outbound calls that are forwarded to agents.
zeroNum	Integer	Number of calls with zero CDRs.

Table 4 Parameters of resultDesc/resultDatas

Parameter	Value Type	Description
preQueueWaitTime	Integer	Waiting duration of a predicted outbound call in a queue, in seconds.
preQueueWaitNum	Integer	Number of queuing times for predicted outbound calls.
preViewedCallSuccNum	Integer	Number of previewed outbound calls that are connected.
preOccupiedCallNum	Integer	Total number of preempted outbound calls.
preOccupiedAgentSuccNum	Integer	Number of preempted outbound calls that are answered by agents.
agentTransferNum	Integer	Number of manual outbound calls that are transferred out.
autoCallNum	Integer	Number of system-level IVR outbound calls.
hangUpReleaseNum	Integer	Number of manual-to-IVR calls that are released.
preTransferNum	Integer	Number of predicted outbound calls that are transferred out.
preTotalTime	Integer	Total duration of predicted outbound calls, in seconds.
preFailQueueWaitNum	Integer	Number of queuing times for lost predicted outbound calls.
preViewedSuccAckTime	Integer	Waiting duration before a previewed outbound call is connected, in seconds.
preFailQueueWaitTime	Integer	Waiting duration of a lost predicted outbound call in a queue, in seconds.
preAbortQueueWaitNum	Integer	Number of queuing times for predicted outbound calls that are released.
preNoAckNum	Integer	Number of predicted outbound calls that are answered by users but not answered by agents.
succWaitAnsTime	Integer	Duration of answering a successful call, in seconds.
outVdnTime	Integer	Total duration of VCC occupation by outbound calls, in seconds.
preViewedCallNum	Integer	Total number of previewed outbound calls.
preCallSuccNum	Integer	Number of predicted outbound calls that are connected.
preOccupiedTransferNum	Integer	Number of preempted outbound calls that are transferred by an agent.
preCallNum	Integer	Number of predicted outbound calls.
preOccupiedAgentSuccTime	Integer	Manual call duration of preempted outbound calls, in seconds.
autoTransferNum	Integer	Number of system-level IVR outbound calls that are transferred out.
outCallCostMinute	Integer	Charging duration of a manual call, in minutes.

Table 4 Parameters of resultDesc/resultDatas

Parameter	Value Type	Description
ivrCallSuccNum	Integer	Number of IVR outbound calls.
ivrCallTime	Integer	IVR outbound call duration, in seconds.
ivrOccupyNum	Integer	Number of IVR outbound call requests.
ivrOnlyNum	Integer	Number of outbound calls connected using only the IVR.
ivrOnlySuccNum	Integer	Number of successful calls that are connected using only the IVR.
ivrOnlyTime	Integer	Duration of outbound calls connected using only the IVR, in seconds.
outCallAllCostMinute	Integer	Total charging duration of each call, in minutes, including the charging duration of manual calls and IVR calls.
reserved_col1- reserved_col20	Integer	Reserved field. The service meaning is determined based on the configuration in the t_cms_custom_indicator_config table. By default, the reserved field 3 indicates the number of outbound calls whose call duration is longer than or equal to 15 seconds.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "queryParam": {
    "callerNos": ["120002"],
    "beginLogDay": "2021-10-26",
    "endLogDay": "2021-10-26",
    "callTypes": [7],
    "skillIds": [1,6],
    "subCcNo": 1,
    "vdn": 32
  }
}
```

- Response parameters

```
{
  "result": "0",
  "resultCode": "0100000",
```

```
"resultDatas": [  
  {  
    "failTimeGt10": 0,  
    "outSuccWaitAnsNum": 2,  
    "callType": 7,  
    "callerNo": "120002",  
    "maxCallTime": 38,  
    "callTimeLt60": 2,  
    "preViewedTransferNum": 0,  
    "failTimeLt10": 0,  
    "succAckTimeGt10": 1,  
    "minWaitAnsTime": 1,  
    "autoCallSuccNum": 0,  
    "hangUpNum": 0,  
    "preOccupiedSuccNum": 0,  
    "preAgentNum": 0,  
    "zeroNum": 0,  
    "subCcNo": 1,  
    "ivrOnlyTime": 0,  
    "succAckTimeLt30": 2,  
    "serviceNo": 1,  
    "maxWaitAnsTime": 15,  
    "obsServiceNo": "-1",  
    "failTimeLt5": 0,  
    "reservedCol3": 2,  
    "ivrOnlyNum": 0,  
    "reservedCol2": 0,  
    "failTimeLt3": 0,  
    "reservedCol1": 0,  
    "reservedCol7": 0,  
    "reservedCol6": 0,  
    "reservedCol5": 0,  
    "reservedCol4": 0,  
    "preOccupiedAgentSuccNum": 0,  
    "autoCallNum": 0,  
    "hangUpReleaseNum": 0,  
    "preTotalTime": 0,  
    "preFailQueueWaitTime": 0,  
    "preAbortQueueWaitNum": 0,  
    "failTimeGt60": 0,  
    "orgCcNo": 1,  
    "succWaitAnsTime": 16,  
    "callTimeGt60": 0,  
  }  
]
```

```
"outVdnTime": 105,  
"callTimeLt5": 0,  
"callTimeLt3": 0,  
"succAckTimeLt5": 1,  
"ivrCallSuccNum": 0,  
"preCallNum": 0,  
"succAckTimeLt10": 1,  
"outCallSuccNum": 2,  
"succAckTimeLt3": 1,  
"callTimeGt10": 2,  
"vdn": 32,  
"preAgentSuccNum": 0,  
"reservedCol10": 0,  
"outOccupyNum": 2,  
"ivrOccupyNum": 0,  
"beginLogDay": "2021-10-26",  
"reservedCol18": 0,  
"reservedCol17": 0,  
"reservedCol16": 0,  
"reservedCol15": 0,  
"reservedCol14": 0,  
"reservedCol13": 0,  
"reservedCol12": 0,  
"failTimeLt60": 0,  
"reservedCol11": 0,  
"hangUpTime": 0,  
"preOccupiedSuccAckTime": 0,  
"ivrOnlySuccNum": 0,  
"reservedCol19": 0,  
"autoAnsTime": 0,  
"callTimeLt10": 0,  
"outWaitAnsTime": 16,  
"currentSkillId": 1,  
"failQueueWaitNum": 0,  
"reservedCol20": 0,  
"reservedCol9": 0,  
"reservedCol8": 0,  
"outCallCostMinute": 2,  
"currentAgentId": 40002,  
"succAckTimeGt60": 0,  
"agentHangUpLt3": 0,  
"preQueueWaitTime": 0,  
"preQueueWaitNum": 0,
```



```

    "preViewedCallSuccNum": 0,
    "minCallTime": 26,
    "preOccupiedCallNum": 0,
    "agentTransferNum": 1,
    "succAckTimeLt60": 2,
    "preTransferNum": 0,
    "preViewedSuccAckTime": 0,
    "preFailQueueWaitNum": 0,
    "endLogDay": "2021-10-26",
    "preNoAckNum": 0,
    "ivrCallTime": 0,
    "failTimeLt30": 0,
    "preViewedCallNum": 0,
    "callTimeLt30": 1,
    "serCcNo": 1,
    "preCallSuccNum": 0,
    "outCallTime": 64,
    "preOccupiedTransferNum": 0,
    "failQueueWaitTime": 0,
    "preOccupiedAgentSuccTime": 0,
    "outCallAllCostMinute": 1,
    "autoTransferNum": 0
  },
  {.....}
],
"resultDesc": [
  {
    "failTimeGt10": 0,
    "outSuccWaitAnsNum": 2,
    "callType": 7,
    "callerNo": "120002",
    "maxCallTime": 38,
    "callTimeLt60": 2,
    "preViewedTransferNum": 0,
    "failTimeLt10": 0,
    "succAckTimeGt10": 1,
    "minWaitAnsTime": 1,
    "autoCallSuccNum": 0,
    "hangUpNum": 0,
    "preOccupiedSuccNum": 0,
    "preAgentNum": 0,
    "zeroNum": 0,
    "subCcNo": 1,

```

```
"ivrOnlyTime": 0,  
"succAckTimeLt30": 2,  
"serviceNo": 1,  
"maxWaitAnsTime": 15,  
"obsServiceNo": "-1",  
"failTimeLt5": 0,  
"reservedCol3": 2,  
"ivrOnlyNum": 0,  
"reservedCol2": 0,  
"failTimeLt3": 0,  
"reservedCol1": 0,  
"reservedCol7": 0,  
"reservedCol6": 0,  
"reservedCol5": 0,  
"reservedCol4": 0,  
"preOccupiedAgentSuccNum": 0,  
"autoCallNum": 0,  
"hangUpReleaseNum": 0,  
"preTotalTime": 0,  
"preFailQueueWaitTime": 0,  
"preAbortQueueWaitNum": 0,  
"failTimeGt60": 0,  
"orgCcNo": 1,  
"succWaitAnsTime": 16,  
"callTimeGt60": 0,  
"outVdnTime": 105,  
"callTimeLt5": 0,  
"callTimeLt3": 0,  
"succAckTimeLt5": 1,  
"ivrCallSuccNum": 0,  
"preCallNum": 0,  
"succAckTimeLt10": 1,  
"outCallSuccNum": 2,  
"succAckTimeLt3": 1,  
"callTimeGt10": 2,  
"vdn": 32,  
"preAgentSuccNum": 0,  
"reservedCol10": 0,  
"outOccupyNum": 2,  
"ivrOccupyNum": 0,  
"beginLogDay": "2021-10-26",  
"reservedCol18": 0,  
"reservedCol17": 0,
```

```
"reservedCol16": 0,  
"reservedCol15": 0,  
"reservedCol14": 0,  
"reservedCol13": 0,  
"reservedCol12": 0,  
"failTimeLt60": 0,  
"reservedCol11": 0,  
"hangUpTime": 0,  
"preOccupiedSuccAckTime": 0,  
"ivrOnlySuccNum": 0,  
"reservedCol19": 0,  
"autoAnsTime": 0,  
"callTimeLt10": 0,  
"outWaitAnsTime": 16,  
"currentSkillId": 1,  
"failQueueWaitNum": 0,  
"reservedCol20": 0,  
"reservedCol9": 0,  
"reservedCol8": 0,  
"outCallCostMinute": 2,  
"currentAgentId": 40002,  
"succAckTimeGt60": 0,  
"agentHangUpLt3": 0,  
"preQueueWaitTime": 0,  
"preQueueWaitNum": 0,  
"preViewedCallSuccNum": 0,  
"minCallTime": 26,  
"preOccupiedCallNum": 0,  
"agentTransferNum": 1,  
"succAckTimeLt60": 2,  
"preTransferNum": 0,  
"preViewedSuccAckTime": 0,  
"preFailQueueWaitNum": 0,  
"endLogDay": "2021-10-26",  
"preNoAckNum": 0,  
"ivrCallTime": 0,  
"failTimeLt30": 0,  
"preViewedCallNum": 0,  
"callTimeLt30": 1,  
"serCcNo": 1,  
"preCallSuccNum": 0,  
"outCallTime": 64,  
"preOccupiedTransferNum": 0,
```

```

    "failQueueWaitTime": 0,
    "preOccupiedAgentSuccTime": 0,
    "outCallAllCostMinute": 1,
    "autoTransferNum": 0
  },
  {...}
]
}

```

4.2.12 Obtaining Agent Rest Monitoring Indicators of the Current Day or the Previous Day

Scenario

This interface is invoked to obtain agent rest monitoring indicators of the current day or the previous day.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Request URL

`https://ip.port/rest/cmsapp/v1/openapi/hisindex/agentbreak`

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the request body

Parameter	Type	Mandatory	Description
cclid	Integer	No	Call center ID. The value ranges from 1 to 65535.

Table 2 Parameters in the request body

Parameter	Type	Mandatory	Description
			In the integration environment, the default value is the value of Call Center ID on the Configuration Center > System Management > CMS System Configuration page after the system administrator signs in to the AICC. If the value is empty, the data whose sys_key is default_cc_id and sys_type is cms_system_config in the t_cms_sysparam database table is used. In the independent deployment environment, the default value is the value of Default Call Center ID on the Configuration Center > System Management > Parameter Settings page after the system administrator signs in to the AICC.
vdn	Integer	Yes	ID of a VCC. The value ranges from 1 to 5000.
agentIds	List<Integer>	Yes	Array of agent IDs. The array contains a maximum of 100 characters. The agent ID is an integer ranging from 0 to 999999.
timeGrain	String	No	This parameter can be used to collect statistics on historical monitoring indicators by minute. The options are 15MIN , 30MIN , and 60MIN . If this parameter is not passed, statistics on the current day in the time zone where the tenant is located are collected.
enableOffset	String	No	The options are Y and N . If Y is transferred, the data of the previous day is queried. If N is transferred, the data of the current day is queried. If no value is transferred, the default value N is used.

Response Message

Table 3 Parameters in the message body

Parameter	Type	Description
resultCode	String	Query result. The options are as follows: 0100000: success Other values: failure For details about the failure causes, see Error Code Reference .
resultDesc	List<JSONObject>	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 .

Table 4 Parameters of resultDesc

Parameter	Type	Description
agentId	String	Agent ID.
restData	List<JSONObject>	Rest cause statistics.
restCause	String	Rest cause code.

Table 4 Parameters of resultDesc

Parameter	Type	Description
restCount	JSONObject	Key-value pair of the time segment (tenant time zone) and the number of rest times.
restDuration	JSONObject	Key-value pair of the time segment (tenant time zone) and the rest duration.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "ccId":1,
  "vdn":34,
  "agentIds":[295],
  "timeGrain":"15MIN"
}
```

- Response parameters

```
{
  "resultCode": "0100000",
  "resultDesc": [
    {
      "agentId": 295
      "restData": [
        {
          "restDuration": {
            "2020-08-24 10:30:00.0": "15",
            "2020-08-24 10:45:00.0": "19"
          },
          "restCount": {
            "2020-08-24 10:30:00.0": "15",
            "2020-08-24 10:45:00.0": "19"
          },
          "restCause": "15"
        },
        {
          "restDuration": {
```

```

        "2020-08-24 10:30:00.0": "15",
        "2020-08-24 10:45:00.0": "19"
    },
    "restCount": {
        "2020-08-24 10:30:00.0": "15",
        "2020-08-24 10:45:00.0": "19"
    },
    "restCause": "2"
}
]
}

```

4.3 Configuration Data Query

4.3.1 Querying Information about All VDNs in a Specified Call Center

Scenario

This interface is invoked to query information about all VDNs in a specified call center.

Note: This interface supports only the independent deployment environment and can be invoked only by the global AK and SK.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Request URL

`https://ip:port/rest/cmsapp/v1/openapi/configures/vdns`

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameter in the request body

No.	Parameter	Value Type	Mandatory	Description
1	cclid	Integer	Yes	ID of a call center. The value ranges from 1 to 65535.

Response Description

Table 3 Parameters in the response body

No.	Parameter	Value Type	Description
1	result	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
2	resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
3	resultDatas	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
4	resultDesc	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 .

Table 4 Parameters of resultDesc/resultDatas

No.	Parameter	Value Type	Description
2.1	vdn	Integer	ID of a VDN.
2.2	vdnName	String	VDN name.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
```

```
Authorization: *****
```

- Request parameters

```
{
  "queryParam":
  {
    "ccId":1
  }
}
```

- Response parameters

```
{
  "result": "0",
  "resultCode": "0100000",
  "resultDatas": [
    {
      "vdn": 1,
      "vdnName": "social cc"
    },
    {
      "vdn": 2,
      "vdnName": "Credit card"
    },
    {
      "vdn": 3,
      "vdnName": "95558_2"
    },
    {
      "vdn": 4,
      "vdnName": "VDN4"
    },
    {
```

```

        "vdn": 5,
        "vdnName": "Test VDN"
    }
],
"resultDesc": [
    {
        "vdn": 1,
        "vdnName": "social cc"
    },
    {
        "vdn": 2,
        "vdnName": "Credit card"
    },
    {
        "vdn": 3,
        "vdnName": "95558_2"
    },
    {
        "vdn": 4,
        "vdnName": "VDN4"
    },
    {
        "vdn": 5,
        "vdnName": "Test VDN"
    }
]
}

```

4.3.2 Querying Information About All Agents by VDN ID

Scenario

This interface is invoked to query information about all agents by VDN ID. A tenant corresponds to a VDN after being created and configured.

Interface Method

The request method must be set to GET. This interface supports only the GET method.

Internal Encapsulation Interface

BMS interface: /ccbms/ws/monitor/siteagentcallandskillinfoobyagentids

Request URL

`https://ip:port/rest/cmsapp/v1/openapi/configures/agents/{ccld}/{vdn}/{offset}/{limit}`

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server. *{ccld}* indicates the call center ID. The value ranges from 1 to 65535. *{vdn}* indicates the VDN ID. The value ranges from 1 to 5000. *{offset}* indicates the query start offset and is optional. *{limit}* indicates the number of query records on each page and is optional. The maximum value is **100**. *{offset}* and *{limit}* take effect only when they are both passed.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Response Description

Table 2 Parameters in the response message body

Parameter	Value Type	Description
result	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .

Table 2 Parameters in the response message body

Parameter	Value Type	Description
count	Integer	Total number of agents. This parameter is returned only when <i>{offset}</i> and <i>{limit}</i> in the request take effect.
resultDatas	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 3 . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
resultDesc	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 3 .

Table 3 Parameters of resultDesc/resultDatas

Parameter	Value Type	Description
agentNo	String	Agent ID.
agentName	String	Agent name.
roleId	String	Agent role.
skillId	String	Skill ID set. Multiple skill IDs are separated by semicolons (;), for example, 1;3;10;11;25;31;32;40 .

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

None

- Response parameters

```
{
  "result": "0",
  "resultCode": "0100000",
  "count": 101,
  "resultDatas": [
    {
      "agentNo": "10000",
      "agentName": "z1",
```

```

        "roleId": "1",
        "skillId": ""
    },
    {
        "agentNo": "101",
        "agentName": "z2",
        "roleId": "1",
        "skillId": "2;3"
    }
],
"resultDesc": [
    {
        "agentNo": "10000",
        "agentName": "z1",
        "roleId": "1",
        "skillId": ""
    },
    {
        "agentNo": "101",
        "agentName": "z2",
        "roleId": "1",
        "skillId": "2;3"
    }
]
}

```

4.3.3 Obtaining All Called Routes in a VDN

Scenario

This interface is invoked to obtain all called routes in a VDN.

Interface Method

The request method must be set to GET. This interface supports only the GET method.

Request URL

<https://ip:port/rest/cmsapp/v1/openapi/configures/calledconfigs/{ccId}/{vdn}>

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server. *{ccld}* indicates the call center ID. The value ranges from 1 to 65535. *{vdn}* indicates the VDN ID. The value ranges from 1 to 5000.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Response Description

Table 2 Parameters in the response message body

No.	Parameter	Value Type	Description
1	result	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
2	resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
3	resultDatas	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 3 . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
4	resultDesc	Array	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see Table 3 .

Table 3 Parameters of resultDesc/resultDatas

No.	Parameter	Value Type	Description
2.1	cclId	Integer	ID of a call center.
2.2	vdnId	Integer	VDN ID.
2.3	accessCode	String	Access code.
2.4	subCode	String	Subsequent access code. The value contains a maximum of 100 characters. The value can contain 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, asterisks (*), and number signs (#).
2.5	deviceType	String	Device type. For details, see Table 1 . The returned value is a 4-byte integer.
2.6	deviceName	String	Device name. If deviceType is set to 1 , indicating a skill queue, the value is the skill queue name. When deviceType is set to 3 , indicating an IVR automatic flow, the value is the flow description.
2.7	deviceNo	String	Device ID.
2.8	dnid	String	Configuration ID.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

None

- Response parameters

```
{
  "result": "0",
  "resultCode": "0100000",
  "resultDatas": [
    {
      "cclId": 1,
      "vdnId": 1,
      "accessCode": "6045",
      "subCode": "1110",
      "deviceType": "1",
```

```
    "deviceName": "test",
    "deviceNo": "22",
    "dnid": "22"
  },
  ],
  "resultDesc": [
    {
      "ccId": 1,
      "vdnId": 1,
      "accessCode": "6045",
      "subCode": "1110",
      "deviceType": "1",
      "deviceName": "test",
      "deviceNo": "22",
      "dnid": "22"
    }
  ]
}
```

4.3.4 Querying Information About All Agents by Tenant Space ID

Scenario

This interface is invoked to query information about all agents of a tenant based on the tenant space ID.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Request URL

`https://ip:port/rest/cmsapp/v1/openapi/tenant/querytenantdetail`

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the message body

Parameter	Type	Mandatory	Description
tenantId	String	Yes	Tenant space ID.
ccId	Integer	No	ID of a call center. In the integration environment, the default value is the value of Call Center ID on the Configuration Center > System Management > CMS System Configuration page after the system administrator signs in to the AICC. If this parameter is empty, the data whose sys_key is default_cc_id and sys_type is cms_system_config in the t_cms_sysparam database table is used. In the independent deployment environment, the default value is the value of Default Call Center ID on the Configuration Center > System Management > Parameter Settings page after the system administrator signs in to the AICC.
limit	Integer	No	Number of query records on each page. The value ranges from 1 to 10000. The default value is 10000. NOTE: The pagination applies only to the response parameter agents. Currently, the pagination is supported only in the integration environment.
offset	Integer	No	offset is used to determine the start page of the query. Value of offset = Value of limit x (Value of page – 1). The value of offset is greater than or equal to 0. The default value is 0. page indicates the start page to be queried. NOTE: The pagination applies only to the response parameter agents. Currently, the pagination is supported only in the integration environment.

Response Description

Table 3 Parameters in the response message body

Parameter	Type	Description
resultCode	String	Query result. The options are as follows: 0100000: success Other values: failure

Table 3 Parameters in the response message body

Parameter	Type	Description
		For details about the failure causes, see Error Code Reference .
resultDesc	Object	Object in the returned result after a successful query. For details about the parameters of this object, see Table 4 .

Table 4 Parameters of resultDesc

Parameter	Type	Description
cclId	Integer	ID of a call center.
tenantId	String	Tenant space ID.
tenantSpaceName	String	Tenant space name.
vdnId	Integer	ID of a VDN.
accessCodes	Array	Access number list. For details about the parameters of the object in this array, see Table 4 Parameters of accessCodes .
agents	Array	Agent list. For details about the parameters of agents , see Table 5 Parameters of agent .

Table 5 Parameters of accessCodes

Parameter	Type	Description
accessCode	String	Access code.
isDefault	Boolean	Default access code ID. This parameter can be set to true , false , or left empty.

Table 6 Parameters of agents

Parameter	Type	Description
workNo	Integer	Agent ID.
sipAccount	String	Softphone number.
accountCode	String	Agent account.
agentType	Integer	Agent type. 4 : voice agent; 5 : multimedia agent; 11 : video agent. This field is returned only during integrated deployment.

Message Example

- Request header

Content-Type: application/json;charset=UTF-8

Authorization: *****

- Request parameters

```
{
  "ccId": 1,
  "tenantId": "202306131208",
  "limit": 10000,
  "offset": 0
}
```

- Response parameters

```
{
  "resultCode": "0100000",
  "resultDesc": {
    "ccId": 3,
    "vdnId": 47,
    "tenantId": "202008142460",
    "accessCodes": [
      {
        "accessCode": "202008191",
        "isDefault": true
      }
    ],
    "tenantSpaceName": "fjw081401",
    "agents": [{
      "workNo": 111,
      "sipAccount": "88880001",
      "accountCode": "123",
      "agentType": 4
    }]
  }
}
```

4.3.5 Querying Skill Groups in a Specified VDN

Description

This interface is invoked to query the information about the skill groups in a specified VDN.

Note: This interface supports only the independent deployment environment and CTI 3.8.

Interface Method

The request method must be set to GET. This interface supports only the GET method.

Request URL

`https://ip:port/rest/cmsapp/v1/openapi/skillgroups/{ccId}/{vdnId}`

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed to application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

No request body exists.

Response Message

Table 2 Parameters in the message body

No.	Parameter	Value Type	Description
1	resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
2	resultDesc	Array	Object array in the returned result after a successful query. For details about the parameters of this object in the array, see Table 3 .

Table 3 Parameters of resultDesc

No.	Parameter	Value Type	Description
2.1	id	Integer	Skill group ID.
2.2	skills	Array	Skill object array. The array contains skill objects. Table 4 describes the definition of a single skill object.
2.3	groupName	String	Skill group name. The value contains a maximum of 64 bytes without any special characters.

Table 4 Parameters of skills

No.	Parameter	Value Type	Description
2.2.1	skillId	Integer	Skill queue ID. The value is an integer. Enter an actual skill queue ID.
2.2.2	skillPower	Integer	Skill power corresponding to the skill queue. The value is an integer ranging from 1 to 100.
2.2.3	agentPower	Integer	Agent power corresponding to the skill queue. The value is an integer ranging from 1 to 300.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

None

- Response parameters

```
{
  "resultCode": "0100000",
  "resultDesc": [
    {
      "id": 2,
      "skills": [
        {
          "skillId": 2,
          "skillPower": 1,
          "agentPower": 1
        }
      ],
      "groupName": "skillgroup2"
    }
  ]
}
```

4.3.6 Querying Information About All Tenants Accessible to an Access User

Description

This interface is invoked by an access user to query information about all accessible tenants based on the AK/SK.

NOTE

This interface supports only the private AK/SK.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Request URL

`https://ip:port/rest/cmsapp/v1/openapi/tenant/querytenants`

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed to application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the message body

Parameter	Type	Mandatory	Description
ccld	Integer	No	Call center ID. The value ranges from 1 to 65535. In the integration environment, the default value is the value of Call Center ID on the Configuration Center > System Management > CMS System Configuration page after a system administrator signs in to the AICC. If this parameter is empty, the data whose sys_key is default_cc_id and sys_type is cms_system_config in the t_cms_sysparam database table is used. In the independent deployment environment, the default value is the value of Default Call Center ID on the Configuration Center > System Management > Parameter Settings page after a system administrator signs in to the AICC.

Response Message

Table 3 Parameters in the response message body

Parameter	Type	Description
resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
resultDesc	Array	Object array in the returned result after a successful query. For details about the parameters of this object in the array, see Table 4 .

Table 4 Parameters of resultDesc

Parameter	Type	Description
cclId	Integer	ID of a call center.
tenantId	String	Tenant space ID.
tenantSpaceName	String	Tenant space name.
vdnId	Integer	ID of a VDN.
accessCodes	Array	Access number list. For details about the parameters of this object in the array, see Table 5 .

Table 5 Parameters of accessCodes

Parameter	Type	Description
accessCode	String	Access code.
isDefault	Boolean	Default access code ID. This parameter can be set to true , false , or left empty.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "cclId": 1
}
```

- Response parameters

```
{
  "resultCode": "0100000",
```

```

"resultDesc": [
  {
    "ccld": 3,
    "vdnId": 47,
    "tenantId": "202008142460",
    "accessCodes": [
      {
        "accessCode": "202008191",
        "isDefault": true
      }
    ],
    "tenantSpaceName": "fjw081401"
  }
]
}

```

4.4 Inspection Relationship Maintenance Interfaces

Note: The following interfaces support only the scenario where the CC-CMS is independently deployed.

4.4.1 Querying a User Group Member

Scenario

This interface is invoked to query members in a specified user group based on **vdn**, **groupId**, and **workNo**.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Request URL

`https://ip:port/rest/cmsapp/v1/openapi/qcrelationconf/querygroupmemberbycondition`

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the message body

Parameter	Value Type	Mandatory	Description
cclid	Integer	No	Call center ID. The value ranges from 1 to 65535. In the independent deployment environment, the default value is the value of Default Call Center ID on the Configuration Center > System Management > Parameter Settings page after the system administrator signs in to the AICC.
vdn	Integer	Yes	ID of a VCC.
groupId	String	Yes	User group ID.
workNo	Integer	No	Agent ID. The value ranges from 0 to 999999.
limit	Integer	Yes	Maximum number of query records on each page (≤ 100).
offset	Integer	Yes	Offset.

Response Description

Table 3 Parameters in the message body

Parameter	Value Type	Description
returnCode	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure

Table 3 Parameters in the message body

Parameter	Value Type	Description
		For details about the failure causes, see Error Code Reference .
description	Array	Object array in the returned result after a successful query. For details about the parameters of the object, see Table 4 . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
resultDesc	Array	Object array in the returned result after a successful query. For details about the parameters of the object, see Table 4 .

Table 4 Parameters in the message body

Parameter	Value Type	Description
workNo	Integer	Agent ID.
accountCode	String	Account code.
groupName	String	Group name.
accountName	String	Account name.
ccRoleId	Integer	Role ID.
id	Integer	Primary key ID.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "ccId":1,
  "vdn":113,
  "groupId":"159282396714302232563636876268",
  "limit":10,
  "offset":0
}
```

- Response parameters

```
{
```

```
"returnCode": "0",
"resultCode": "0100000",
"description": [
  {
    "workNo": 1117,
    "accountCode": "HW513331117",
    "groupName": "Inspector group",
    "accountName": "HW972571117",
    "ccRoleId": 3,
    "id": "160526521924802293467767451224"
  },
  {
    "workNo": 1118,
    "accountCode": "HW340521118",
    "groupName": "Inspector group",
    "accountName": "HW972571118",
    "ccRoleId": 3,
    "id": "160526521924824882258144059275"
  }
],
"resultDesc": [
  {
    "workNo": 1117,
    "accountCode": "HW513331117",
    "groupName": "Inspector group",
    "accountName": "HW972571117",
    "ccRoleId": 3,
    "id": "160526521924802293467767451224"
  },
  {
    "workNo": 1118,
    "accountCode": "HW340521118",
    "groupName": "Inspector group",
    "accountName": "HW972571118",
    "ccRoleId": 3,
    "id": "160526521924824882258144059275"
  }
]
}
```

4.4.2 Deleting Members from a Group in Batches

Scenario

This interface is invoked to delete user group members in batches based on **vdn** and **workNos**.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Request URL

`https://ip:port/rest/cmsapp/v1/openapi/qcrelationconf/batchdeletegroupmember`

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the message body

Parameter	Value Type	Mandatory	Description
cclid	Integer	No	Call center ID. The value ranges from 1 to 65535. In the independent deployment environment, the default value is the value of Default Call Center ID on the Configuration Center > System Management > Parameter Settings page after the system administrator signs in to the AICC.
workNos	Array	No	Agent ID set of user group members. If this parameter is left blank, all group members under the tenant corresponding to the VCC are deleted. Agent ID. The value ranges from 0 to 999999.
vdn	Integer	Yes	ID of a VCC.

Response Description

Table 3 Parameters in the message body

Parameter	Value Type	Description
returnCode	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
resultCode	String	Execution result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
description	String	Description of the result. NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
resultDesc	String	Description of the result.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "cclId":1,
  "workNos":[57005,57006],
  "vdn":113
}
```

- Response parameters

```
{
  "returnCode": "0",
  "resultCode": "0100000",
  "description": "Deleted successfully"
```

```
"resultDesc": "Deleted successfully"
}
```

4.4.3 Adding Members to a Group in Batches

Scenario

This interface is invoked to add members to a specified user group in batches.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Request URL

`https://ip:port/rest/cmsapp/v1/openapi/qcrelationconf/batchaddgroupmember`

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the message body

Parameter	Value Type	Mandatory	Description
cclid	Integer	No	Call center ID. The value ranges from 1 to 65535. In the independent deployment environment, the default value is the value of Default Call Center ID on the Configuration Center > System Management > Parameter Settings page after the system administrator signs in to the AICC.
vdn	Integer	Yes	ID of a VCC.

Table 2 Parameters in the message body

Parameter	Value Type	Mandatory	Description
workNos	Array	Yes	Agent ID set. Agent ID. The value ranges from 0 to 999999.
groupId	String	Yes	User group ID.

Response Description

Table 3 Parameters in the message body

Parameter	Value Type	Description
returnCode	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
resultCode	String	Execution result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
description	String	Description of the result. NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
resultDesc	String	Description of the result.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
```

```

    "ccld":1,
    "vdn":113,
    "workNos":[57005,57006],
    "groupId":"159774518399110207461019283363"
  }

```

- Response parameters

```

{
  "returnCode": "0",
  "resultCode": "0100000",
  "description": "Added successfully"
  "resultDesc": "Added successfully"
}

```

4.4.4 Querying All User Groups

Description

This interface is invoked to query all user groups in a VDN based on the VDN.

Interface Method

The request method must be set to GET. This interface supports only the GET method.

Request URL

`https://ip:port/rest/cmsapp/v1/openapi/qcrelationconf/queryallccgroup/{ccld}/{vdn}`

The **ccld** parameter is optional. If it is left empty, in the independent deployment environment, the default value is the value of **Default Call Center ID** on the **Configuration Center > System Management > Parameter Settings** page after a system administrator signs in to the AICC.

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed to application/json; charset=UTF-8 .

Table 1 Parameters in the request header

No.	Parameter	Type	Mandatory	Description
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Response Message

Table 2 Parameter description

Parameter	Type	Description
returnCode	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
description	Array	Object array in the returned result after a successful query. For details about the parameters of the object, see Table 3 . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
resultDesc	Array	Object array in the returned result after a successful query. For details about the parameters of the object, see Table 3 .

Table 3 Parameter description

Parameter	Type	Description
groupName	String	User Group Name
groupType	Integer	User Group Type
remark	String	Remarks.
id	String	Primary key ID of a user group.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
```

```
Authorization: *****
```

- No request message exists.

- Response parameters

```
{
  "returnCode": "0",
  "resultCode": "0100000",
  "description": [
    {
      "groupName": "ceshi001",
      "groupType": 3,
      "id": "159253664007418170345218084185"
    },
    {
      "groupName": "ceshi002",
      "groupType": 2,
      "id": "159253664765739060254053779940"
    },
    {
      "groupName": "ceshi003",
      "groupType": 3,
      "id": "159253666192354492596520647666"
    },
    {
      "groupName": "ceshi004",
      "groupType": 2,
      "id": "159253667128968076722272915560"
    },
    {
      "groupName": "ceshi005",
      "groupType": 3,
      "id": "159289723573509154577931092749"
    },
    {
      "groupName": "ceshi445566",
      "remark": "Inspector group",
      "groupType": 3,
      "id": "159832671551117675139418726647"
    }
  ],
}
```

```
"resultDesc": [  
  {  
    "groupName": "ceshi001",  
    "groupType": 3,  
    "id": "159253664007418170345218084185"  
  },  
  {  
    "groupName": "ceshi002",  
    "groupType": 2,  
    "id": "159253664765739060254053779940"  
  },  
  {  
    "groupName": "ceshi003",  
    "groupType": 3,  
    "id": "159253666192354492596520647666"  
  },  
  {  
    "groupName": "ceshi004",  
    "groupType": 2,  
    "id": "159253667128968076722272915560"  
  },  
  {  
    "groupName": "ceshi005",  
    "groupType": 3,  
    "id": "159289723573509154577931092749"  
  },  
  {  
    "groupName": "ceshi445566",  
    "remark": "Inspector group",  
    "groupType": 3,  
    "id": "159832671551117675139418726647"  
  }  
]
```

4.4.5 Adding a User Group

Scenario

This interface is invoked to add a user group.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Request URL

`https://ip:port/rest/cmsapp/v1/openapi/qcrelationconf/createusergroup`

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the message body

Parameter	Value Type	Mandatory	Description
cclid	Integer	No	Call center ID. The value ranges from 1 to 65535. In the independent deployment environment, the default value is the value of Default Call Center ID on the Configuration Center > System Management > Parameter Settings page after the system administrator signs in to the AICC.
vdn	Integer	Yes	ID of a VCC.
groupName	String	Yes	User group name.
groupType	Integer	Yes	User group type. The options are 2 (agent group) and 3 (inspector group).
remark	String	No	User group information.
workNos	Array	No	Agent ID set. If this parameter is set, group members are added when a user group is added. Agent ID. The value ranges from 0 to 999999.

Response Description

Table 3 Parameters in the message body

Parameter	Value Type	Description
returnCode	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
resultCode	String	Execution result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
description	String	Description of the result. NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
resultDesc	String	Description of the result.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "ccld":1,
  "vdn":113,
  "groupName":"ceshipostman",
  "groupType":3,
  "remark": "Inspector group",
  "workNos":[50041,50042]
}
```

- Response parameters

```
{
  "returnCode": "0",
```

```

"resultCode": "0100000",
"description": "Added successfully"
"resultDesc": "Added successfully"
}

```

4.4.6 Deleting a User Group

Description

This interface is invoked to delete a user group based on the VDN and user group ID.

Note: If a user group has members and inspection relationships are configured for the user group, the members and configured inspection relationships are deleted with the user group.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Request URL

`https://ip:port/rest/cmsapp/v1/openapi/qcrelationconf/deleteusergroup`

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed to application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the message body

Parameter	Type	Mandatory	Description
cclid	Integer	No	Call center ID. The value ranges from 1 to 65535. In the independent deployment environment, the default value is the value of Default Call Center ID on the Configuration Center > System Management > Parameter Settings page after a system administrator signs in to the AICC.

Table 2 Parameters in the message body

Parameter	Type	Mandatory	Description
vdn	Integer	Yes	ID of a VDN.
id	String	Yes	User group ID.

Response Message

Table 3 Parameter description

Parameter	Type	Description
returnCode	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
resultCode	String	Execution result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
description	String	Description of the result. NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
resultDesc	String	Description of the result.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "ccld":1,
  "id":"159832684658001135254942692203",
  "vdn":113
```

```
}

```

- Response parameters

```
{
  "returnCode": "0",
  "resultCode": "0100000",
  "description": "Deleted successfully"
  "resultDesc": "Deleted successfully"
}
```

4.4.7 Updating User Group Information

Description

This interface is invoked to update user group information.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Request URL

https://ip:port/rest/cmsapp/v1/openapi/qcrelationconf/updategroup

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed to application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the message body

Parameter	Type	Mandatory	Description
cclid	Integer	No	Call center ID. The value ranges from 1 to 65535.

Table 2 Parameters in the message body

Parameter	Type	Mandatory	Description
			In the independent deployment environment, the default value is the value of Default Call Center ID on the Configuration Center > System Management > Parameter Settings page after a system administrator signs in to the AICC.
vdn	Integer	Yes	ID of a VDN.
id	String	Yes	User group ID.
groupName	String	Yes	New user group name.
remark	String	No	User group information.

Response Message

Table 3 Parameters in the message body

Parameter	Type	Description
returnCode	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
resultCode	String	Execution result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
description	String	Description of the result. NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
resultDesc	String	Description of the result.

Message Example

- Request header

Content-Type: application/json;charset=UTF-8

```
Authorization: *****
```

- Request parameters

```
{
  "ccId":1,
  "groupName":"ceshi445566",
  "remark": "Inspector group",
  "id":"159832671551117675139418726647",
  "vdn":113
}
```

- Response parameters

```
{
  "returnCode": "0",
  "resultCode": "0100000",
  "description": "Updated successfully"
  "resultDesc": "Updated successfully"
}
```

4.4.8 Setting Inspection Relationships

Description

This interface is invoked to set the inspection relationships between a specified inspector group and a user group.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Request URL

`https://ip:port/rest/cmsapp/v1/openapi/qcrelationconf/configqcrelation`

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed to application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the message body

Parameter	Type	Mandatory	Description
cclId	Integer	No	Call center ID. The value ranges from 1 to 65535. In the independent deployment environment, the default value is the value of Default Call Center ID on the Configuration Center > System Management > Parameter Settings page after a system administrator signs in to the AICC.
qcGroupId	Integer	Yes	ID of an inspector group.
inspectedGroupIds	Array	Yes	Agent group ID set.
vdn	Integer	Yes	ID of a VDN.

Response Message

Table 3 Parameter description

Parameter	Type	Description
returnCode	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
resultCode	String	Execution result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
description	String	Description of the result. NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.

Table 3 Parameter description

Parameter	Type	Description
resultDesc	String	Description of the result.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "ccId":1,
  "qcGroupId":"159282396714302232563636876268",
  "inspectedGroupIds":["159253666192354492596520647666"],
  "vdn":113
}
```

- Response parameters

```
{
  "returnCode": "0",
  "resultCode": "0100000",
  "description": "Configured successfully"
  "resultDesc": "Configured successfully"
}
```

4.4.9 Deleting Configured Inspection Relationships

Description

This interface is invoked to delete inspection relationships based on the IDs of the inspector group and inspected party group.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Request URL

<https://ip:port/rest/cmsapp/v1/openapi/qcrelationconf/deleteqcrelationconf>

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed to application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the message body

Parameter	Type	Mandatory	Description
cclid	Integer	No	Call center ID. The value ranges from 1 to 65535. In the independent deployment environment, the default value is the value of Default Call Center ID on the Configuration Center > System Management > Parameter Settings page after a system administrator signs in to the AICC.
qcGroupId	Integer	Yes	ID of an inspector group.
inspectedGroupIds	Array	Yes	Agent group ID set.
vdn	Integer	Yes	ID of a VDN.

Response Message

Table 3 Parameter description

Parameter	Type	Description
returnCode	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
resultCode	String	Execution result. The options are as follows: 0100000 : success

Table 3 Parameter description

Parameter	Type	Description
		Other values: failure For details about the failure causes, see Error Code Reference .
description	String	Description of the result. NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
resultDesc	String	Description of the result.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "cclId":1,
  "qcGroupId":"159282396714302232563636876268",
  "inspectedGroupIds":["159253666192354492596520647666"],
  "vdn":113
}
```

- Response parameters

```
{
  "returnCode": "0",
  "resultCode": "0100000",
  "description": "Deleted successfully"
  "resultDesc": "Deleted successfully"
}
```

For details about the parameters in the response message body of this interface, see [Response Message](#).

4.4.10 Adding an Agent to a User Group

Scenario

The interface is invoked to add an agent to a user group on the CMS.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Request URL

`https://ip:port/rest/cmsapp/v1/openapi/qcrelationconf/joinusergroup`

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the message body

Parameter	Value Type	Mandatory	Description
cclid	Integer	No	Call center ID. The value ranges from 1 to 65535. In the independent deployment environment, the default value is the value of Default Call Center ID on the Configuration Center > System Management > Parameter Settings page after the system administrator signs in to the AICC.
vdnId	Integer	Yes	ID of a VCC. The value ranges from 1 to 5000.
agentId	Integer	Yes	ID of an agent who joins a user group. The value ranges from 0 to 999999.
groupId	String	No	ID of the user group that the user joins. If the group ID is not carried, a group is created using the name specified by groupName , and then the agent is added to the new group. If the agent already exists in another group, the agent is automatically removed from it. If the group ID is carried, groupName is ignored, and the agent is added to the group corresponding to groupId . If the agent already exists in another group, the agent is automatically removed from it.

Table 2 Parameters in the message body

Parameter	Value Type	Mandatory	Description
groupName	String	No	Name of the group. (The value cannot contain invalid characters and can contain a maximum of 100 characters.)
groupType	Integer	No	User group type. The default value is 2. The options are 2 (agent group) and 3 (inspector group). Other values are invalid.

Response Description

Table 3 Parameters in the message body

Parameter	Value Type	Description
message	String	Description.
retcode	String	Error code. The options are as follows: 0: success Other values: failure For details about error code parameters, see Table 5 .
result	Object	Object in the returned result after an agent is successfully added to a user group. For details about the parameters of the object, see Table 4 .

Table 4 Parameters of result

Parameter	Value Type	Description
groupId	String	ID of the user group to which an agent is added.
groupName	String	Name of the user group to which an agent is added. (The value cannot contain invalid characters and can contain a maximum of 20 characters.)

Table 5 Error end codes

Error End Code	Description	Comments
300-010	ccid and vdnid of the VCC user do not match.	Check whether the input parameters are correct.
300-016	The agent does not exist in the VCC.	Check whether the agent exists in the VCC. If yes, try again 1 minute later.
300-017	The inspector group name already exists.	Change the name of the inspector group.

Table 5 Error end codes

Error End Code	Description	Comments
300-018	The name of the inspector group is empty.	Check whether the groupName parameter is carried or left empty.
300-019	The inspector group does not exist.	Check whether the inspector group corresponding to groupId exists.
1	The parameter is invalid or the system reports an error.	Check whether the input parameters are correct.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "cclId":1,
  "vdnId":4,
  "agentId":667,
  "groupId":"159282396714302232563636876268",
  "groupName":"Agent group",
  "groupType":""
}
```

- Response parameters

```
{
  "result": {
    "groupName": "Agent group A",
    "groupId": "161960203515344228641565404173"
  },
  "message": "success",
  "retcode": "0"
}
```

4.4.11 Updating Agent Names

This interface supports only the scenario where the CC-CMS is independently deployed.

Scenario

This interface is invoked to update an agent name.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Request URL

`https://ip:port/rest/cmsapp/v1/openapi/qcrelationconf/updateagentgroup`

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the message body

Parameter	Value Type	Mandatory	Description
cclid	Integer	No	Call center ID. The value ranges from 1 to 65535. In the independent deployment environment, the default value is the value of Default Call Center ID on the Configuration Center > System Management > Parameter Settings page after the system administrator signs in to the AICC.
vdnId	Integer	Yes	ID of a VCC. The value ranges from 1 to 5000.
agentId	Integer	Yes	Agent ID. The name of the VCC administrator cannot be changed, that is, the agent ID of the VCC administrator cannot be transferred. The value ranges from 0 to 999999.
agentName	String	Yes	Agent name. (The value cannot contain invalid characters.)

Response Description

Table 3 Parameters in the message body

Parameter	Value Type	Description
message	String	Description.
retcode	String	Error code. The options are as follows: 0 : success 1 : failure
result	Object	Object in the returned result after an agent is successfully added to a user group. For details about the parameters of the object, see Table 4 .

Table 4 Parameters of result

Parameter	Value Type	Description
cclId	Integer	ID of a call center.
vdnId	Integer	ID of a VCC.
agentId	Integer	New agent ID.
agentName	String	New agent name.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "cclId":1,
  "vdnId":4,
  "agentId":667,
  "agentName":"Zhang San"
}
```

- Response parameters

```
{
  "result": {
    "agentId": "6667",
    "cclId": 1,
    "vdnId": 4,
    "agentName": "Li Si"
  },
}
```

```

"message": "success",
"retcode": "0"
}

```

4.5 Querying Outbound Call Task Lists

4.5.1 Querying the Outbound Call Task List of a Tenant

Description

This interface is invoked to query the outbound call task list of a tenant and return task information.

Note: The iSales service module must be deployed for this interface.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Request URL

`https://ip:port/rest/cmsapp/v1/openapi/fromisales/querytasks`

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed to application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the message body

Parameter	Type	Mandatory	Description
cclid	Integer	No	Call center ID. The value ranges from 1 to 65535. In the integration environment, the default value is the value of Call Center ID on the Configuration Center > System Management > CMS System Configuration page after a system administrator signs in to the AICC. If this parameter is empty, the data whose sys_key is default_cc_id and sys_type is cms_system_config in the t_cms_sysparam database table is used.

Table 2 Parameters in the message body

Parameter	Type	Mandatory	Description
			In the independent deployment environment, the default value is the value of Default Call Center ID on the Configuration Center > System Management > Parameter Settings page after a system administrator signs in to the AICC.
vdn	Integer	Yes	ID of a VCC. The value ranges from 1 to 5000.
limit	Integer	No	Page size. The default value is 10 . A maximum of 100 records can be queried.
offset	Integer	No	Pagination start position. If this parameter is not set, the default value 0 is used.

Response Message

Table 3 Parameters in the message body

Parameter	Type	Description
resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
resultDesc	Array	Object array in the returned result after a successful query. For details about the parameters of the response message body, see Table 4 .

Table 4 Parameters in the response message body

Parameter	Type	Description
count	Integer	Total number of tasks.
tasks	JSONArray	Task list.

For details about the parameters of **tasks**, see [Table 5](#).

Table 5 Parameters of tasks

Parameter	Type	Description
taskId	String	Task ID.
taskName	String	Task name.
taskType	String	Task type. 1: Intelligent outbound call 2: Predicted outbound calls

Table 5 Parameters of tasks

Parameter	Type	Description
		3: Automatic outbound call 4: Agent outbound call 5: Previewed outbound call
beginDate	String	Start time, which is written by the CC-iSales frontend user. The value is a timestamp.
endDate	String	End time, which is written by the CC-iSales frontend user. The value is a timestamp.
status	String	Task status. 2: Being executed 3: Suspended 4: Completed

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "ccId": "1",
  "vdn": 34,
  "limit": 10,
  "offset": 0
}
```

- Response parameters

```
{
  "resultCode": "0100000",
  "resultDesc": {
    "count": 14,
    "tasks": [
      {
        "beginDate": "1735357466000",
        "taskType": "4",
        "endDate": "1735443866000",
        "taskName": "1",
        "taskId": "1088825198",
        "status": "1"
      },
      {
```

```

        "beginDate": "1734629077000",
        "taskType": "4",
        "endDate": "1734715477000",
        "taskName": "2",
        "taskId": "1312452402",
        "status": "1"
    }
    .....
    ]
}
}

```

4.5.2 Querying the Daily Settlement Statistics of a Specified Task

Scenario

This interface is invoked to query the daily settlement statistics of a specified task and return the daily settlement statistics indicators.

Note: The iSales service module must be deployed for this interface.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Request URL

`https://ip.port/rest/cmsapp/v1/openapi/fromisales/dailystatistic`

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the message body

Parameter	Value Type	Mandatory	Description
ccld	Integer	No	Call center ID. The value ranges from 1 to 65535. In the integration environment, the default value is the value of Call Center ID on the Configuration Center > System Management > CMS System Configuration page after the system administrator signs in to the AICC. If this parameter is empty, the data whose sys_key is default_cc_id and sys_type is cms_system_config in the t_cms_sysparam database table is used. In the independent deployment environment, the default value is the value of Default Call Center ID on the Configuration Center > System Management > Parameter Settings page after the system administrator signs in to the AICC.
vdn	Integer	Yes	ID of a VCC. The value ranges from 1 to 5000.
taskId	String	Yes	Task ID. The value contains a maximum of 30 characters.
beginDayTime	String	Yes	Start date for statistics collection, which is in the time zone of the tenant. Format: YYYY-MM-DD
endDayTime	String	Yes	End date for statistics collection, which is in the time zone of the tenant. Format: YYYY-MM-DD

Response Description

Table 3 Parameters in the response message

Parameter	Value Type	Description
resultCode	String	Query result. The options are as follows: 0100000: success Other values: failure For details about the failure causes, see Error Code Reference.
resultDesc	Array	Object array in the returned result after a successful query. Table 4 describes the parameters in the response message body of this interface.

Table 4 Parameters in the response body

Parameter	Value Type	Description
statDate	String	Date of the daily settlement, in UTC format. The value is a timestamp.
totalCount	Integer	Number of daily outbound calls.
succCount	Integer	Number of successful calls (number of connected calls)
succRate	String	Success rate (call connection rate).

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "vdn": 222,
  "ccId": 89,
  "taskId": "1176499974",
  "beginDayTime": "2021-01-21",
  "endDayTime": "2021-01-23"
}
```

- Response parameters

```
{
  "resultCode": "0100000",
  "resultDesc": [
    {
      "statDate": "1734566400000",
      "totalCount": 200,
      "succCount": 101,
      "succRate": "50.5%"
    },
    {
      "statDate": "1737556400000",
      "totalCount": 320,
      "succCount": 248,
      "succRate": "77.5%"
    }
  ]
}
```

4.5.3 Querying the Real-Time Statistics of a Specified Task

Description

This interface is invoked to query the real-time statistics of a specified task and return the real-time statistics indicators.

Note: The iSales service module must be deployed for this interface.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Request URL

https://ip:port/rest/cmsapp/v1/openapi/fromisales/realstatistic

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed to application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the message body

Parameter	Type	Mandatory	Description
cclid	Integer	No	Call center ID. The value ranges from 1 to 65535. In the integration environment, the default value is the value of Call Center ID on the Configuration Center > System Management > CMS System Configuration page after a system administrator signs in to the AICC. If this parameter is empty, the data whose sys_key is default_cc_id and sys_type is cms_system_config in the t_cms_sysparam database table is used. In the independent deployment environment, the default value is the value of Default Call Center ID on the Configuration Center > System Management > Parameter Settings page after a system administrator signs in to the AICC.
vdn	Integer	Yes	ID of a VDN. The value ranges from 1 to 5000.

Table 2 Parameters in the message body

Parameter	Type	Mandatory	Description
taskId	Integer	Yes	Task ID. The value contains a maximum of 30 characters.

Response Message

Table 3 Parameters in the message body

Parameter	Type	Description
resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
resultDesc	Array	Object array in the returned result after a successful query. Table 4 describes the parameters in the response message body of this interface.

Table 4 Parameters in the response message body

Parameter	Type	Description
totalCount	Integer	Number of daily outbound calls.
succCount	Integer	Number of successful calls (number of connected calls).
failCount	Integer	Number of failed calls.
succRate	String	Success rate (call connection rate).
compCount	Integer	Number of completed calls.
waitCount	Integer	Number of waiting calls.
execCount	Integer	Number of calls that are being processed.
compRate	String	Completion rate.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "ccId": "89",
```

```
"vdn":92,  
"taskId":"1122104910"  
}
```

- Response parameters

```
{  
  "resultCode": "0100000",  
  "resultDesc": {  
    "compCount": "4",  
    "failCount": "2",  
    "compRate": "50%",  
    "execCount": "0",  
    "succCount": "2",  
    "succRate": "50%",  
    "totalCount": "4",  
    "waitCount": "0"  
  }  
}
```

4.6 Interfaces for Querying System Indicators

If the environment is in pool mode, query the information about two UAPs. If the environment is in single mode, query the information about a single UAP. The environment mode is set when the CMS is deployed. For details, see the CMS deployment document.

4.6.1 Querying UAP Performance Information

Prerequisites

You need to create the **Incoming Office measurement** task on the performance management tool. For details, see "Maintenance > Performance Specifications" in *UAP9600 Product Documentation*. This interface is invoked only for global AK/SK authentication.

Note: This interface supports only the independent deployment environment.

Description

This interface is invoked to query the performance data of the UAP in the latest 5 minutes, including the trunk usage information, such as the number of idle circuits, number of available circuits, and circuit usage.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Request URL

https://ip:port/rest/cmsapp/v1/openapi/uap/queryUapPerfInfo

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed to application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the request message body

Parameter	Type	Mandatory	Description
ofcNo	Array	No	Array of inbound numbers. If the array is null, all inbound numbers are queried. You can run the MML command LST OFC to obtain the inbound numbers.
cclId	Integer	No	ID of a call center. In the independent deployment environment, the default value is the value of Default Call Center ID on the Configuration Center > System Management > Parameter Settings page after a system administrator signs in to the AICC.

Response Message

Table 3 Parameters in the response message body

Parameter	Type	Description
resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
resultDesc	List	Object list in the returned result after a successful query. For details about the parameters of the object, see the following table.
10.xx.xx.xx#0	String	10.xx.xx.xx : UAP IP address. # : fixed separator.

Table 3 Parameters in the response message body

Parameter	Type	Description
		0: UAP ID.

Table 4 Parameters of resultDesc

Parameter	Type	Description
RET	String	End time of data collection.
RNAT12	String	No answer times.
busy circuits (TIMES)	String	Number of busy circuits.
ASD32	String	Average occupation duration, in seconds.
ITAR19	String	Incoming trunk available ratio.
CLBT40	String	The called party (local call) is busy.
MEID	String	Device ID.
UBC43	String	Number of blocked circuits.
CCT34	String	Number of failed calls due to congestion at the peer end.
LCOICT7	String	Call loss due to home office internal congestion.
IOCS44	String	Inbound CAPS value.
MOIID	String	Measurement object ID.
CCT3	String	Number of connected calls.
CDA9	String	Congestion duration auxiliary quantity.
AART11	String	Number of calls released before ringing, that is, the number of times that the calling party hangs up before the calls are connected or ended.
TAT6	String	Number of answered transfer calls. An answered transfer call indicates that the user who answers the call is not a local user.
Period	String	Period.
TCAT5	String	Intra-office answer times. Intra-office answer indicates that a user who answers the call is a local user.
AR21	String	Answer rate.
TCAT2	String	Times of transfer seizure attempts.
TCAT1	String	Times of intra-office seizure attempts.

Table 4 Parameters of resultDesc

Parameter	Type	Description
ICN13	String	Number of installed circuits.
ECT35	String	Number of times that an unallocated number is dialed.
incoming direction	String	0(anonymous): 0 indicates an inbound call ID, and anonymous indicates a call name.
IBCN16	String	Number of installed bi-directional circuits.
ACN14	String	Number of available circuits.
TCT42	String	Transferring call seizure times.
IAT36	String	Invalid address times.
Credible	String	Trusted or not.
CFORT39	String	The caller party does not have permission.
RT	String	Time of data collection.
ST26	String	Seized call traffic volume.
Time	String	Count.
average answer duration (SECOND)	String	Average connection duration.
AT30	String	Answered call traffic.
TUT37	String	Terminal incompatibility times.
BCN15	String	Number of blocked circuits.
BBCR23	String	Bi-directional circuit block ratio.
CTBT41	String	Toll busy times of the called party.
BTST24	String	Traffic occupied by a bi-directional trunk.
BTSTA25	String	Traffic auxiliary quantity occupied by a bi-directional trunk.
ABCN17	String	Number of available bi-directional circuits.
CTA29	String	Connection traffic auxiliary quantity.
AT4	String	Number of answered calls.
EROC46	String	Circuit usage.
TET38	String	Terminal error times.
CT28	String	Connected call traffic.

Table 4 Parameters of resultDesc

Parameter	Type	Description
SOIC45	String	Number of idle circuits.
ST0	String	Seizure times.
BCR22	String	Block ratio.
ABRT33	String	Abandon times before ringing.
CR20	String	Connection rate.
BBCN18	String	Number of blocked bi-directional circuits.
CD8	String	Congestion duration.
ATA31	String	Answering traffic auxiliary quantity.
CBT10	String	Busy called party.
STA27	String	Seized traffic auxiliary quantity.
average connection duration (MILLISECOND)	String	Average connection duration.
answer duration aux (TIMES)	String	Total duration auxiliary quantity of answered sessions.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "ofcNo": [0,1],
  "ccld": 1
}
```

- Response parameters

```
{
  "resultCode": "0100000",
  "resultDesc": {
    "10.xx.xx.xx#0": [
      {
        "RET": "2021-02-04 10:05:00 +00:00",
        "RNAT12": "0",
```



```
"busy circuits (TIMES)": "0",
"ASD32": "0.000000",
"ITAR19": "0.000000",
"CLBT40": "-1",
"MEID": "5",
"UBC43": "-1",
"CCT34": "-1",
"LCOICT7": "0",
"IOCS44": "0",
"MOIID": "0",
"CCT3": "0",
"CDA9": "-1",
"AART11": "0",
"TAT6": "0",
"Period": "5",
"TCAT5": "0",
"AR21": "0.000000",
"TCAT2": "0",
"TCAT1": "0",
"ICN13": "0",
"ECT35": "0",
"incoming direction": "0(anonymous)",
"IBCN16": "0",
"ACN14": "0",
"TCT42": "-1",
"IAT36": "0",
"Credible": "Credible",
"CFORT39": "-1",
"RT": "2021-02-04 10:00:00 +00:00",
"ST26": "0.000000",
"Time": "1000",
"average answer duration (SECOND)": "0",
"AT30": "0.000000",
"TUT37": "-1",
"BCN15": "0",
"BBCR23": "0.000000",
"CTBT41": "-1",
"BTST24": "0.000000",
"BTSTA25": "-1",
"ABCN17": "0",
"CTA29": "-1",
"AT4": "0",
"EROC46": "0.000000",
```

```

    "TET38": "-1",
    "CT28": "0.000000",
    "SOIC45": "0",
    "ST0": "0",
    "BCR22": "0.000000",
    "ABRT33": "0",
    "CR20": "0.000000",
    "BBCN18": "0",
    "CD8": "0",
    "ATA31": "-1",
    "CBT10": "0",
    "STA27": "-1",
    "average connection duration (MILLISECOND)": "0",
    "answer duration aux (TIMES)": "-1"
  }
]
}
}

```

4.6.2 Querying Key Information About Control Blocks

NOTICE

The number of concurrent requests for invoking the interface cannot exceed 10. Otherwise, the interface invoking times out due to deteriorated performance.

Prerequisites

This interface is invoked only for global AK/SK authentication.

Note: This interface supports only the independent deployment environment.

Description

This interface is invoked to query the running states of specified control blocks.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Request URL

`https://ip:port/rest/cmsapp/v1/openapi/uap/dsppidkey`

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed to application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the request message body

Parameter	Type	Mandatory	Description
MODTYP	String	Yes	It specifies the type of a module. Currently, CCU and MSU are supported.
PID	String	Conditionally mandatory	CCU module ID. This parameter is valid only when the module type is set to CCU . It specifies the module ID of a CCU.
PIDMSU	String	Conditionally mandatory	MSU module ID. This parameter is valid only when the module type is set to MSU . It specifies the module ID of an MSU.
MN	String	Conditionally mandatory	CCU module number. This parameter is valid only when the module type is set to CCU and the CCU module ID is specified. It specifies the module number of a CCU. You can run the LST APP command to query the module number of a CCU.
BICCTYPE	String	Conditionally mandatory	BICC type. This parameter is valid only when the module type is set to CCU and the CCU module ID is set to BICC . It specifies the type of a BICC module.
MSUMN	String	Conditionally mandatory	MSU module number. This parameter is valid only when the module type is set to MSU and the MSU module ID is specified. It specifies the module number of an MSU. You can run the LST APP command to query the module number of an MSU.
cclId	Integer	No	ID of a call center. In the independent deployment environment, the default value is the value of Default Call Center ID on the Configuration Center > System Management > Parameter Settings page after a system administrator signs in to the AICC.

Response Message

Table 3 Parameters in the response message body

Parameter	Type	Description
resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
resultDesc	List	Object list in the returned result after a successful query. For details about the parameters of the object, see the following table.
10.xx.xx.xx#0	String	10.xx.xx.xx : UAP IP address. # : fixed separator. 0 : UAP ID.

Table 4 Parameters of the resultDesc object (Set the module type to CCU, CCU module ID to CCB, and CCU module number to 57.)

Parameter	Type	Description
IDC switch index	String	Online indication.
Null CB number	String	Number of control blocks in the NULL state.
Booked CB number	String	Number of busy circuits.
Info receiving status number	String	Number of control blocks that are receiving information.
Wait overlap CB number	String	Number of control blocks in the WAITOVERLAP state.
Overlap transmit number	String	Number of control blocks in the Overlap transmit state.
Call proceeding number	String	Number of control blocks in the CALLPROCEEDIN state.
Alerting CB number	String	Number of control blocks in the ALERTING state.
Connection request CB number	String	Number of control blocks that are requesting connections.
Active CB number	String	Number of active control blocks.
Release guard CB number	String	Number of control blocks in the Release guard state.
Release waiting CB number	String	Number of control blocks that are waiting to be released.
Temply hold CB number	String	Number of control blocks in the Temply hold state.
Wait tone CB number	String	Number of control blocks that are in the waiting tone state.
Null after combine CB number	String	Number of control blocks in the NULLAFTERCOMBINE state.

Table 4 Parameters of the resultDesc object (Set the module type to CCU, CCU module ID to CCB, and CCU module number to 57.)

Parameter	Type	Description
Call park CB number	String	Number of control blocks in the Call park state.

Table 5 Parameters of the resultDesc object (Set the module type to CCU, CCU module ID to INTESS, and CCU module number to 56.)

Parameter	Type	Description
Last allocated INTESS number	String	ID of the INTess control block that was allocated last time.
Error stat flag	String	Error cause statistics switch.
Error stat interval	String	Error cause statistics period.
Message statistics flag	String	Message statistics switch.
Message statistics interval	String	Message statistics period.
Message statistics mask	String	Message statistics mask.
Idle CB number	String	Number of control blocks in the IDLE state.
Overlap CB number	String	Number of control blocks in the OVERLAP state.
Call delivered CB number	String	Number of control blocks in the CALLDELIVERED state.
Call present CB number	String	Number of control blocks in the CALLPRESENT state.
Call received	String	Number of control blocks in the CALLRECEIVED state.
Connection request CB number	String	Number of control blocks in the connection request state.
Active CB number	String	Number of control blocks in the ACTIVE state.
SPACTIVE CB number	String	Number of control blocks in the SPACTIVE state.
CB number in INTESS_spy state	String	Number of control blocks in the INTESS_SPY state.
Wait REL CB number	String	Number of control blocks in the WAITREL state.

Table 6 Parameters of the resultDesc object (Set the module type to MSU, PID to BSI, and MSUMN module number to 472.)

Parameter	Type	Description
Total number of connections	String	Total number of connections.
Total no of connection in IDLE state	String	Total number of idle connections.
Total no of connection in CREATE CONNECT state	String	Total number of connections that are being created.

Table 6 Parameters of the resultDesc object (Set the module type to MSU, PID to BSI, and MSUMN module number to 472.)

Parameter	Type	Description
Total no of connection in CONNECT state	String	Total number of connected connections.
Total no of connection in DELETE CONNECT state	String	Total number of deleted connections.
Total no of connection in MODIFY CONNECT state	String	Total number of modified connections.
Total no of connection having service type as IVR	String	Total number of IVR business connections.
Total no of connection having service type as CONF	String	Total number of conference site business connections.
Total no of connection having service type as CONF CHAN	String	Total number of business connections of conference site channels.
Total no of conf connection in IDLE state	String	Total number of idle conference site connections.
Total number of Conf connections	String	Total number of conference site connections.
Total no of conf connection in CREATE CONF state	String	Total number of conference site connections that are being created.
Total no of conf connection in CONF USING state	String	Total number of conference site connections in use.
Total no of conf connection in CONF DELETE state	String	Total number of deleted conference site connections.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "MODTYP":"CCU",
  "PID":"CCB",
  "MN":"56",
  "ccld":1
}
```

- Response parameters

```
{
```

```

"resultCode": "0100000",
"resultDesc": {
  "10.xx.xx.xx#0": {
    "Alerting CB number": "0",
    "Null after combine CB number": "0",
    "IDC switch index": "1",
    "Wait tone CB number": "0",
    "Release guard CB number": "0",
    "Release waiting CB number": "0",
    "Booked CB number": "0",
    "Temply hold CB number": "0",
    "Active CB number": "0",
    "Info receiving status number": "3",
    "Call park CB number": "0",
    "Null CB number": "9997",
    "Overlap transmit number": "0",
    "Call proceeding number": "0",
    "Wait overlap CB number": "0",
    "Connection request CB number": "0"
  }
}
}

```

4.6.3 Querying the Registration Status of a Terminal

NOTICE

The number of concurrent requests for invoking the interface cannot exceed 10. Otherwise, the interface invoking times out due to deteriorated performance.

Prerequisites

This interface is invoked only for global AK/SK authentication.

Note: This interface supports only the independent deployment environment.

Description

This interface is invoked to query the registration status of a terminal.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Request URL

`https://ip:port/rest/cmsapp/v1/openapi/uap/queryterminalstatus`

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed to application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the request message body

Parameter	Type	Mandatory	Description
EID	String	Yes	It specifies the ID of the multimedia device to be operated. The rules for setting the parameter are as follows: For a multimedia device that adopts the SIP, the value is a string, which generally contains up to 32 digits, for example, 8780001. The asterisk (*) and percent (%) can match zero or multiple characters. For example, 6666* and 6666% match all character strings starting with 6666. The question mark (?) matches only one character. For example, 6666? matches only five-character strings starting with 6666. The combination of the asterisk (*), question mark (?), and percent sign (%) is supported. For example, %3? matches all character strings whose penultimate character is 3. NOTICE: The number of wildcard matching results must be less than 200. Otherwise, UAP services are affected.
limit	Integer	False and offset appear in pairs.	Query quantity. The default value is 100 .
offset	Integer	False and limit appear in pairs.	Start position of the query. The default value is 0 .
ccld	Integer	No	ID of a call center. In the independent deployment environment, the default value is the value of Default Call Center ID on the Configuration Center > System Management > Parameter Settings page after a system administrator signs in to the AICC.

Response Message

Table 3 Parameters in the response message body

Parameter	Type	Description
resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
resultDesc	List	Object list in the returned result after a successful query. For details about the parameters of the object, see the following table.
10.xx.xx.xx#1	String	10.xx.xx.xx : UAP IP address. # : fixed separator. 1 : UAP ID.

Table 4 Parameters of resultDesc

Parameter	Type	Description
data	List	Data result set.
count	Integer	Total number of records.

Table 5 Parameters of data

Parameter	Type	Description
ccuModuleNumber	String	CCU module number.
terminationType	String	Terminal type.
equipmentId	String	Equipment ID.
registerStatus	String	Registration status. The value is registered or not register .
socketStatus	String	Socket status. The value is fault , normal , connecting , or uninstall .
ipAndPort	String	IP address and port number.
monitorFlag	String	Agent status monitoring flag. The value is unstarted or started .
protocolType	String	Protocol type.
roamIpAndPort	String	Roaming IP address and port number.
tlsStatus	String	TSL status. The value is unconnect , connect , or uninstall .
ipType	String	IP address type.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "ccId":1,
  "EID": "8888*",
  "limit":"100",
  "offset":"0"
}
```

- Response parameters

```
{
  "resultCode": "0100000",
  "resultDesc": {
    "data":
    {
      "10.xx.xx.xx#1": [
        {
          "socketStatus": "uninstall",
          "terminationType": "sip",
          "ipAndPort": "0.0.0.0:0",
          "ccuModuleNumber": "56",
          "registerStatus": "not register",
          "monitorFlag": "unstarted",
          "protocolType": "invalid protocol",
          "equipmentId": "88881757",
          "roamIpAndPort": "0.0.0.0:0",
          "tlsStatus": "uninstall",
          "ipType": "IPv4"
        }
      ],
      "10.xx.xx.xx#2": [
        {
          "socketStatus": "uninstall",
          "terminationType": "sip",
          "ipAndPort": "0.0.0.0:0",
          "ccuModuleNumber": "56",
          "registerStatus": "not register",
          "monitorFlag": "unstarted",

```

```

        "protocolType": "invalid protocol",
        "equipmentId": "88881757",
        "roamIpAndPort": "0.0.0.0:0",
        "tlsStatus": "uninstall",
        "ipType": "IPv4"
    }
]
},
"count": 1
}
}

```

4.6.4 Interface for Querying UAP Performance Indicator Statistics

Prerequisites

This interface is invoked only for global AK/SK authentication.

Note: This interface supports only the independent deployment environment.

Description

This interface is invoked to query UAP performance indicator statistics.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Request URL

`https://ip:port/rest/cmsapp/v1/openapi/uap/perfindex`

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed to application/json; charset=UTF-8 .

Table 1 Parameters in the request header

No.	Parameter	Type	Mandatory	Description
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the request message body

Parameter	Type	Mandatory	Description
cclId	Integer	No	ID of a call center. In the independent deployment environment, the default value is the value of Default Call Center ID on the Configuration Center > System Management > Parameter Settings page after a system administrator signs in to the AICC.
uapIP	String	No	IP address of the UAP.
ofcNo	Integer	No	Array of inbound call numbers.
beginDate	String	Yes	Query start time. The UTC time zone is used. The interval between the start time and end time cannot exceed 31 days (rounded up).
endDate	String	Yes	Query end time. The UTC time zone is used.
limit	Integer	No	This parameter is optional. The default value is 100 .
offset	Integer	No	This parameter is optional. The default value is 0 .

Response Message

Table 3 Parameters in the response message body

Parameter	Type	Description
resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
resultDesc	List	Object list in the returned result after a successful query. For details about the parameters of the object, see the following table.

Table 4 Parameters of resultDesc

Parameter	Type	Description
beginTime	String	Statistics start time, which is a timestamp in milliseconds (start time of a 5-minute statistical period).

Table 4 Parameters of resultDesc

Parameter	Type	Description
cclId	Integer	ID of a call center.
uapIp	String	IP address of the UAP.
ofcNo	Integer	Array of inbound call numbers.
systemInBound	Integer	Number of system inbound calls.
systemInSucc	Integer	Number of system connected calls.
systemCallRate	Integer	System call connection rate.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "cclId":1,
  "uapIP":"1.1.1.1",
  "ofcNo":[0, 1],
  "beginDate":"2021-09-09 00:00:00",
  "endDate":"2021-09-11 23:00:00",
  "limit":"100",
  "offset":"0"
}
```

- Response parameters

```
{
  "resultCode": 0,
  "resultDesc": {
    "data": [
      {
        "beginTime": "1631594100000",
        "cclId": 1,
        "uapIp": "1.1.1.1",
        "ofcNo": 1,
        "systemInBound": 1,
        "systemInSucc": 1,
        "systemCallRate": 1
      }
    ]
  }
}
```

```
    }
  ],
  "count": 1
}
```

4.6.5 Interface for Querying UAP Terminal Statistics

Prerequisites

This interface is invoked only for global AK/SK authentication.

Note: This interface supports only the independent deployment environment.

Scenario

This interface is invoked to query UAP terminal statistics.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Request URL

`https://ip.port/rest/cmsapp/v1/openapi/uap/terminalindex`

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the request message body

Parameter	Type	Mandatory	Description
cclId	Integer	No	ID of a call center.

Table 2 Parameters in the request message body

Parameter	Type	Mandatory	Description
			In the independent deployment environment, the default value is the value of Default Call Center ID on the Configuration Center > System Management > Parameter Settings page after a system administrator signs in to the AICC.
uapIP	String	No	Array of VDN IDs. The maximum length of the array is 100.
beginDate	String	Yes	Query start time. The UTC time zone is used. The interval between the start time and end time cannot exceed 31 days (rounded up).
endDate	String	Yes	Query end time. The UTC time zone is used.
limit	Integer	No	This parameter is optional. The default value is 100 .
offset	Integer	No	This parameter is optional. The default value is 0 .

Response Message

Table 3 Parameters in the response message body

Parameter	Type	Description
resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
resultDesc	List	Object array in the returned result after a successful query. For details about the parameters of the object in this array, see the following table.

Table 4 Parameters of resultDesc

Parameter	Type	Description
beginTime	String	Statistics start time, which is a timestamp in milliseconds (start time of a 5-minute statistical period).
ccId	Integer	ID of a call center.
uapIp	String	IP address of the UAP.
agentOnlineCount	Integer	Number of registered softphone clients.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
```

```
Authorization: *****
```

- Request parameters

```
{
  "ccId":1,
  "uapIP":"1.1.1.1",
  "beginDate":"2021-09-09 00:00:00",
  "endDate":"2021-09-11 23:00:00",
  "limit":"100",
  "offset":"0"
}
```

- Response parameters

```
{
  "resultCode": 0,
  "resultDesc": {
    "data": [
      {
        "beginTime": "1631594100000",
        "ccId": 1,
        "uapIp": "1.1.1.1",
        "agentOnlineCount": 1
      }
    ],
    "count": 1
  }
}
```

4.6.6 Interface for Querying CTI System Indicators

Prerequisites

This interface is invoked only for global AK/SK authentication.

Note: This interface supports only the independent deployment environment.

Description

This interface is invoked to query CTI system indicators.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Request URL

`https://ip:port/rest/cmsapp/v1/openapi/cti/sysindex`

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed to application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameters in the request message body

Parameter	Type	Mandatory	Description
cclid	Integer	No	ID of a call center. In the independent deployment environment, the default value is the value of Default Call Center ID on the Configuration Center > System Management > Parameter Settings page after a system administrator signs in to the AICC.
vdn	List	No	Array of VDN IDs. The maximum length of the array is 100.
beginDate	String	Yes	Query start time. The UTC time zone is used. The interval between the start time and end time cannot exceed 31 days (rounded up).
endDate	String	Yes	Query end time. The UTC time zone is used.
limit	Integer	No	This parameter is optional. The default value is 100 .
offset	Integer	No	This parameter is optional. The default value is 0 .

Response Message

Table 3 Parameters in the response message body

Parameter	Type	Description
resultCode	String	Query result. The options are as follows: 0100000 : success Other values: failure

Table 3 Parameters in the response message body

Parameter	Type	Description
		For details about the failure causes, see Error Code Reference .
resultDesc	List	Object list in the returned result after a successful query. For details about the parameters of the object, see the following table.

Table 4 Parameters of resultDesc

Parameter	Type	Description
beginTime	String	Statistics start time, which is a timestamp in milliseconds (start time of a 5-minute statistical period).
ccId	Integer	ID of a call center.
vdn	Integer	ID of a VDN.
onlineAgentNums	Integer	Number of online agents (including agents who do not answer inbound calls).
talkingNums	Integer	Number of talking agents (excluding the agents who do not answer inbound calls).
ideaAgentNum	Integer	Number of idle agents (excluding agents who do not answer inbound calls).
busyNums	Integer	Number of agents who indicate busy (excluding the number of agents who do not answer inbound calls).
restNums	Integer	Number of agents who take a rest (excluding agents who do not answer inbound calls).
inCallNums	Integer	Number of inbound calls within five minutes.
rateNum	Integer	Number of manual inbound calls connected within 5 minutes.
callRate	Integer	Manual inbound call connection rate within 5 minutes.
callOutNums	Integer	Number of outbound calls within five minutes.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "ccId":1,
  "vdn":[1,2,3],
  "beginDate":"2021-09-09 00:00:00",
```

```
"endDate": "2021-09-11 23:00:00",  
"limit": "100",  
"offset": "0"  
}
```

- Response parameters

```
{  
  "resultCode": 0,  
  "resultDesc": {  
    "data": [  
      {  
        "beginTime": "1631594100000",  
        "ccId": 1,  
        "vdn": 1,  
        "onlineAgentNums": 1,  
        "talkingNums": 1,  
        "ideaAgentNum": 1,  
        "busyNums": 1,  
        "restNums": 1,  
        "inCallNums": 1,  
        "rateNum": 1,  
        "callRate": 1,  
        "callOutNums": 1  
      }  
    ],  
    "count": 6  
  }  
}
```

4.6.7 Querying the Time of the CTI Platform Server

Scenario

This interface is invoked to query the time of the CTI platform server. (CTI 3.6 does not support this interface.)

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Internal Encapsulation Interface

BMS interface: /ccbms/ws/monitor/queryctiservertime

Request URL

https://ip:port/rest/cmsapp/v1/openapi/configures/ctiservertime

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameter in the request body

No.	Parameter	Value Type	Mandatory	Description
1	cclid	Integer	Yes	ID of a call center. The value ranges from 1 to 65535 .

Response Description

Table 3 Parameters in the response body

No.	Parameter	Value Type	Description
1	result	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
2	resultCode	String	Query result. The options are as follows:

Table 3 Parameters in the response body

No.	Parameter	Value Type	Description
			0100000 : success Other values: failure For details about the failure causes, see Error Code Reference .
3	resultDatas	Long	UTC timestamp of the server where the CTI platform is located. NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
4	resultDesc	Long	UTC timestamp of the server where the CTI platform is located.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "cclId":1
}
```

- Response parameters

```
{
  "result": "0",
  "resultCode": "0100000",
  "resultDatas": 1605057110000,
  "resultDesc": 1605057110000
}
```

4.6.8 Querying the Current License Usage in the System

Scenario

This interface is invoked to query the current license usage in the system. (CTI 3.6 does not support this interface.)

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Internal Encapsulation Interface

BMS interface: /ccbms/ws/monitor/querysystemlicensestatus

Request URL

https://ip:port/rest/cmsapp/v1/openapi/configures/querysystemlicensestatus

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameter in the request message body

No.	Parameter	Value Type	Mandatory	Description
1	cclid	Integer	Yes	ID of a call center. The value ranges from 1 to 65535 .

Response Description

Table 3 Parameters in the response message body

No.	Parameter	Value Type	Description
1	result	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference . NOTE:

Table 3 Parameters in the response message body

No.	Parameter	Value Type	Description
			This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
2	resultCode	String	Query result. The options are as follows: 01000000 : success Other values: failure For details about the failure causes, see Error Code Reference .
3	resultDatas	Object	Object in the returned result after a successful query. For details about the parameters of the object, see Table 4 . NOTE: This field is reserved for compatibility with interface response parameters in 8.13 and earlier versions.
4	resultDesc	Object	Object in the returned result after a successful query. For details about the parameters of the object in this array, see Table 4 .

Table 4 Parameters of resultDesc/resultDatas

No.	Parameter	Value Type	Description
2.1	licenseMaxAgentNum	Long	Maximum number of sign-in agents supported by the license
2.2	licenseMaxVideoAgentNum	Long	Maximum number of sign-in video agents supported by the license
2.3	licenseMaxVdnNum	Long	Maximum number of VDNs supported by the license
2.4	licenseMaxIvrCallNum	Long	Maximum number of automatic business calls supported by the license
2.5	licenseMaxVoiceCallNum	Long	Maximum number of voice calls supported by the license
2.6	licenseMaxWebCallNum	Long	Maximum number of web calls supported by the license
2.7	loginAgentNum	Long	Number of currently signed-in agents
2.8	loginVideoAgentNum	Long	Number of currently signed-in video agents
2.9	vdnNum	Long	Number of VDNs that are currently configured.
2.10	ivrCallNum	Long	Number of current automatic business calls.
2.11	voiceCallNum	Long	Number of current voice calls.
2.12	webCallNum	Long	Number of current web calls.

Table 4 Parameters of resultDesc/resultDatas

No.	Parameter	Value Type	Description
2.13	historyMaxAgentNum	Long	Maximum number of sign-in agents in history.
2.14	historyMaxVideoAgentNum	Long	Maximum number of sign-in video agents in history.
2.15	historyMaxIvrCallNum	Long	Maximum number of automatic business calls in history.
2.16	historyMaxVoiceCallNum	Long	Maximum number of voice calls in history.
2.17	historyMaxWebCallNum	Long	Maximum number of web calls in history.
2.18	licenseStartDate	Long	License start time.
2.19	licenseEndDate	Long	License expiration time.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "ccId":1
}
```

- Response parameters

```
{
  "result": "0",
  "resultCode": "0100000",
  "resultDatas": {
    "licenseMaxAgentNum": 4,
    "licenseMaxVideoAgentNum": 4294967295,
    "licenseMaxVdnNum": 10,
    "licenseMaxIvrCallNum": 4,
    "licenseMaxVoiceCallNum": 6,
    "licenseMaxWebCallNum": 4294967295,
    "loginAgentNum": 1,
    "loginVideoAgentNum": 4294967295,
    "vdnNum": 1,
    "ivrCallNum": 0,
    "voiceCallNum": 1,
    "webCallNum": 4294967295,
  }
}
```



```

    "historyMaxAgentNum": 4294967295,
    "historyMaxVideoAgentNum": 4294967295,
    "historyMaxIvrCallNum": 4294967295,
    "historyMaxVoiceCallNum": 4294967295,
    "historyMaxWebCallNum": 4294967295,
    "licenseStartDate": 0,
    "licenseEndDate": 0
  },
  "resultDesc": {
    "licenseMaxAgentNum": 4,
    "licenseMaxVideoAgentNum": 4294967295,
    "licenseMaxVdnNum": 10,
    "licenseMaxIvrCallNum": 4,
    "licenseMaxVoiceCallNum": 6,
    "licenseMaxWebCallNum": 4294967295,
    "loginAgentNum": 1,
    "loginVideoAgentNum": 4294967295,
    "vdnNum": 1,
    "ivrCallNum": 0,
    "voiceCallNum": 1,
    "webCallNum": 4294967295,
    "historyMaxAgentNum": 4294967295,
    "historyMaxVideoAgentNum": 4294967295,
    "historyMaxIvrCallNum": 4294967295,
    "historyMaxVoiceCallNum": 4294967295,
    "historyMaxWebCallNum": 4294967295,
    "licenseStartDate": 0,
    "licenseEndDate": 0
  }
}

```

4.7 Checking Cluster Health Status

Description

This interface is invoked by a third-party system to check whether the CC-CMS runs properly.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

URL

https://ip:port/rest/cmsapp/v1/ping

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed to application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameter in the message body

Parameter	Type	Mandatory	Description
say (can be modified)	Object	Yes	Request packet. The code does not process the packet and writes it back to the response message body.

Response Description

Table 3 If the response code is **200**, it indicates that the CMS runs properly. The following table describes the parameters in the response message.

Parameter	Type	Description
status	String	If the response is successful, UP is returned.
request	Object	Input parameter object returned when the response is successful.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
Authorization: *****
```

- Request parameters

```
{
  "say": "Hello world!"
}
```

- Response parameters

```
{
  "request": {
    "say": "Hello world!"
  },
  "status": "UP"
}
```

4.8 Interface for Uploading Logs

Description

After the IP address and port number of the NSLB are configured on the OpenEye, an HTTPS interface of the CMS can be invoked to upload logs for fault information collection.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

URL

`https://ip:port/rest/${service.name}/v1/ClientLog/uploadClientLog`

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at multipart/form-data; boundary=<calculated when="" request="" is="" sent=""> .
2	Authorization	String	True	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .
3	FileType	String	True	File type. Currently, only 0 is supported.

Table 2 Parameters in the message body

Parameter	Type	Mandatory	Description
file	MultipartFile	True	Request packet, including log file content and file names.

Response Description

Table 3 If the response code is **200**, it indicates that the CMS runs properly. The following table describes the parameters in the response message.

Parameter	Type	Description
status	string	If the response is successful, 200 is returned.

Message Example

- Request header

```
Content-Type:multipart/form-data; boundary=-----327397481665620642032364
Authorization:*****
FileType:0
```

- Request parameters

```
{
  "file": OELOG_99991093_10.234.63.214_20250226162127.zip
}
```

- Response parameters

```
None
```

4.9 Downloading Inspection Recording Files

Scenario

This interface is invoked to download recording files by call IDs.

If no recording file is found, an error is returned. If there are multiple files, the files are compressed into a ZIP package.

This interface is provided for third-party systems to invoke in the backend.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Request URL

`https://ip:port/rest/cmsapp/v1/openapi/hisindex/downloadRecordFile`

In a public cloud environment, *ip* indicates the IP address of the CC-CMS server, and *port* indicates the HTTPS port number of the CC-CMS server. Contact the administrator to obtain the values.

In a non-public cloud environment where the NSLB service is configured, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Value Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed at application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

Table 2 Parameter in the request body

No.	Parameter	Value Type	Mandatory	Description
1	callID	String	Yes	Call SN.
2	workNo	String	No	Agent ID.
3	beginTime	String	No	Call start time. Format: <i>yyyyMMddHHmmss</i>
4	callerNo	String	No	Calling number.
5	calleeNo	String	No	Called number.

Response Description

Table 3

No.	Parameter	Value Type	Mandatory	Description
1	resultCode	String	Yes	Result code returned for a request. 0 : success; other values: failure
2	resultDesc	String	Yes	Request result description.
3	resultData	Object	Yes	Binary data of an inspection recording file.

Note: If the download fails, the three parameters are displayed. If the download is successful, only binary data is returned.

Message Example

- Request header

```
Content-Type: application/json;charset=UTF-8
```

Authorization: *****

• Request parameters

```
{
  "callID": "1748587841-77268",
  "workNo": "",
  "beginTime": "",
  "callerNo": "",
  "calleeNo": ""
}
```

• Response parameters

```
{
  "resultData": "",
  "resultCode": "1",
  "resultDesc": "Record List is empty"
}
```

5 Appendix

5.1 Indicator Code Reference

5.1.1 Skill Queue Monitoring Indicators

Table 1 Historical monitoring indicators of skill queues

Code	Indicator Name	Description
IDX_COUNT_SKILL_REQUEST	Number of manual inbound call requests	Number of times that an inbound call requests the manual service. The number increases in the following cases: Inbound calls allocated to agents. (An inbound call that is transferred to an agent is counted repeatedly.) Inbound call hangs up while queuing. (If a call is transferred from the current queue to another queue, the call is counted only once.)
IDX_COUNT_SKILL_INCOMING_CALL NOTE: IDX_COUNT_SKILL_INCOMING_CALL is a new indicator in AICC 22.100.0. For the interfaces for obtaining historical monitoring indicators of skills in a specified date and	Number of manual inbound call requests	Both IDX_COUNT_SKILL_INCOMING_CALL and IDX_COUNT_SKILL_REQUEST collect statistics on the number of manual inbound calls, but their statistics logic is different. IDX_COUNT_SKILL_INCOMING_CALL

Table 1 Historical monitoring indicators of skill queues

Code	Indicator Name	Description
month, data in versions earlier than AICC 22.100.0 cannot be collected.		L collects statistics only on the number of manual inbound calls in voice skill queues. The number increases in the following cases: If an inbound call has been allocated to a skill queue, the number of skill queues to which the inbound call is first allocated increases by 1, and other skill queues to which the inbound call is transferred subsequently are not counted.
IDX_COUNT_SKILL_CONNECTED	Number of connected manual inbound call requests	Number of calls queuing in a skill queue and answered by agents in this skill queue or calls transferred to other skill queues and answered by agents in those skill queues. The duration of a call with an agent must be greater than 0.
IDX_COUNT_SKILL_ABANDON	Number of lost manual inbound call requests	Number of lost manual inbound call requests = Number of manual inbound call requests – Number of connected manual inbound call requests
IDX_COUNT_SKILL_USER_ABANDON	Number of abandoned manual inbound calls	Number of calls proactively abandoned by users during queuing and ringing.
IDX_COUNT_SKILL_AGENT_ABANDON	Number of manual inbound calls hung up by agents	Number of calls proactively hung up by agents during ringing.
IDX_COUNT_SKILL_QUEUE_DISCONNECTED	Number of manual inbound calls that are released in queues	Number of times that calls are disconnected by the CTI platform when the calls are queuing.

Table 1 Historical monitoring indicators of skill queues

Code	Indicator Name	Description
IDX_COUNT_SKILL_SYSTEM_LOST	Number of interrupted manual inbound calls on the platform	Number of platform interruptions caused by reasons such as agent response timeout.
IDX_COUNT_SKILL_LOST_RING	Number of manual inbound calls that are not answered	Number of calls that are not answered after being allocated to agents. For the calls, the answer duration is greater than 0 and the conversation duration is 0.
IDX_COUNT_SKILL_TIMEOUT_QUEUE	Number of manual inbound calls that are transferred to a queue due to timeout	Number of calls transferred from the original skill queue to other skill queues due to queuing timeout.
IDX_COUNT_SKILL_TIMEOUT_AGENT	Number of manual inbound calls that are transferred to an agent due to timeout	Number of calls transferred from the original skill queue to other agents due to queuing timeout.
IDX_COUNT_SKILL_OVER_QUEUE	Number of manual inbound calls that are transferred to a queue due to overflow	Number of calls transferred from the original skill queue to other queues due to overflow.
IDX_COUNT_SKILL_OVER_AGENT	Number of manual inbound calls that are transferred to an	Number of calls transferred from the original skill queue to other agents due to overflow.

Table 1 Historical monitoring indicators of skill queues

Code	Indicator Name	Description
	agent due to overflow	
IDX_COUNT_SKILL_CONNECTED_IN_(N)	Number of calls connected manually within <i>N</i> seconds	The value of <i>N</i> can be 3, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, or 60 .
IDX_COUNT_SKILL_ABANDON_IN_(N)	Number of calls abandoned manually within <i>N</i> seconds	The value of <i>N</i> can be 3, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, or 60 . Query by minute is not supported (that is, the input parameter cannot contain timeGrain).
IDX_COUNT_SKILL_CONNECTED_AFTER_(N)	Number of calls connected after <i>N</i> seconds	The value of <i>N</i> can be 3, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, or 60 . Query by minute is not supported (that is, the input parameter cannot contain timeGrain).
IDX_COUNT_SKILL_ABANDON_AFTER_(N)	Number of calls abandoned manually after <i>N</i> seconds	The value of <i>N</i> can be 3, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, or 60 . Query by minute is not supported (that is, the input parameter cannot contain timeGrain).
IDX_COUNT_SKILL_USER_ABANDON_QUEUE	Number of abandoned manual inbound calls during queuing	Number of calls proactively abandoned by users during queuing.
IDX_COUNT_SKILL_USER_ABANDON_AGENT	Number of abandoned manual inbound calls during ringing	Number of calls proactively abandoned by users during ringing.
IDX_COUNT_SKILL_WAIT_FAIL_CONNECTED_QUEUE	Number of Waits for Lost Calls	Number of waits for lost calls on queue devices.

Table 1 Historical monitoring indicators of skill queues

Code	Indicator Name	Description
IDX_COUNT_SKILL_TIMEOUT_IVR	Number of Calls Transferred to the IVR Due to Queuing Timeout	Number of calls that are transferred to IVR due to queuing timeout.
IDX_COUNT_SKILL_OVER_IVR	Number of Calls Transferred to IVR Due to Queue Overflow	Number of times that a user transfers a call to the IVR in the queue overflow.
IDX_RATE_SKILL_CONNECTED	Connected manual inbound call requests (%)	Percentage of the number of connected manual inbound calls to the number of manual inbound call requests. This indicator shows the call processing capability of a skill queue.
IDX_RATE_SKILL_ABANDON	Lost manual inbound call requests (%)	Lost manual inbound call requests (%) = 100% - Connected manual inbound call requests (%). This indicator shows the percentage of calls that are abandoned when they are waiting to be answered. A high call loss rate indicates poor service quality. In this case, improve the agent usage, and add queues or agents.
IDX_RATE_SKILL_RING_CONNECT	Manual inbound call answer rate (%)	Ring rate of manual inbound answered calls = Number of ringing times of answered calls/Total number of ringing times
IDX_DURATION_SKILL_AVG_CONNECTED	Average duration of connected manual inbound calls (s)	Average voice call duration. Average duration of connected manual inbound calls = Duration of connected manual inbound calls/Number of connected manual inbound calls
IDX_DURATION_SKILL_CONNECTED	Manual inbound call duration (s)	Total voice call duration.

Table 1 Historical monitoring indicators of skill queues

Code	Indicator Name	Description
IDX_DURATION_SKILL_AVG_QUEUE_ACK	Average waiting duration of manual inbound calls (s)	Average waiting duration of voice calls. This indicator shows the average waiting duration (queuing and ring duration) of all calls.
IDX_DURATION_SKILL_AVG_CONNECTED_ACK	Average ring duration of connected manual inbound calls (s)	Average ring duration of all connected calls in a queue.
IDX_DURATION_SKILL_AVG_ABANDON_ACK	Average ring duration of lost manual inbound calls (s)	Average ring duration of all unanswered calls.
IDX_DURATION_SKILL_AVG_CONNECTED_QUEUE	Average queueing duration of connected manual inbound calls (s)	Average queueing duration of all connected calls in a queue.
IDX_DURATION_SKILL_AVG_ABANDON_QUEUE	Average queueing duration (s) of manual inbound calls connected but not answered	Average queueing duration of all calls connected but not answered in a queue.
IDX_DURATION_SKILL_MAX_QUEUE_ACK	Maximum waiting duration of manual inbound calls (s)	This indicator shows the maximum waiting duration (queuing and ring duration) of all calls.
IDX_DURATION_SKILL_MIN_QUEUE_ACK	Minimum waiting duration of	This indicator shows the minimum waiting duration (queuing and ring duration) of all calls.

Table 1 Historical monitoring indicators of skill queues

Code	Indicator Name	Description
	manual inbound calls (s)	
IDX_DURATION_SKILL_USER_ABANDON_QUEUE	Queuing duration (s) of a call hung up by a user in the queue	Queuing duration when a user hangs up in a queue.
IDX_DURATION_SKILL_USER_ABANDON_AGENT	Waiting duration (s) of a call hung up by a user during agent ringing	Waiting duration for a user to hang up a call when the phone rings (queuing and ringing).
IDX_DURATION_SKILL_ABANDON_QUEUE	Ring Duration of Lost Calls (s)	Queuing duration of all calls connected but not answered in a queue.
IDX_DURATION_SKILL_ABANDON_ACK	Queuing Duration of Lost Calls (s)	Ring duration of all unanswered calls.
IDX_DURATION_SKILL_WAIT_SUCCESS_CONNECTED_QUEUE	Waiting Duration Before Connection	Waiting duration of calls in a skill queue.
IDX_DURATION_SKILL_SUCCESS_ACK	Ringing Duration Before Connection	Ringing duration of all calls connected in a skill queue.
IDX_DURATION_SKILL_ALL_ACK	Total Ringing Durations	Ringing duration of calls in a skill queue.
IDX_DURATION_SKILL_ALL_QUEUE	Total queuing duration	Total queuing duration of calls in a skill queue.



All indicators that are used to query data by minute or on the current day (the input parameter does not contain **timeGrain**) in [Table 1](#) are real-time indicators.

Table 2 Real-time monitoring indicators of skill queues

Code	Indicator Name	Description
IDX_COUNT_REAL_SKILL_QUEUE	Number of queuing calls	Number of queuing calls.
IDX_COUNT_REAL_SKILL_HANDLE	Number of calls being processed	Number of calls being processed.
IDX_DURATION_SKILL_MAX_QUEUE_ACK	Maximum waiting duration in a queue	Maximum waiting duration in a queue.
IDX_COUNT_20S_CALL_FOR_5MIN	Number of calls connected in 20 seconds in the last 5 minutes	Number of calls connected in 20 seconds in the last 5 minutes.
IDX_COUNT_REAL_AGENT_SIGN_IN	Number of sign-in agents	Number of sign-in agents.
IDX_COUNT_REAL_AGENT_CONNECT	Number of agents in talking state	Number of agents in talking state.
IDX_COUNT_REAL_AGENT_SET_IDLE	Number of agents in idle state	Number of agents in idle state.
IDX_COUNT_REAL_AGENT_USABLE	Number of available agents	Number of available agents.
IDX_COUNT_REAL_AGENT_SET_BUSY	Number of agents in busy state	Number of agents in busy state.
IDX_COUNT_REAL_AGENT_ACW	Number of agents in wrap-up state	Number of agents in wrap-up state.
IDX_COUNT_REAL_AGENT_REST	Number of agents in rest state	Number of agents in rest state.
IDX_COUNT_REAL_AGENT_STUDY	Number of agents in learning state	Number of agents in learning state.

Table 3 Customized indicator of a skill queue

Code	Indicator Name	Description
IDX_SKILL_RESERVED_COL(N)	Customized indicator <i>N</i> of a skill queue	The value ranges from 1 to 20.

5.1.2 Agent Monitoring Indicators

Table 1 Historical agent monitoring indicators

Code	Indicator Name	Description
IDX_COUNT_AGENT_SIGN_IN	Number of sign-in times	Number of agent sign-in times. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are non-real-time indicators.

Table 1 Historical agent monitoring indicators

Code	Indicator Name	Description
IDX_COUNT_AGENT_CONNECTED	Number of inbound calls	Number of manual inbound calls. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are real-time indicators.
IDX_COUNT_AGENT_CALL_OUT	Number of outbound calls	Number of outbound calls. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are real-time indicators.
IDX_COUNT_AGENT_COLLATION	Number of wrap-up times after calls	Number of wrap-up times after calls. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are non-real-time indicators.
IDX_COUNT_AGENT_COLLATION_CALL_IN	Number of wrap-up times after inbound calls	Number of wrap-up times after inbound calls. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are non-real-time indicators.
IDX_COUNT_AGENT_COLLATION_CALL_OUT	Number of wrap-up times after outbound calls	Number of wrap-up times after outbound calls. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are non-real-time indicators.
IDX_COUNT_AGENT_REST	Number of times that an agent enters the rest state	Number of times that an agent enters the rest state. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are non-real-time indicators.

Table 1 Historical agent monitoring indicators

Code	Indicator Name	Description
IDX_COUNT_AGENT_HOLD	Number of hold times	Number of times that an agent holds a call. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are non-real-time indicators.
IDX_COUNT_AGENT_SET_BUSY	Number of times that an agent enters the busy state	Number of times that an agent enters the busy state. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are non-real-time indicators.
IDX_COUNT_AGENT_MOVE_INSIDE	Number of internal transfers	Number of times that an agent transfers internal calls. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are non-real-time indicators.
IDX_COUNT_AGENT_MOVE_OUT	Number of transfer-out times	Number of calls that an agent transfers out. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are non-real-time indicators.
IDX_COUNT_AGENT_MOVE_IVR	Number of calls that an agent transfers to the IVR in hang-up transfer mode	Number of calls that an agent transfers to the IVR in hang-up transfer mode. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are non-real-time indicators.
IDX_COUNT_AGENT_3WC	Number of three-party call times	Number of times that an agent performs three-party conversations. Indicators that are used to query data by minute or on the current day (the input

Table 1 Historical agent monitoring indicators

Code	Indicator Name	Description
		parameters do not contain timeGrain) are non-real-time indicators.
IDX_COUNT_AGENT_INSIDE_CALL	Number of internal calls	Number of internal calls that an agent makes. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are non-real-time indicators.
IDX_COUNT_AGENT_INSIDE_HELP_SEEKING	Number of internal help requests	Number of times that an agent initiates internal help requests. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are non-real-time indicators.
IDX_COUNT_AGENT_REQUEST	Number of requests	Number of manual inbound call requests. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are real-time indicators.
IDX_COUNT_AGENT_ABANDON	Number of lost calls	Number of lost manual inbound calls. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are real-time indicators.
IDX_COUNT_AGENT_WAIT_ANS	Number of calls to be answered after ringing	Number of manual inbound calls to be answered after ringing. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are real-time indicators.

Table 1 Historical agent monitoring indicators

Code	Indicator Name	Description
IDX_COUNT_AGENT_LOST_RING	Number of calls that are not answered after ringing	Number of manual inbound calls that are not answered after ringing. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are real-time indicators.
IDX_COUNT_AGENT_NO_ACK	Number of calls that are not answered by an agent for a long time	Number of manual inbound calls that are not answered by agents for a long time. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are real-time indicators.
IDX_COUNT_AGENT_HANG_UP	Number of hang-up calls (in answering state)	Number of hang-up manual inbound calls when they are answered by agents. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are real-time indicators.
IDX_COUNT_AGENT_CONNECTED_IN_(N)	Number of calls connected manually within <i>N</i> seconds	The value of <i>N</i> can be 3, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, or 60 . Query by minute is not supported (that is, the input parameters do not contain timeGrain). Indicators that are used to query data on the current day (the input parameters do not contain timeGrain) are real-time indicators.
IDX_COUNT_TOTAL_AGENT_CALL_OUT	Total number of outbound calls	Total number of outbound calls, including the number of connected outbound calls and the number of failed outbound calls. Indicators that are used to query data by minute or on the current day (the input parameters do not contain

Table 1 Historical agent monitoring indicators

Code	Indicator Name	Description
		timeGrain) are real-time indicators.
IDX_DURATION_AGENT_SIGN_IN	Sign-in duration	Duration for an agent to sign in. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are non-real-time indicators.
IDX_DURATION_AGENT_CALL_IN	Conversation duration of inbound calls	Conversation duration of manual inbound calls. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are real-time indicators.
IDX_DURATION_AGENT_AVG_CALL_IN	Average conversation duration of inbound calls	Average conversation duration of manual inbound calls. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are real-time indicators.
IDX_DURATION_AGENT_CALL_OUT	Conversation duration of outbound calls	Conversation duration of outbound calls. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are real-time indicators.
IDX_DURATION_AGENT_AVG_CALL_OUT	Average conversation duration of outbound calls	Average conversation duration of outbound calls. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are real-time indicators.
IDX_DURATION_AGENT_COLLATION	Wrap-up duration after a call ends	Wrap-up duration after a call ends. Indicators that are used to query data by minute or on the current day (the input

Table 1 Historical agent monitoring indicators

Code	Indicator Name	Description
		parameters do not contain timeGrain) are non-real-time indicators.
IDX_DURATION_AGENT_COLLATION_CALL_IN	Wrap-up duration after an inbound call ends	Wrap-up duration after an inbound call ends. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are non-real-time indicators.
IDX_DURATION_AGENT_AVG_COLLATION_CALL_IN	Average wrap-up duration after an inbound call ends	Average wrap-up duration after an inbound call ends. (Statistics cannot be collected by minute.) Indicators that are used to query data on the current day (the input parameters do not contain timeGrain) are non-real-time indicators.
IDX_DURATION_AGENT_COLLATION_CALL_OUT	Wrap-up duration after an outbound call ends	Wrap-up duration after an outbound call ends. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are non-real-time indicators.
IDX_DURATION_AGENT_AVG_COLLATION_CALL_OUT	Average wrap-up duration after an outbound call ends	Average wrap-up duration after an outbound call ends. (Statistics cannot be collected by minute.) Indicators that are used to query data on the current day (the input parameters do not contain timeGrain) are non-real-time indicators.
IDX_DURATION_AGENT_AVG_COLLATION	Average wrap-up duration after a call ends	Average wrap-up duration after a call ends. (Statistics cannot be collected by minute.) Indicators that are used to query data on the current day (the input parameters do not contain timeGrain) are non-real-time indicators.
IDX_DURATION_AGENT_REST	Rest duration	Duration for an agent to take a rest.

Table 1 Historical agent monitoring indicators

Code	Indicator Name	Description
		Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are non-real-time indicators.
IDX_DURATION_AGENT_AVG_REST	Average rest duration	Average rest duration of an agent. (Statistics cannot be collected by minute.) Indicators that are used to query data on the current day (the input parameters do not contain timeGrain) are non-real-time indicators.
IDX_DURATION_AGENT_HOLD	Hold duration	Duration for an agent to hold a call. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are non-real-time indicators.
IDX_DURATION_AGENT_AVG_HOLD	Average hold duration	Average duration for an agent to hold a call. (Statistics cannot be collected by minute.) Indicators that are used to query data on the current day (the input parameters do not contain timeGrain) are non-real-time indicators.
IDX_DURATION_AGENT_SET_BUSY	Busy duration	Duration for an agent to show busy. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are non-real-time indicators.
IDX_DURATION_AGENT_AVG_SET_BUSY	Average busy duration	Average duration for an agent to show busy. (Statistics cannot be collected by minute.) Indicators that are used to query data on the current day (the input parameters do not contain timeGrain) are non-real-time indicators.

Table 1 Historical agent monitoring indicators

Code	Indicator Name	Description
IDX_DURATION_AGENT_RING	Ring duration	Ring duration of manual inbound calls. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are real-time indicators.
IDX_DURATION_AGENT_FREE	Idle duration	Duration for an agent to show idle. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are non-real-time indicators.
IDX_DURATION_AGENT_AVG_RING	Ring duration	Average ring duration of manual inbound calls. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are real-time indicators.
IDX_DURATION_AGENT_MAX_CALLING	Maximum call duration	Maximum conversation duration of manual inbound calls. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are real-time indicators.
IDX_DURATION_AGENT_MIN_CALLING	Minimum call duration	Minimum conversation duration of manual inbound calls. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are real-time indicators.
IDX_RATE_AGENT_CONNECTED	Call connection rate	Ratio of the number of successful manual inbound calls to the number of manual inbound call requests. Indicators that are used to query data by minute or on the current day (the input

Table 1 Historical agent monitoring indicators

Code	Indicator Name	Description
		parameters do not contain timeGrain) are real-time indicators.
IDX_RATE_AGENT_SUCC_CALL_OUT	Outbound call success rate	Ratio of the number of outbound calls to the total number of outbound calls. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are real-time indicators.
IDX_COUNT_AGENT_ARRANGE	Number of wrap-up times	Number of times that an agent enters the wrap-up state. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are non-real-time indicators.
IDX_DURATION_AGENT_ARRANGE	Wrap-up duration	Duration for an agent to be in wrap-up state. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are non-real-time indicators.
IDX_COUNT_AGENT_PLAY	Number of voice playback times	Number of times that an agent plays voice. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are non-real-time indicators.
IDX_DURATION_AGENT_PLAY	Voice playback duration	Duration for an agent to show idle. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are non-real-time indicators.

Table 2 Real-time agent monitoring indicators

Code	Indicator Name	Description
IDX_AGENT_PHONE	Phone number	Agent phone number. An agent needs to sign in.
IDX_AGENT_CURRENT_STATE	Agent status	Agent status. For details, see CTI Platform Agent Status Code Reference .
IDX_DURATION_AGENT_CURRENT_STATE	Current status duration	Duration of the current agent status, in seconds.
IDX_AGENT_LOGIN_SKILLS	List of Skills Signed In or Owned	Array of skill queue IDs.

Table 3 Customized agent indicator

Code	Indicator Name	Description
IDX_AGENT_RESERVED_COL(N)	Customized indicator <i>N</i> for agents	The value ranges from 1 to 20.

5.1.3 VDN Monitoring Indicators

[Table 1](#) describes the historical monitoring indicators by VDN.

Table 1 Historical monitoring indicators by VDN

Code	Indicator Name	Description
IDX_COUNT_TOTAL_CALL	Total number of calls	Total number of inbound calls + Total number of outbound calls. Only query by minute is supported. The input parameters must contain timeGrain . Indicators that are used to query data by minute are real-time indicators.
IDX_COUNT_CONNECTED_CALL	Total number of connected calls	Total number of connected inbound calls + Total number of connected outbound calls. Only query by minute is supported. The input parameters must contain timeGrain . Indicators that are used to query data by minute are real-time indicators.
IDX_COUNT_ABANDONED_CALL	Total number of lost calls	Total number of calls – Total number of connected calls. Only query by minute is supported. The input parameters must contain timeGrain .

Table 1 Historical monitoring indicators by VDN

Code	Indicator Name	Description
		Indicators that are used to query data by minute are real-time indicators.
IDX_COUNT_INBOUND_CALL	Total number of inbound calls	Query by minute is supported. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are real-time indicators.
IDX_COUNT_INBOUND_CONNECTED	Total number of connected inbound calls	Query by minute is supported. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are real-time indicators.
IDX_COUNT_OUTBOUND_CALL	Total number of outbound calls	Query by minute is supported. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are real-time indicators.
IDX_COUNT_OUTBOUND_CONNECTED	Total number of connected outbound calls	Query by minute is supported. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are real-time indicators.
IDX_COUNT_AGENT_INBOUND	Number of manual inbound calls	Query by minute is supported. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are real-time indicators.
IDX_COUNT_AGENT_INBOUND_CONNECTED	Number of connected manual inbound calls	Query by minute is supported. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are real-time indicators.

Table 1 Historical monitoring indicators by VDN

Code	Indicator Name	Description
IDX_COUNT_IVR_INBOUND	Total number of IVR inbound calls	Query by minute is supported. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are real-time indicators.
IDX_COUNT_IVR_INBOUND_CONNECTED	Number of connected IVR inbound calls	Query by minute is supported. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are real-time indicators.
IDX_COUNT_IVR_OUTBOUND	Number of connected IVR outbound calls	Query by minute is supported. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are real-time indicators.
IDX_UNIQUE_COUNT_IVR_INBOUND	Number of IVR inbound call requests (by call)	One call is calculated only once (Query by minute is supported.) Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are real-time indicators.
IDX_UNIQUE_COUNT_IVR_INBOUND_CONNECTED	Number of connected IVR inbound calls (by call)	One call is calculated only once (minute-granularity query is supported). Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are real-time indicators.
IDX_UNIQUE_COUNT_IVR_OUTBOUND	Number of IVR outbound call requests (Call Dimension)	One call is calculated only once (minute-granularity query is supported). Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are real-time indicators.

Table 1 Historical monitoring indicators by VDN

Code	Indicator Name	Description
IDX_UNIQUE_COUNT_IVR_OUTBOUND_CONNECTED	Number of connected IVR outbound calls (by call)	One call is calculated only once (minute-granularity query is supported). Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are real-time indicators.
IDX_COUNT_AGENT_OUTBOUND_CONNECTED	Number of connected manual outbound calls	Query by minute is supported. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are real-time indicators.
IDX_COUNT_INBOUND_ABANDON_CONNECTED	Number of abandoned inbound calls on the current day	Number of inbound calls on the current day – Number of connected inbound calls on the current day. Query by minute is not supported. Indicators that are used to query data on the current day (the input parameters do not contain timeGrain) are real-time indicators.
IDX_RATE_IVR_INBOUND_CONNECTED	IVR inbound call connection rate	Query by minute is supported. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are real-time indicators.
IDX_RATE_INBOUND_CONNECTED	Total inbound call connection rate	Query by minute is supported. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are real-time indicators.
IDX_RATE_OUTBOUND_CONNECTED	Total outbound call connection rate	Query by minute is supported. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are real-time indicators.

Table 1 Historical monitoring indicators by VDN

Code	Indicator Name	Description
IDX_RATE_AGENT_INBOUND_CONNECTED	Manual inbound call connection rate	Query by minute is supported. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are real-time indicators.
IDX_RATE_AGENT_INBOUND_ABANDON	Manual inbound call abandoned rate	Query by minute is not supported. Indicators that are used to query data on the current day (the input parameters do not contain timeGrain) are real-time indicators.
IDX_RATE_PHONE_INBOUND_ABANDON	Inbound call abandoned rate on the current day	Query by minute is not supported. Indicators that are used to query data on the current day (the input parameters do not contain timeGrain) are real-time indicators.
IDX_RATE_ABANDON_IN_(N)	Rate of calls abandoned within <i>N</i> seconds. <i>N</i> can be 3, 5, 10, 30, or 60.	Query by minute is not supported. Indicators that are used to query data on the current day (the input parameters do not contain timeGrain) are real-time indicators.
IDX_RATE_AGENT_CONNECTED_IN_(N)	Rate of calls connected manually in <i>N</i> seconds. <i>N</i> can be 3, 5, 10, 30, or 60.	Query by minute is not supported. Indicators that are used to query data on the current day (the input parameters do not contain timeGrain) are real-time indicators.
IDX_RATE_AGENT_CONNECTED_AFTER_(N)	Rate of calls connected manually after <i>N</i> seconds. <i>N</i> can be 10 or 60.	Query by minute is not supported. Indicators that are used to query data on the current day (the input parameters do not contain timeGrain) are real-time indicators.
IDX_DURATION_AVG_VDN_INBOUND	Average duration that inbound calls occupy the VDN (s)	Query by minute is supported. Indicators that are used to query data by minute or on the current day (the input

Table 1 Historical monitoring indicators by VDN

Code	Indicator Name	Description
		parameters do not contain timeGrain) are real-time indicators.
IDX_DURATION_AVG_VDN_OUTBOUND	Average duration that outbound calls occupy the VDN (s)	Query by minute is supported. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are real-time indicators.
IDX_DURATION_AVG_CONNECTED	Average inbound call duration (s)	Query by minute is not supported. Indicators that are used to query data on the current day (the input parameters do not contain timeGrain) are real-time indicators.
IDX_DURATION_AVG_WAIT	Average inbound call waiting duration (s)	Query by minute is not supported. Indicators that are used to query data on the current day (the input parameters do not contain timeGrain) are real-time indicators.
IDX_RATE_AGENT_CALL_IN_CONNECTED	Manual inbound call connection rate	Number of connected manual inbound calls/Number of manual requests. Requests for transfer to another agent are counted repeatedly. Query by minute is not supported. Indicators that are used to query data on the current day (the input parameters do not contain timeGrain) are real-time indicators.
IDX_RATE_AGENT_CALL_IN_CONNECTED_IN_20	Rate of manual inbound calls connected within 20 seconds	Number of manual inbound calls connected within 20 seconds/Total number of connected inbound calls. Query by minute is not supported. Indicators that are used to query data on the current day (the input parameters do not contain timeGrain) are real-time indicators.

Table 1 Historical monitoring indicators by VDN

Code	Indicator Name	Description
IDX_DURATION_AVG_AGENT_CALL_IN_CONNECTED	Average manual inbound call duration	Query by minute is not supported. Indicators that are used to query data on the current day (the input parameters do not contain timeGrain) are real-time indicators.
IDX_DURATION_AVG_AGENT_CALL_OUT_CONNECTED	Average manual outbound call duration	Query by minute is not supported. Indicators that are used to query data on the current day (the input parameters do not contain timeGrain) are real-time indicators.
IDX_COUNT_IVR_ONLY_INBOUND	Number of inbound calls that are connected using only the IVR	Query by minute is supported. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are real-time indicators.
IDX_COUNT_IVR_ONLY_INBOUND_CONNECTED	Number of successful calls that are connected using only the IVR	Query by minute is supported. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are real-time indicators.
IDX_DURATION_IVR_ONLY_INBOUND	Total duration of inbound calls that are connected using only the IVR, in seconds	Query by minute is supported. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are real-time indicators.
IDX_COUNT_SYSTEM_INBOUND	Number of inbound calls on the platform	Query by minute is supported. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are real-time indicators.
IDX_COUNT_SYSTEM_IN_SUCC	Number of connected calls on the platform	Query by minute is supported. Indicators that are used to query data by minute or on the current day (the input

Table 1 Historical monitoring indicators by VDN

Code	Indicator Name	Description
		parameters do not contain timeGrain) are real-time indicators.
IDX_COUNT_AGENT_ASSIGNED	Number of manual connected calls on the platform	Query by minute is supported. Indicators that are used to query data by minute or on the current day (the input parameters do not contain timeGrain) are real-time indicators.

[Table 2](#) describes the real-time monitoring indicators by VDN.

Table 2 VDN real-time monitoring indicator codes (unavailable in CTI 3.6)

Code	Indicator Name	Description
IDX_COUNT_REAL_VDN_CONNECT	Number of VDN connected calls	-
IDX_COUNT_REAL_VDN_CALL	Number of VDN calls	-
IDX_COUNT_REAL_IVR_HANDLE	Number of calls processed by IVR	-
IDX_COUNT_REAL_SKILL_QUEUE	Number of queuing calls	-
IDX_COUNT_REAL_AGENT_SIGN_IN	Number of sign-in agents	-
IDX_COUNT_REAL_AGENT_SET_IDLE	Number of agents in idle state	-
IDX_COUNT_REAL_AGENT_SET_BUSY	Number of agents in busy state	-
IDX_COUNT_REAL_AGENT_CONNECT	Number of agents in talking state	-
IDX_COUNT_REAL_AGENT_STUDY	Number of agents in learning state	-
IDX_COUNT_REAL_AGENT_ACW	Number of agents in wrap-up state	-
IDX_COUNT_REAL_AGENT_ANSWER	Number of agents in answering state	-
IDX_COUNT_REAL_AGENT_PREOCCUPY	Number of preoccupied agents	-
IDX_COUNT_REAL_AGENT_OCCUPY	Number of occupied agents	-
IDX_COUNT_REAL_AGENT_NOT_SIGN_IN	Number of agents who have not signed in	-
IDX_COUNT_REAL_AGENT_REST	Number of agents in rest state	-

5.1.4 IVR Monitoring Indicators

Table 1 IVR historical monitoring indicator codes

Code	Indicator Name	Description
IDX_COUNT_IVR_INBOUND_CALL	Number of IVR inbound call requests	Query by minute is supported.
IDX_COUNT_IVR_INBOUND_CONNECTED	Number of connected IVR inbound calls	Query by minute is supported.
IDX_COUNT_IVR_ABANDONED_CALL	Number of lost IVR calls	Query by minute is supported.
IDX_COUNT_IVR_INBOUND_FLOWOUT	Number of transferred IVR inbound calls	Query by minute is supported.
IDX_COUNT_IVR_INBOUND_USER_ABORT	Number of released IVR inbound calls	Query by minute is supported.
IDX_RATE_IVR_INBOUND_CONNECTED	IVR inbound call connection rate	Query by minute is supported.
IDX_DURATION_AVG_IVR_CONNECTED	Average IVR call duration	Query by minute is supported.
IDX_COUNT_IVR_TO_AGENT	Number of IVR-to-manual calls	Query by minute is supported.

 NOTE

All indicators that are used to query data by minute or on the current day (the input parameter does not contain **timeGrain**) in [Table 1](#) are real-time indicators.

5.2 Error Code Reference

When an error is reported during interface invoking, the values of **resultCode** and **resultDesc** are returned. The following is an example:

```
{
  "resultCode": "0100002",
  "resultDesc": "Invalid parameter"
}
```

Error Code	Description
0100000	The interface is invoked successfully.
0100001	The interface fails to be invoked.
0100002	The parameter is invalid.
0100003	The current interface supports only CTI V300R008. The current CTI version is V300R006.

Error Code	Description
0100999	Unknown error.
0101001	The inspection relationship interface is abnormal.
0101002	The number to be queried does not exist on the UAP.
0101003	Insufficient permission.

5.3 CTI Platform Agent Status Code Reference

No.	Description
0	Not signed in.
1	Idle.
2	Preoccupied.
3	Occupied.
4	Answering.
5	Talking.
6	Working.
7	Showing busy.
8	Rest
9	Study
10	Adjusting.

5.4 Description of Call Media Types

Table 1 Call media types

No.	Media stream type.	Description
1	MEDIA_TYPE_CHAT	Text chat.
2	MEDIA_TYPE_WEBPHONE	Setting Up a CTD Call
3	MEDIA_TYPE_ESCORT	Escorted browsing and form sharing.
4	MEDIA_TYPE_CALLBACK	Callback request call
5	MEDIA_TYPE_PHONE	Common voice call
6	MEDIA_TYPE_EMAIL	Email call. NOTE:

Table 1 Call media types

No.	Media stream type.	Description
		The current version does not support this type.
7	MEDIA_TYPE_FAX	Fax call. NOTE: The current version does not support this type.
8	MEDIA_TYPE_VIDEO	IP video call (H.323).
9	MEDIA_TYPE_WB	Electronic whiteboard.
10	MEDIA_TYPE_APP_SHARE	Application sharing.
11	MEDIA_TYPE_FILE_TRANSFER	File transfer.
12	MEDIA_TYPE_VIDEO_2B1D	2B+D ISDN video call.
13	MEDIA_TYPE_VIDEO_6B1D	6B+D ISDN video call
14	MEDIA_TYPE_OPS	OPS call
15	MEDIA_TYPE_PREDICT_OUTBOUND	Predicted outbound call.
16	MEDIA_TYPE_PREVIEW_OUTBOUND	Previewed outgoing call
17	MEDIA_TYPE_MSG	Message media.
18	MEDIA_TYPE_WEBPHONE_VIDEO	Click-to-dial video call
19	MEDIA_TYPE_PHONE_VIDEO	Common video call.
20	MEDIA_TYPE_WEB_LEAVE_MESSAGE	Non-real-time call
21	MEDIA_TYPE_DESKTOP_SHARE	Desktop Sharing Media Type
22	MEDIA_TYPE_VC_CALL	Calling Number of an Incoming Call in a Skill Queue
23	CLASSIC_MEDIA_TYPE_NUM	Number of contact media types
50	MEDIA_TYPE_MULTI_MEDIA_EMAILCHAT	Email chat
51	MEDIA_TYPE_MULTI_MEDIA_WEBCHAT	Network Chat
52	MEDIA_TYPE_MULTI_MEDIA_SMSCHAT	SMS chat
53	MEDIA_TYPE_MULTI_MEDIA_SOCIALCHAT	Text chat
54	MEDIA_TYPE_MULTI_MEDIA_FAXCHAT	Fax Chat
55	MEDIA_TYPE_MULTI_MEDIA_H5CHAT	H5 video.

5.5 Description of Call Submedia Types

Table 1 Call submedia types (corresponding to call media type No. 53)

No.	Submedia Type	Description
1	SUB_MEDIA_TYPE_WEB	Web
2	SUB_MEDIA_TYPE_WHATSAPP	WhatsApp
3	SUB_MEDIA_TYPE_LINE	LINE
4	SUB_MEDIA_TYPE_WECHAT	WeChat
5	SUB_MEDIA_TYPE_FACEBOOK	FaceBook Messenger
6	SUB_MEDIA_TYPE_TWITTER	Twitter
7	SUB_MEDIA_TYPE_EMAIL	Email
8	SUB_MEDIA_TYPE_SMS	SMS
9	SUB_MEDIA_TYPE_TELEGRAM	Telegram
10	SUB_MEDIA_TYPE_INSTAGRAM	Instagram

5.6 Description of Call Types

Indicates the call type.

Table 1 Cal type

No.	Cal type	Description	Call mode
0	SP_CALL_NORMAL	Common inbound call.	Indicates that an incoming call is received.
1	SP_CALL_SPECIAL	Special Incoming Call	Indicates that an incoming call is received.
2	KE11_CALL	VMS call (obtaining voice information)	Outgoing Call
3	KE15_CALL	VMS call (deleting voice information)	Outgoing Call
4	BP_CALL	VMS call (sending information to the BP)	Outgoing Call

Table 1 Cal type

No.	Cal type	Description	Call mode
5	SP_CALL_LONG	Toll incoming call	Indicates that an incoming call is received.
6	INTER_CALL	internal call An internal call is a call that is made by an agent to another agent and is successfully answered by the called agent. Internal calls cannot be transferred out or muted.	Outgoing Call
7	SP_CALL_OUT	common outbound call It refers to outgoing calls made by agents.	Outgoing Call
8	OP_CALL_OUT	Agent outgoing call type	Outgoing Call
9	OP_PRI_OUT	PRI outgoing call	Outgoing Call
10	IVR_CALL_OUT	IVR outgoing call	Outgoing Call
11	SELF_CALLOUT	Outgoing call made by a phone agent NOTE: This parameter is valid only when the function of recording PC+PHONE call CDRs is enabled. To set this parameter, choose Web Configuration Console > VDN1 Configuration > Basic Information Configuration > Configure CDR Writing as a VDN administrator.	Outgoing Call
12	IVR_PRI_CALL	IVR PRI outgoing call	Outgoing Call
13	NIRC_CALL_IN	Incoming call of the network call center	Indicates that an incoming call is received.
14	NIRC_CALL_OUT	Outgoing call of the network call center	Outgoing Call
15	NIRC_SPY_CALL_IN	Incoming inspection call of the network call center	Indicates that an incoming

Table 1 Cal type

No.	Cal type	Description	Call mode
			call is received.
16	NIRC_SPY_CALL_OUT	Outgoing inspection call of the network call center	Outgoing Call
17	NIRC_INTER_CALL_IN	Internal incoming network call to a virtual agent	Indicates that an incoming call is received.
18	NIRC_INTER_CALL_OUT	Internal outgoing network call to a virtual agent	Outgoing Call
20	ICD_SP_OPS_AIDERCALL	Auxiliary call	Outgoing Call
21	ICD_SP_OPS_INCALL	Incoming call (international agent)	Indicates that an incoming call is received.
22	ICD_SP_OPS_PASSCALL	Relayed call	Outgoing Call
23	ICD_SP_OPS_GIVECALL	Outgoing call (international agent)	Outgoing Call
30	ICD_SP_VCP_ISDNCALL_IN	Actual incoming video call (initiated by the ACD)	Indicates that an incoming call is received.
31	ICD_SP_VCP_ISDNCALL_OUT	Actual outgoing video call (to the ACD)	Outgoing Call
32	ICD_SP_VCP_VIRTUALCALL_IN_2BD	Incoming 2B+D virtual video call (two calls are integrated into one call)	Indicates that an incoming call is received.
33	ICD_SP_VCP_VIRTUALCALL_IN_6BD	Incoming video 6B+D virtual call (six calls are integrated into one call)	Indicates that an incoming call is received.
34	ICD_SP_VCP_VIRTUALCALL_OUT_2BD	Outgoing 2B+D Virtual Video Call (Initiated by the CCS)	Outgoing Call

Table 1 Cal type

No.	Cal type	Description	Call mode
35	ICD_SP_VCP_VIRTUALCALL_OUT_6BD	6B+D virtual outgoing video call (initiated by the CCS)	Outgoing Call
40	OUTBOUND_CALL_OUT	Reserved outgoing call The system queries the reserved outgoing call database regularly. After obtaining the reserved calls whose time is earlier than the current computer time, the system forcibly makes the calls.	Outgoing Call
41	ICD_SP_OUTBOUND_PRE_OCCUPY	Preoccupied outbound voice calling The system occupies an idle agent and then makes an outgoing call. If the outgoing call succeeds, the idle agent handles the call.	Outgoing Call
42	ICD_SP_OUTBOUND_PRE_CONNECT	Pre-connected outgoing call The system makes an outgoing call to a customer. If the outgoing call succeeds, the system connects the call to an agent after the phone of the customer rings.	Outgoing Call
43	ICD_SP_OUTBOUND_VIRTUAL_CALLIN	Predicted outgoing call The system makes an outgoing call to a customer. If the outgoing call succeeds, the system connects the call to an agent after the customer picks up the phone.	Outgoing Call
44	ICD_SP_OUTBOUND_PREVIEW	Previewed outgoing call The system sends information about an outgoing call to an agent. The agent makes the outgoing call after previewing the outgoing call information.	Outgoing Call
45	ICD_SP_OUTBOUND_CALLBACK	Callback Request If a web user needs to contact a call center when browsing web pages on the Internet, the web user can use the callback request service. An agent in the call center then calls the phone number specified by the web user. After the call is connected, the web user can receive voice guidance of the agent and enjoy other services such as the escorted browsing service at the same time.	Outgoing Call
46	ICD_SP_IDD	International incoming toll call	Indicates that an incoming

Table 1 Cal type

No.	Cal type	Description	Call mode
			call is received.
50	ICD_SP_CBRT_CALL	RBT business call	Outgoing Call
51	ICD_SP_INTERNAL_TWO_HELP	Two-party help call When answering an incoming call, an agent can ask for internal help. In the case of a two-party help call, the call of the customer is held, and the agent talks with the requested agent.	Outgoing Call
52	ICD_SP_INTERNAL_THREE_HELP	Three-party help call Attendants can ask for internal help when answering incoming calls. In the case of a three-party help call, the customer, agent, and requested agent can talk with each other. A two-party call for internal help can be switched to a three-party call.	Outgoing Call
60	ICD_SP_OUTBOUND_PRE_OCCUPY_PRI_CALL	Preoccupied PRI outgoing call	Outgoing Call
61	ICD_SP_OUTBOUND_PRE_CONNECT_PRI_CALL	Pre-connected PRI outgoing call	Outgoing Call
62	ICD_SP_OUTBOUND_VIRTUAL_CALLIN_PRI_CALL	Virtual incoming and outgoing calls in the PRI mode	Outgoing Call
63	ICD_SP_OUTBOUND_PREVIEW_PRI_CALL	Previewed PRI outgoing call	Outgoing Call
64	ICD_SP_OUTBOUND_CALLBACK_PRI_CALL	PRI callback request	Outgoing Call
80	SP_SUBSCRIBE_CALLIN	SUBSCRIBE incoming call	Indicates that an incoming call is received.
81	SP_SUBSCRIBE_CALLOUT	SUBSCRIBE outgoing call	Outgoing Call
82	SP_VOICE_OPEN_CALLIN	Incoming Internet Voice Call	Indicates that an incoming call is received.

Table 1 Cal type

No.	Cal type	Description	Call mode
83	SP_VOICE_OPEN_CALLOUT	Internet voice outgoing call	Outgoing Call
90	SP_ROUTEOUT_CALLOUT	Routing an Outgoing Call to an External DN	Outgoing Call
92	BRIDGE_PHONE_CALL_IN	Incoming call of the agent phone bridge call	Indicates that an incoming call is received.
93	BRIDGE_PHONE_CALL_OUT	Outgoing Call of an Agent Phone Bridge Call	Outgoing Call
94	AGENT_CONF_CALL	Manual Conference Occupation Call	Outgoing Call
95	OFFLINE_OUT	Offline Outbound Call	Outgoing Call
96	SMC_PROXY_CALL	Proxy Call for Video-based Incident Reporting	Outgoing Call

5.7 Agent Type Description

Agent type.

Table 1 Agent type.

Value	Agent type.	Description
0	AGENT_TYPE_EXTENSION	Extension type.
1	AGENT_TYPE_USERLINE	Type of a subscriber line
2	AGENT_TYPE_1B1D	1B+D agents (including TUAs and always-off-hook agents)
3	AGENT_TYPE_SINGLE_PHONE	Single and non-permanently held phone agent
4	AGENT_TYPE_PC_PHONE	PC+Phone agent
5	AGENT_TYPE_ISDN_2BD	2B+D video agent
6	AGENT_TYPE_ISDN_6BD	6B+D video agent
7	AGENT_TYPE_1B1D_PHONE	Phone agent, which is similar to the 1B+1D agent on the CTI side
8	AGENT_TYPE_MANAGER	Configuration management agent

Table 1 Agent type.

Value	Agent type.	Description
9	AGENT_TYPE_NB1D	nB+D agent
10	AGENT_TYPE_NET	network agent
11	AGENT_TYPE_PC_PHONE_VIDEO	Video agent (with audio and video capabilities)
12	AGENT_TYPE_SINGLE_PHONE_VIDEO	Single-phone video agents
13	AGENT_TYPE_EMEDIA	Multimedia Agent

5.8 Call Features

Table 1 Call features

Value	Description
0	Call from a common customer
1	Call from an agent
2	Toll call from a customer
3	Routing initiated after the CTI receives a routed network call
4	International inbound toll call
40	Reserved outbound call
41	Preoccupied outbound call
42	Pre-connected outbound call
43	Virtual inbound or outbound call
44	Previewed outbound call
45	Callback request
51	Internal help call

5.9 Reasons for Leaving the Device of the Call

Table 1 Reasons for leaving the device of the call

Cause code	Description
0	Normal access/transfer.
1	Overflow transfer.

Table 1 Reasons for leaving the device of the call

Cause code	Description
2	Timeout transfer.
3	Transfer on no agent.
4	Transfer on queue cancellation.
5	Transfer because the agent does not answer the call.
6	Resume a suspended call.
7	Connecting a held call.
8	Three-party help.
9	Picking a held call.
10	Three-party conversation.
11	The party requesting internal help releases the call.
12	The call is intercepted.
13	The call is picked up.
14	The analysis of the called is not configured.
15	The user releases the call.
16	The agent releases the call.
17	The call is transferred out.
18	The call is held.
19	An internal processing error occurs.
20	The called party is busy.
22	The mobile phone is switched off.
23	The mobile phone is out of the service area.
24	The called number does not exist.
25	The ACD cannot distribute any resources for the call.
26	The user does not answer the call.
27	The called party rejects the call.
28	The call is successfully transferred to a skill queue.

Table 1 Reasons for leaving the device of the call

Cause code	Description
29	The three-party conversation is transferred to a skill queue.
30	Transferring the call in the success transfer mode times out.
31	Transferring the three-party conversation to a skill queue times out.
34	If call timeout or queuing on busy or overflow occurs, the network call fails.
35	The network call is directed to its original call center for queuing.
36	The queued call is re-routed.
37	The call accesses the CDN through call routing.
38	The call accesses the CDN through agent consultation.
39	The CDN properly returns a routing result.
40	CDN routing times out.
41	An agent cancels the consultation call, or the routing fails.
42	The call accesses the PGR because CDN routing times out.
43	The call is properly returned by the CDN and is routed to the PGR.
44	The call accesses the PGR through inbound routing.
45	The call accesses the PGR through agent consultation.
46	The agent is unreachable. The CDN reroutes the call.
47	The call is waiting in multiple queues.
48	The primary-queue call is answered by numbers in the secondary queue.
50	The agent rejects the call.
51	A user call joins in a conference call.
52	A conference chairperson proactively releases a conference, or the conference is released due to a CCS exception.
53	The participant is removed from the conference.
54	The call is transferred when the call completion rate is less than the threshold.
88	After the agent answers the call, the user hangs up.
89	After the agent answers the call, the user hangs up.

5.10 Cause Code

Reason for call release. For details, see [Call_release_cause_EN.xlsx](#).

5.11 Device Types

Table 1 Device type

Device type ID	Device type
1	Skill queue
2	Agent
3	IVR service flow
5	Call
7	Temporary routing device
8	Virtual device
10	CDN device
11	MELCAS device
14	PGR (default CDN route controlled by the SCAI)
15	Conference call

5.12 Agent Operation Types

Table 1 Types of operations

Operation Type	Description	Operation Object	Operation Reason
NOTE: In the following, - filled in the operation object and reason columns indicates that the corresponding operation does not write the agent operation bill. US_NO_USE= 0xFFFF; 65535			
0	The agent logs out of the CCS.	US_NO_USE	1-255: Causes are transmitted by clients. 256: An agent console is disconnected from the call center platform. 257: An agent is forced by a QC inspector to log out of the call center platform. 258: An agent proactively logs out of the call center platform. 259: An agent logs out of the call center platform before re-login.

Table 1 Types of operations

Operation Type	Description	Operation Object	Operation Reason
			260: The handshake for agent control blocks between the media server and CCS times out. 261: The CCS detects the time-out of the handshake for agent control blocks between the CCS and UIS. 262: An agent logs out of the call center platform because the media server is disconnected from the CCS. 263: The handshake between the UIS and agent console times out. 264: The agent ID is being used to forcibly log in on another client. As a result, the current agent is forced to log out.
1	The agent logs in to the CCS.	US_NO_USE	US_NO_USE
2	The agent indicates idle.	-	-
3	The agent indicates busy.	US_NO_USE	1: The agent indicates busy. 2: The agent indicates busy forcibly for not answering the call when the phone rings for a long time. 3: The agent being inspected is forced to indicate busy. 4: The agent has a rest. 5: The agent phone is unavailable. 7: The agent rejects the call. 200–250: reason codes defined by the service side
4	Conversation status.	US_NO_USE	1: The agent does not really exit the conversation status. Bills are generated every five minutes starting from 00:00 to 23:55. 0: The agent fully exits the conversation status. NOTE: The conversation status in these bills refers to only text conversations.
5	Waiting for answer from an agent after distributing a call to the agent.	-	-
6	The agent waits for the connection success message after answering a call or	-	-

Table 1 Types of operations

Operation Type	Description	Operation Object	Operation Reason
	making an outgoing call successfully.		
7	The agent releases a call.	-	-
8	The agent requests disconnecting.	-	-
9	The agent waits for the called to ring when making an outgoing call.	-	-
10	Three-party conversation.	Work ID	US_NO_USE
11	Agent channel fails or the agent is down.	-	-
12	Agent channel or agent is restored.	-	-
13	The call transfers from the manual flow to the automatic flow. The agent is suspended.	US_NO_USE	US_NO_USE
14	The agent is listened in to or intruded.	-	-
15	The agent is not listened in to or intruded.	-	-
16	The agent is recorded.	-	-
17	The agent is not recorded.	-	-
18	The agent is monitored.	-	-
19	The agent is not monitored.	-	-
20	The agent is listening in or intruding or whispering.	Work ID	0: listening in to another agent
		US_NO_USE	2: listening in to a trunk
21	The agent is not listening in or intruding or whispering.	-	-
22	Voice playing.	US_NO_USE	US_NO_USE
23	Stopping the voice playing.	-	-
24	Monitoring.	-	-

Table 1 Types of operations

Operation Type	Description	Operation Object	Operation Reason
25	Stopping monitoring.	-	-
26	The agent rests.	Work ID	The reason for agent rest is specified when the client invokes CCC_REQUEST_REST_EX_Msg (The default value is 0.)
27	Work status.	US_NO_USE	0: Adjusting after the call The service system sets that the agent automatically enters the Working state. 0-255: Adjusting The agent invokes an interface to enter the Working state. The default value is 0.
28	Listening in to and intruding a call.	-	-
29	Forcible logout.	Work ID	US_NO_USE
30	Forcible idle.	Work ID	US_NO_USE
31	Forcible busy.	Work ID	US_NO_USE
32	The agent is mute to the user.	US_NO_USE	US_NO_USE
33	The agent cancels the mute status to the user.	-	-
34	The agent enters the study status.	US_NO_USE	US_NO_USE
35	The agent enters the work status.	-	-
36	The agent enters the adjusting status.	-	-
37	The agent ends the work status.	-	-
38	The agent holds the call.	Work ID of a QC inspector	1: The call is held by a QC inspector.
		US_NO_USE	
39	The agent picks the held call.	-	-
40	The agent requests an internal help.	Work ID	0: Help between uncertain parties

Table 1 Types of operations

Operation Type	Description	Operation Object	Operation Reason
			1: Requesting help between two parties 2: Requesting help between three parties 3: The call is transferred. The called party talks to the agent first after answering the call. The agent connects the caller and the called while hooking on. 4: The call is transferred among three parties. The called party talks to the agent first after answering the call. The agent connects the caller and the called while hooking on. 5: Three-party conversation 6: Connecting the call
41	The agent makes an internal call.	Work ID	US_NO_USE
42	The agent connects a held call.	-	-
43	The agent intercepts a call.	Work ID	US_NO_USE
44	The agent halts the voice playing.	US_NO_USE	US_NO_USE
45	The agent transfers the call internally.	Work ID	US_NO_USE
46	The agent transfers a call out.	US_NO_USE	US_NO_USE
47	The agent logs in to the media server.	US_NO_USE	US_NO_USE
50	Video image screening.	US_NO_USE	US_NO_USE
51	Video and audio screening.	US_NO_USE	US_NO_USE
52	The agent logs out of the media server.	US_NO_USE	US_NO_USE
53	The agent transfers the call which is queuing in the common queue.	US_NO_USE	US_NO_USE
54	Release transfer.	US_NO_USE	US_NO_USE
55	The call is transferred to another device.	US_NO_USE	US_NO_USE

Table 1 Types of operations

Operation Type	Description	Operation Object	Operation Reason
56	Configure agent status.	US_NO_USE	Agent status code defined by the service side.
57	Switch the call held during internal-help request processing.	US_NO_USE	Internal-help request switching.
58	The agent is in the conference state.	Work ID	-
59	Multimedia collaboration.	Work ID	0: Initiating a multimedia conference 1: Joining a multimedia conference
60	Conversation status.	US_NO_USE	1: The agent does not really exit the conversation status. Bills are generated every five minutes starting from 00:00 to 23:55. 0: The agent fully exits the conversation status. NOTE: The conversation status covers all call types. Whether to record the conversation status bills is controlled by the license.

5.13 Interfaces of Earlier Versions (Versions Earlier Than 8.13.0)

NOTE

The interfaces provided in CEC 8.13.0 or later have better performance than those described in this section. Therefore, the interfaces in this section are not recommended in CEC 8.13.0 or later.

5.13.1 Querying Details of a Specified Agent

Description

This interface is invoked to query details of a specified agent.

Request Method

The request method must be set to POST. This interface supports only the POST method.

Internal Encapsulation Interfaces

BMS interface: /ccbms/ws/monitor/siteagentinfoobyagentids

Request URL

https://ip:port/rest/cmsapp/v1/openapi/agentgroup/agentbyworkno

In the URL, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed to application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

- Message header

Content-Type:application/json; charset=UTF-8

- Message body

The following provides an example of the request message body of this interface:

```
{
  "queryParam":
  {
    "ccld":1,
    "vdn":1,
    "agentId":101
  }
}
```

[Table 2](#) describes the parameters in the request message body of this interface.

Table 2 Parameters in the message body

Parameter	Type	Mandatory	Description
ccld	Integer	Yes	ID of a call center. The value ranges from 1 to 65535.
vdn	Integer	Yes	ID of a VDN. The value ranges from 1 to 5000.
agentId	Integer	Yes	Agent ID. The value ranges from 101 to 59999.

Response Message

The following provides an example of the response message body of this interface:

```
{
  "result": "0",
  "resultDatas": {
    "workno": 101,
    "vdnid": 1,
    "citStatus": 0,
    "name": "HW35154101",
    "phonenumber": null,
    "logindate": 4294967295000,
    "currentstatetime": 436879,
    "mediatype": 0,
    "status": 2
  }
}
```

[Table 3](#) describes the parameters in the response message body of this interface.

Table 3 Parameters in the message body

Parameter	Type	Description
result	String	Query result. The options are as follows: 0: success Other values: failure For details about the failure causes, see Error Code Reference .
resultDatas	Object	Object in the returned result after a successful query. For details about the parameters of this object, see Table 4 .

Table 4 Parameters of resultDatas

Parameter	Type	Description
workno	Integer	Agent ID.
vdnid	Integer	ID of the VDN to which a call belongs.
citStatus	Integer	Platform status of an agent. For details, see CTI Platform Agent Status Code Reference .
name	String	Agent name.
phonenumber	String	Agent phone number.
logindate	Long	Sign-in time.
currentstatetime	Long	Duration in the current state.
mediatype	Integer	Agent media type. For details, see Agent Type Description .

Table 4 Parameters of resultDatas

Parameter	Type	Description
status	Integer	Agent status. The options are as follows: 0: Whether the SGSN and MME are integrated is unknown. The value corresponds to status 9 on the CTT platform. 2: sign-out The value corresponds to status 0 on the CTT platform. 3: busy The value corresponds to status 7 on the CTI platform. 4: idle The value corresponds to status 1 on the CTT platform. 5: wrap-up The value corresponds to status 6 on the CTT platform. 7: calling state. The value corresponds to states 2, 3, 4, and 5 on the CTT platform. 8: rest The value corresponds to status 8 on the CTT platform.

5.13.2 Querying System Status by Agent

Description

This interface is invoked to query system status by agent.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Internal Encapsulation Interfaces

BMS interface: /ccbms/ws/monitor/teamagentstatusinfo

Request URL

https://ip:port/rest/cmsapp/v1/openapi/querymonitor/systemstate

In the URL, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed to application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

- Message header

Content-Type:application/json; charset=UTF-8

• Message body

The following provides an example of the request message body of this interface:

```
{
  "queryParam":
  {
    "ccld":1,
    "vdn":1
  }
}
```

Table 2 describes the parameters in the request message body of this interface.

Table 2 Parameters in the message body

Parameter	Type	Mandatory	Description
ccld	Integer	Yes	ID of a call center. The value ranges from 1 to 65535.
vdn	Integer	Yes	ID of a VDN. The value ranges from 1 to 5000.

Response Message

The following provides an example of the response message body of this interface:

```
{
  "result": "0",
  "resultDesc": {
    "busyagent": 0,
    "talkingnum": 0,
    "Idleagent": 0,
    "totalloginagent": 0
  }
}
```

Table 3 describes the parameters in the response message body of this interface.

Table 3 Parameters in the message body

Parameter	Type	Description
result	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference .

Table 3 Parameters in the message body

Parameter	Type	Description
resultDatas	Object	Object in the returned result after a successful query. For details about the parameters of this object, see Table 4 .

Table 4 Parameters of resultDatas

Parameter	Type	Description
busyagent	Integer	Number of busy agents.
talkingnum	Integer	Number of agents in talking state.
Idleagent	Integer	Number of idle agents.
totalloginagent	Integer	Number of sign-in agents.

5.13.3 Querying the Agent Information in Rest State

Description

This interface is invoked to query agent information in rest state.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Internal Encapsulation Interfaces

BMS interfaces:

- /ccbms/ws/monitor/teamrestinfo
- /ccbms/ws/monitor/teamrestagentinfo

Request URL

https://ip:port/rest/cmsapp/v1/openapi/agentgroup/agentrestinfo

In the URL, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed to application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

- Message header

Content-Type:application/json; charset=UTF-8

- Message body

The following provides an example of the request message body of this interface:

```
{
  "queryParam":
  {
    "ccld":1,
    "vdn":1
  }
}
```

[Table 2](#) describes the parameters in the request message body of this interface.

Table 2 Parameters in the message body

Parameter	Type	Mandatory	Description
ccld	Integer	Yes	ID of a call center. The value ranges from 1 to 65535.
vdn	Integer	Yes	ID of a VDN. The value ranges from 1 to 5000.

Response Message

The following provides an example of the response message body of this interface:

```
{
  "result": "0",
  "resultDatas":
  {
    "allowrestnum": 101,
    "onrestnum": 1,
    "waitrestnum": 0,
    "agentsrestdetails":
    {[
      "agentname": "z1",
```

```

    "workno":55001
    "cantrestreason":1
    "begintime":2345
    "restduration":22333
  }}
}
}

```

[Table 3](#) describes the parameters in the response message body of this interface.

Table 3 Parameters in the message body

Parameter	Type	Description
result	String	Query result. The options are as follows: 0: success Other values: failure For details about the failure causes, see Error Code Reference .
resultDatas	Object	Object in the returned result after a successful query. For details about the parameters of this object, see Table 4 .

Table 4 Parameters of resultDatas

Parameter	Type	Description
allowrestnum	Integer	Number of agents who are allowed to have a rest but still at work.
onrestnum	Integer	Number of agents that are resting.
waitrestnum	Integer	Number of agents who apply for rest.
agentsrestdetails	Array	List of details about agents who take a rest in the VDN. For details about the parameters of this object, see Table 5 .

Table 5 Parameters of agentsrestdetails

Parameter	Type	Description
agentname	String	Agent name.
workno	Integer	Agent ID.
cantrestreason	Integer	Reason why agents are not allowed to have a rest. This parameter is valid only for the agents who wait for a rest. For details about the parameters of this object, see Table 6 .
begintime	Long	Start time. For resting agents, this parameter indicates the rest start time. For agents who are not in rest, this parameter indicates the requested or approved rest start time. The value is in the timestamp format.
restduration	Long	Requested rest duration, in seconds.

Table 6 Reason for not allowing rest

Value of the Reason for Not Allowing Rest	Description
0	The agent is waiting for approval.
1	The agent is in busy state.
2	The number of agents who can take a rest in the VDN reaches the maximum.
3	The agent does not answer any inbound calls, because the agent does not sign in to any skill queues.
4	The number of resting agents in a skill queue the agent signs in reaches the maximum.

5.13.4 Querying Statistics Details of Skill Queues in Batches

Description

This interface is invoked to query statistics details of skill queues in batches.



- To invoke this interface in the integration environment, set **cclid** of the call center to **1**.

Interface Method

The request method must be set to POST. This interface supports only the POST method.

Internal Encapsulation Interfaces

BMS interfaces:

- /ccbms/ws/monitor/querycurrentcallstatinfo
- /ccbms/ws/monitor/queryStatInfoOfEverySkill

Request URL

<https://ip:port/rest/cmsapp/v1/openapi/realindex/skill/stateinfo>

In the URL, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed to application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

- Message header

Content-Type:application/json; charset=UTF-8

- Message body

The following provides an example of the request message body of this interface:

```
{
  "queryParam":
  {
    "ccld":1,
    "vdn":1,
    "skillIds":[1,2]
  }
}
```

[Table 2](#) describes the parameters in the request message body of this interface.

Table 2 Parameters in the request message body

Parameter	Type	Mandatory	Description
ccld	Integer	Yes	ID of a call center. The value ranges from 1 to 65535.
vdn	Integer	Yes	ID of a VDN. The value ranges from 1 to 5000.
skillIds	Array	Yes	Object array of skill queue ID information. The array cannot exceed 1000 characters.

Response Message

The following provides an example of the response message body of this interface:

```
{
  "result": "0",
  "resultDatas":[{
    "skillId":1,
    "agentLoginNums":1,
```

```
"agentTalkingNums":0,
"agentIdleNums":0,
"agentAvailableNums":0,
"agentSetbusyNums":1,
"agentRestNums":0,
"agentMaxIdleTimeIn5":0,
"agentWorkNums":0,
"agentStudyNums":0,
"agentOccupyNums":0,
"agentPreOccupyNums":0,
"totalCallNums":0,
"processingCallNums":0,
"maxCallWaitTimesIn5":0,
"callWaitNums":0,
"evenCallTimeIn5":0,
"evenWaitTimeIn5":0},
{
"skillId":2,
"agentLoginNums":0,
"agentTalkingNums":0,
"agentIdleNums":0,
"agentAvailableNums":0,
"agentSetbusyNums":0,
"agentRestNums":0,
"agentMaxIdleTimeIn5":0,
"agentWorkNums":0,
"agentStudyNums":0,
"agentOccupyNums":0,
"agentPreOccupyNums":0,
"totalCallNums":0,
"processingCallNums":0,
"maxCallWaitTimesIn5":0,
"callWaitNums":0,
"evenCallTimeIn5":0,
"evenWaitTimeIn5":0}
]
}
```

Table 3 describes the parameters in the response message body of this interface.

Table 3 Parameters in the response message body

Parameter	Type	Description
result	String	Query result. The options are as follows:

Table 3 Parameters in the response message body

Parameter	Type	Description
		0 : success Other values: failure For details about the failure causes, see Error Code Reference .
resultDatas	Array	Object array in the returned result after a successful query. For details about the parameters of this object in the array, see Table 4 .

Table 4 Parameters of resultDatas

Parameter	Type	Description
skillId	Integer	Skill queue ID.
agentLoginNums	Integer	Number of sign-in agents.
agentTalkingNums	Integer	Number of agents in talking state.
agentIdleNums	Integer	Number of idle agents.
agentAvailableNums	Integer	Number of available agents. The value may be different from the number of idle agents when an agent can have one-to-many conversations.
agentSetbusyNums	Integer	Number of agents in busy state.
agentRestNums	Integer	Number of agents in rest state.
agentMaxIdleTimeIn5	Integer	Maximum idle duration of agents in 5 minutes.
agentWorkNums	Integer	Number of agents in wrap-up state.
agentStudyNums	Integer	Number of agents in learning state.
agentOccupyNums	Integer	Number of agents in occupied state.
agentPreOccupyNums	Integer	Number of preoccupied agents.
totalCallNums	Integer	Total number of processed calls, including the calls that are being processed.
processingCallNums	Integer	Number of calls that are being processed.
maxCallWaitTimeIn5	Integer	Maximum time that a call can wait in 5 minutes.
callWaitNums	Integer	Number of queuing calls (number of calls that are waiting).
evenCallTimeIn5	Integer	Average call duration in 5 minutes.
evenWaitTimeIn5	Integer	Average call waiting duration in 5 minutes.

5.13.5 Querying Details About Agents in Configured Skill Queues

Description

This interface is invoked to query details about agents in configured skill queues.

Request Method

The request method must be set to POST. This interface supports only the POST method.

Internal Encapsulation Interfaces

BMS interface: /ccbms/ws/monitor/agentstatusinfo

Request URL

https://ip:port/rest/cmsapp/v1/openapi/agentgroup/agentbyconfigedskill

In the URL, *ip* indicates the IP address of the NSLB server, and *port* indicates the HTTPS port number of the CC-CMS service mapped on the NSLB server.

Request Description

Table 1 Parameters in the request header

No.	Parameter	Type	Mandatory	Description
1	Content-Type	String	Yes	The value is fixed to application/json; charset=UTF-8 .
2	Authorization	String	Yes	For details about the generation mode, see C2 Monitoring, System Outbound Call, CDR, and Knowledge Base Interface Authentication .

- Message header

Content-Type:application/json; charset=UTF-8

- Message body

The following provides an example of the request message body of this interface:

```
{
  "queryParam":
  {
    "ccld":1,
    "vdn":1,
    "skillId":1
  }
}
```

```
}

```

Table 2 describes the parameters in the request message body of this interface.

Table 2 Parameters in the message body

Parameter	Type	Mandatory	Description
cclid	Integer	Yes	ID of a call center. The value ranges from 1 to 65535.
vdn	Integer	Yes	ID of a VDN. The value ranges from 1 to 5000.
skillId	Integer	Yes	Skill queue ID. The value ranges from 1 to 2000.

Response Message

The following provides an example of the response message body of this interface:

```
{
  "result": "0",
  "resultDatas": [{
    "workno": 20011,
    "name": "20011",
    "status": 4,
    "ctiStatus": 1,
    "groupid": 1,
    "mediatype": 1,
    "vdnid": 1,
    "phonenum": "70942",
    "currentstatetime": 1505,
    "logindate": 1501125758000
  ]
}
```

Table 3 describes the parameters in the message body of this interface.

Table 3 Parameters in the message body

Parameter	Type	Description
result	String	Query result. The options are as follows: 0 : success Other values: failure For details about the failure causes, see Error Code Reference .
resultDatas	Array	Array in the returned result after a successful query.

Table 3 Parameters in the message body

Parameter	Type	Description
		For details about the parameters of this object in the array, see Table 4 .

Table 4 Parameters of resultDatas

Parameter	Type	Description
workno	Integer	Agent ID.
name	String	Agent name.
status	Integer	Agent status. For details, see Table 5 .
ctiStatus	Integer	Agent platform status. For details, see CTI Platform Agent Status Code Reference .
groupid	Integer	Agent group ID.
mediatype	Integer	Agent media type. For details, see Agent Type Description .
vdnid	Integer	ID of the VDN to which a call belongs.
phonenumber	String	Agent phone number.
currentstatetime	Long	Duration in the current state.
logindate	Long	Sign-in time.

Table 5 Agent status

Value	Description
0	Unknown. The value corresponds to status 9 on the CTI platform.
2	An exception occurs when the interface is invoked.
3	Showing busy. The value corresponds to status 7 on the CTI platform.
4	Idle. The value corresponds to status 1 on the CTI platform.
5	Wrap-up. The value corresponds to status 6 on the CTI platform.
7	Talking. The value corresponds to states 2, 3, 4, and 5 on the CTI platform.
8	Rest. The value corresponds to status 8 on the CTI platform.